MATH REVIEW

APPENDIX **A**

Some knowledge of algebra and trigonometry is essential for a full understanding of this book. In addition, a limited ability to deal with complex numbers and derivatives (a part of calculus) is helpful, although not entirely essential. This appendix is meant as the briefest of summaries of complex numbers and differentiation, preceded by a collection of useful formulas from trigonometry, exponentials, and logarithms. It is not meant as a textbook substitute. For a highly readable self-help book on calculus, we recommend *Quick Calculus*, by D. Kleppner and N. Ramsey, Wiley, 2nd ed., 1985.

A.1 Trigonometry, exponentials, and logarithms

Here is a collection of useful formulas:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

is the solution of the quadratic equation

$$ax^2 + bx + c = 0.$$

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y,$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y,$$

$$\sin 2x = 2 \sin x \cos x,$$

$$\cos x \cos y = \frac{1}{2}\left[\cos(x+y) + \cos(x-y)\right],$$

$$\cos x \sin y = \frac{1}{2}\left[\sin(x+y) - \sin(x-y)\right],$$

$$\sin x \sin y = \frac{1}{2}\left[\cos(x-y) - \cos(x+y)\right]$$

$$e^{x+y} = e^x e^y,$$

$$e^{x-y} = e^x / e^y,$$

$$x^{a/b} = \sqrt[b]{x^a},$$

$$e^{\log_e x} = x,$$

$$\log_e(xy) = \log_e x + \log_e y,$$

$$\log_e(x/y) = \log_e x - \log_e y,$$

$$\log_e x^n = n \log_e x,$$

$$\log_e e^x = x,$$

$$\log_e x = \log_e 10 \log_{10} x \approx 2.3 \log_{10} x,$$

$$a^x = e^{x \log_e a}.$$

A.2 Complex numbers

A complex number is an object of the form

$$\mathbf{N} = a + ib,$$

where a and b are real numbers and i is the square root of -1; a is called the real part, and b is called the imaginary part.[1] Boldface letters or squiggly underlines are sometimes used to denote complex numbers. At other times you're just supposed to *know*!

Complex numbers can be added, subtracted, multiplied, etc., just as real numbers:

$$(a+ib) + (c+id) = (a+c) + i(b+d),$$

$$(a+ib) - (c+id) = (a-c) + i(b-d),$$

$$(a+ib)(c+id) = (ac-bd) + i(bc+ad),$$

$$\frac{a+ib}{c+id} = \frac{(a+ib)(c-id)}{(c+id)(c-id)} = \frac{ac+bd}{c^2+d^2} + \frac{bc-ad}{c^2+d^2}\,i.$$

[1] Electrical engineers depart from the universal convention of $i \equiv \sqrt{-1}$, using instead the symbol j in order to avoid duplicating the use of the symbol i (which designates small-signal current). We follow the EEs in this book, but not in this Math appendix. Were we to do so, we would likely be disowned by our math colleagues.

All these operations are natural, in the sense that you just treat i as something that multiplies the imaginary part, and go ahead with ordinary arithmetic. Note that $i^2 = -1$ (used in the multiplication example) and that division is simplified by multiplying top and bottom by the *complex conjugate*, the number you get by changing the sign of the imaginary part. The complex conjugate is sometimes indicated with an asterisk. If

$$\mathbf{N} = a + ib,$$

then

$$\mathbf{N}^* = a - ib.$$

The magnitude (or *modulus*) of a complex number is a real number with no imaginary part:

$$|\mathbf{N}| = |a+ib| = \sqrt{(a+ib)(a-ib)} = \sqrt{a^2+b^2},$$

i.e.,

$$|\mathbf{N}| = \sqrt{\mathbf{N}\mathbf{N}^*},$$

simply obtained by multiplying by the complex conjugate and taking the square root. The magnitude of the product (or quotient) of two complex numbers is simply the product (or quotient) of their magnitudes.

The real (or imaginary) part of a complex number is sometimes written as

real part of $\mathbf{N} = \mathscr{R}e(\mathbf{N})$,
imaginary part of $\mathbf{N} = \mathscr{I}m(\mathbf{N})$.

You get them by writing out the number in the form $a+ib$, then taking either a or b. This may involve some multiplication or division, since the complex number may be a real mess.

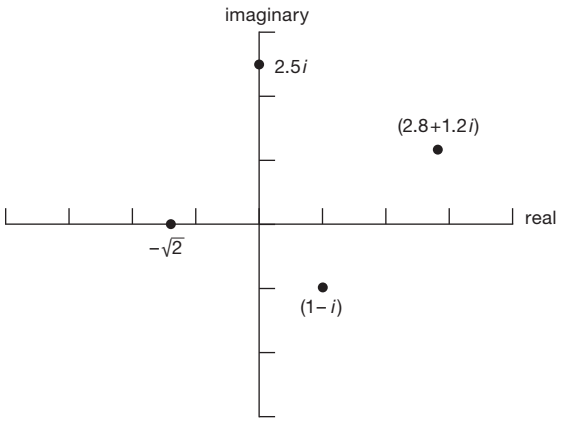


Figure A.1. Complex numbers in the "complex plane."

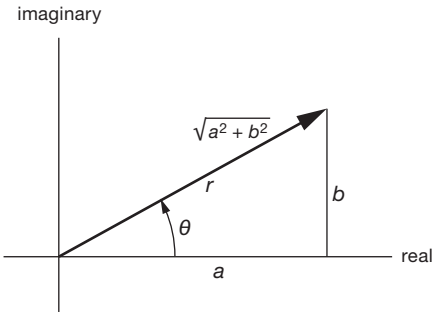


Figure A.2. Complex numbers, as magnitude and angle.

Complex numbers are sometimes represented on the complex plane. It looks just like an ordinary x,y graph, except that a complex number is represented by plotting its real part as x and its imaginary part as y, as shown in Figure A.1. In keeping with this analogy, you sometimes see complex numbers written just like x,y coordinates:

$$a + ib \leftrightarrow (a,b).$$

Just as with ordinary x,y pairs, complex numbers can be represented in polar coordinates; that's known as "magnitude, angle" representation. For example, the number $a+ib$ can also be written as (Figure A.2)

$$a + ib = r\angle\theta,$$

where[2] $r = \sqrt{a^2+b^2}$ and $\theta = \tan^{-1}(b/a)$. This is usually written in a different way, using the astonishing fact that

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

(You can derive the preceding result, known as Euler's[3] formula, by expanding the exponential in a Taylor series.) Thus we have the following equivalents:

$$\begin{aligned} \mathbf{N} &= a + ib = re^{i\theta}, \\ r &= |\mathbf{N}| = \sqrt{\mathbf{N}\mathbf{N}^*} = \sqrt{a^2+b^2}, \\ \theta &= \tan^{-1}(b/a), \end{aligned}$$

i.e., the modulus r and angle θ are simply the polar coordinates of the point that represents the number in the complex plane. Polar form is handy when complex numbers have to be multiplied; you just multiply their magnitudes and add their angles (or, to divide, you divide their magnitudes and subtract their angles):

$$(r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) = r_1 r_2 e^{i(\theta_1+\theta_2)}.$$

[2] Caution: the formula for θ returns values only between $-\pi/2$ and $+\pi/2$; the signs of both a and b, and not merely their quotient, are required for a correct value of θ in all four quadrants.

[3] Leonhard Euler, pronounced like "oiler."

Finally, to convert from polar to rectangular form, just use Euler's formula:

$$re^{i\theta} = r\cos\theta + ir\sin\theta,$$

i.e.,

$$\mathscr{R}e(re^{i\theta}) = r\cos\theta,$$
$$\mathscr{I}m(re^{i\theta}) = r\sin\theta.$$

(These can be used to easily derive the sum and difference of trigonometric functions, so you never have to remember those pesky formulas. Just work out $e^{i(x\pm y)}$.)

If you have a complex number multiplying a complex exponential, just do the necessary multiplications. If

$$\begin{aligned}\mathbf{N} &= a+ib,\\ \mathbf{N}e^{i\theta} &= (a+ib)(\cos\theta+i\sin\theta),\\ &= (a\cos\theta - b\sin\theta),\\ &\quad +i(b\cos\theta + a\sin\theta).\end{aligned}$$

When dealing with circuits and signals, the angular argument θ often takes the form of an evolving wave: $\theta = \omega t = 2\pi ft$; thus, for example, $V(t) = \mathscr{R}e(V_0e^{i\omega t}) = V_0\cos\omega t$, etc.

A.3 Differentiation (Calculus)

We start with the concept of a *function* $f(x)$, i.e., a formula that gives a value $y = f(x)$ for each x. The function $f(x)$ should be *single valued* i.e., it should give a single value of y for each x. You can think of $y = f(x)$ as a graph, as in Figure A.3. The derivative of y with respect to x, written dy/dx ("dee y dee x"), is the *slope* of the graph of y versus x. If you draw a tangent to the curve at some point, its slope is dy/dx *at that point*; i.e., the derivative is itself a function, since it has a value at each point. In Figure A.3 the slope at the point $(1, 1)$ happens to be 2, whereas the slope at the origin is zero (we'll see shortly how to compute the derivative).

In mathematical terms, the derivative is the limiting value of the ratio of the change in y (Δy) to the change in x (Δx), as Δx goes to zero. To quote a song once sung in the hallowed halls of Harvard (by Tom Lehrer and Lewis Branscomb),

> You take a function of x, and you call it y
> Take any x-nought that you care to try
> Make a little change and call it delta x
> The corresponding change in y is what you find nex'
> And then you take the quotient, and now, carefully

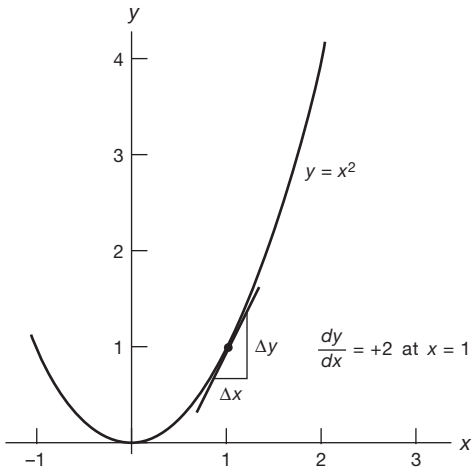


Figure A.3. A single-valued function: $f(x) = x^2$.

> Send delta x to zero, and I think you'll see
> That what the limit gives us (if our work all checks)
> Is what you call $dy/dx\ldots$
> It's just dy/dx.
>
> (*The Derivative Song*, sung to the tune of *There'll Be Some Changes Made*, W. Benton Overstreet).

Differentiation is a straightforward art, and the derivatives of many common functions are tabulated in standard tables and automatically calculated in programs like Mathematica®. Here are some rules (u and v are arbitrary functions of x, and a represents a constant).

A.3.1 Derivatives of some common functions

$$\frac{d}{dx}a = 0$$

$$\frac{d}{dx}ax = a$$

$$\frac{d}{dx}ax^n = anx^{n-1},$$

$$\frac{d}{dx}\sin ax = a\cos ax,$$

$$\frac{d}{dx}\cos ax = -a\sin ax,$$

$$\frac{d}{dx}e^{ax} = ae^{ax},$$

$$\frac{d}{dx}\log_e x = 1/x.$$

A.3.2 Some rules for combining derivatives

Here $u(x)$ and $v(x)$ represent generic functions of x:

$$\frac{d}{dx}au(x) = a\frac{d}{dx}u(x),$$

$$\frac{d}{dx}(u+v) = \frac{du}{dx} + \frac{dv}{dx},$$

$$\frac{d}{dx}uv = u\frac{d}{dx}v + v\frac{d}{dx}u,$$

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\dfrac{du}{dx} - u\dfrac{dv}{dx}}{v^2},$$

$$\frac{d}{dx}\log_e u = \frac{1}{u}\frac{du}{dx}$$

$$\frac{d}{dx}\{u[v(x)]\} = \frac{du}{dv}\frac{dv}{dx}.$$

The last one is very useful and is called the chain rule.

A.3.3 Some examples of differentiation

$$\frac{d}{dx}x^2 = 2x,$$

$$\frac{d}{dx}(1/x^{\frac{1}{2}}) = -\frac{1}{2}x^{-\frac{3}{2}},$$

$$\frac{d}{dx}xe^x = xe^x + e^x \qquad \text{(product rule)},$$

$$\frac{d}{dx}e^{-x^2} = -2xe^{-x^2} \qquad \text{(chain rule)},$$

$$\frac{d}{dx}a^x = \frac{d}{dx}(e^{x\log_e a}) = a^x \log_e a \quad \text{(chain rule)}.$$

Once you have differentiated a function, you often want to evaluate the value of the derivative at some point. Other times you may want to find a minimum or maximum of the function; that's the same thing as having a zero derivative, so you can just set the derivative equal to zero and solve for x. For example, you can easily determine that the slope of the function plotted in Figure A.3 equals 2 at x=1, and that its minimum occurs at x=0 (where its slope is zero).

HOW TO DRAW SCHEMATIC DIAGRAMS

APPENDIX **B**

A well-drawn schematic makes it easy to understand how a circuit works, and it aids greatly in troubleshooting. A poor schematic only creates confusion. By keeping a few rules and suggestions in mind, you can draw a good schematic in no more time than it takes to draw a poor one. In this appendix we dispense advice of three varieties: general principles, rules, and hints. We have also drawn some real knee-slappers to illustrate habits to avoid.

B.1 General principles

- Schematics should be unambiguous. Therefore pin numbers, parts values, reference designators, polarities, etc., should be clearly labeled to avoid confusion.
- A good schematic makes circuit functions clear. Therefore keep functional areas distinct; don't be afraid to leave blank areas on the page, and don't try to fill the page. There are conventional ways to draw functional subunits; for instance, don't draw a differential amplifier as in Figure B.1, because the function won't be easily recognized. Likewise, flip-flops are usually drawn with clock and inputs on the left, set and clear on top and bottom, and outputs on the right.

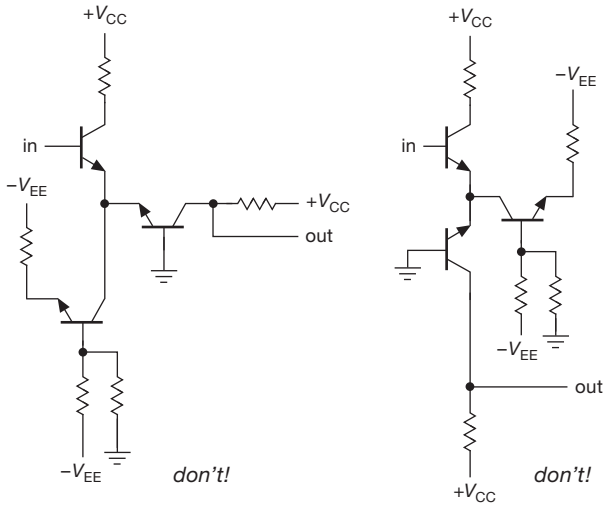


Figure B.1. Arrange components so that the function (here a differential amplifier) is clear. Don't corrupt the presentation to save space.

B.2 Rules

- Wires connecting are indicated by heavy black dots; wires crossing, but not connecting, have no dot (don't use a little half-circular "jog"; it went out in the 1950s).
- Four wires must not connect at a point; i.e., wires must not cross *and* connect. You sometimes see this rule violated, but it's poor practice (because a missing or undersized dot is a different circuit).
- Always use the same symbol for the same device; e.g., don't draw flip-flops in two different ways (exception: assertion-level logic symbols show each gate in two possible ways).
- Wires and components are aligned horizontally or vertically, unless there's a good reason to do otherwise.
- Label pin numbers on the outside of a symbol, signal names on the inside.
- All parts should have values or types indicated; it's best to give all parts a label ("refdes"), too, e.g., R_7 or U_3.

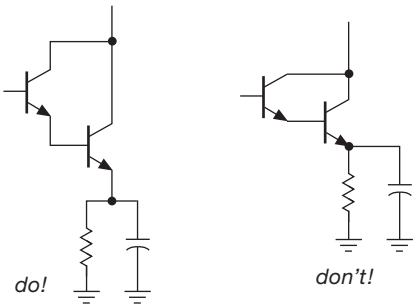
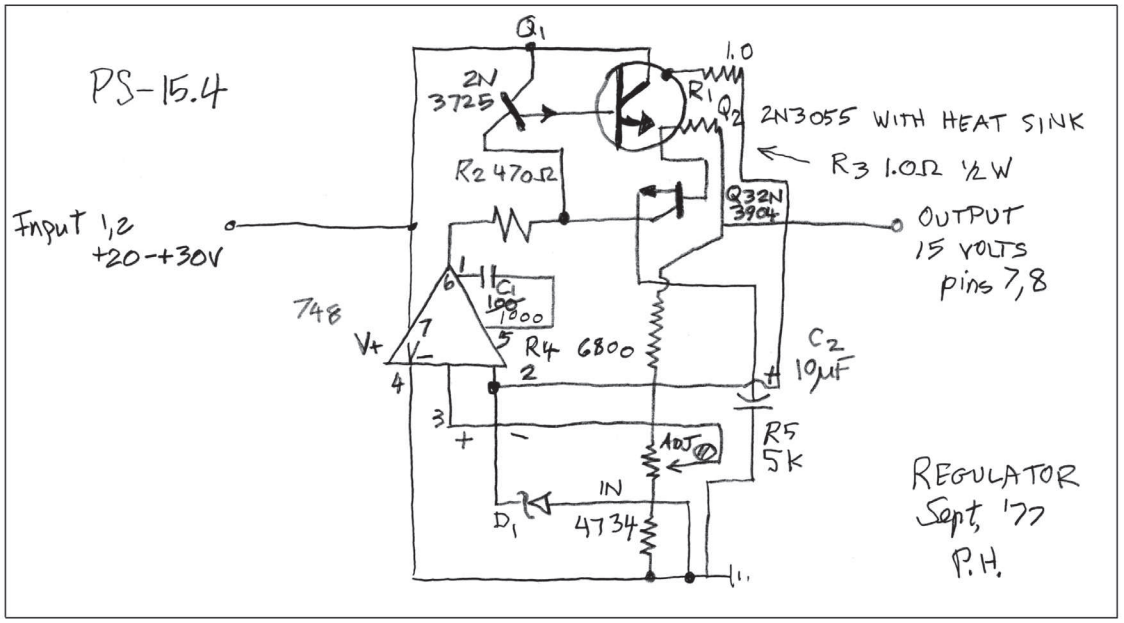


Figure B.2. Bring leads away from component symbols before connecting or jogging.



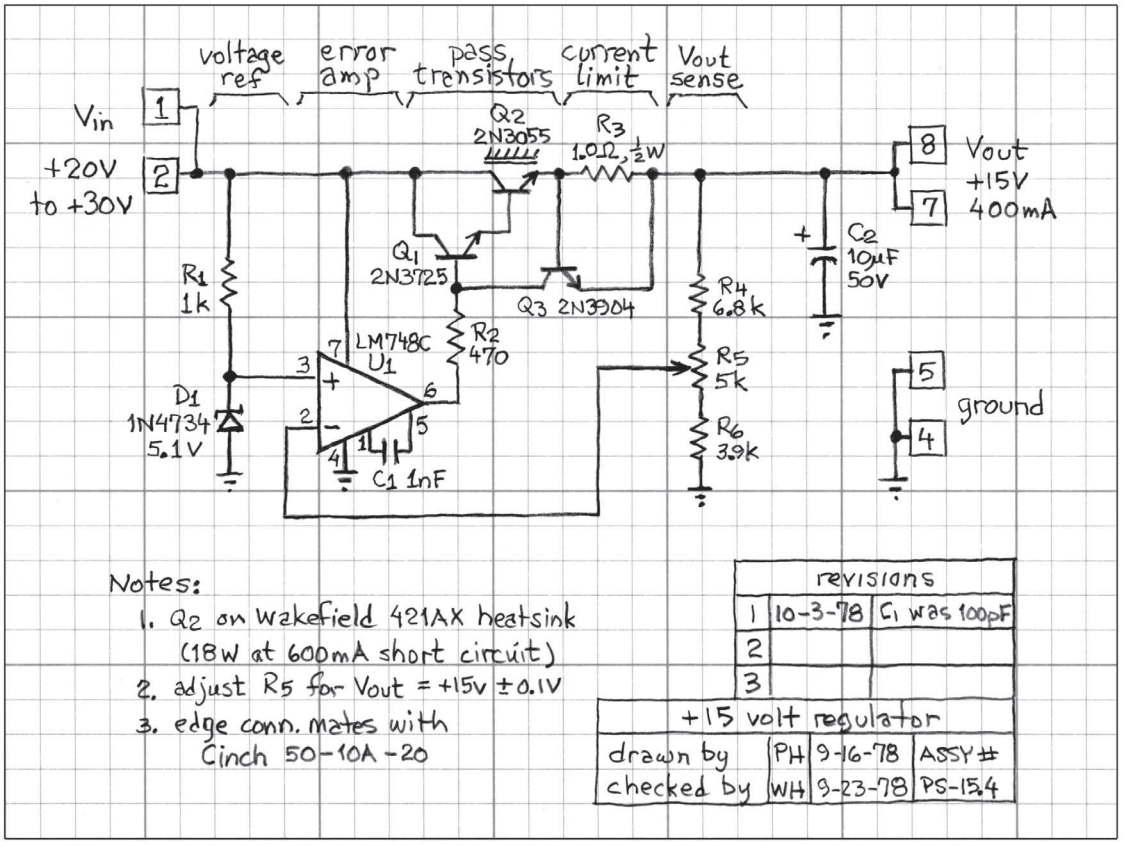


Figure B.3. Good drawing (perhaps too good) on “engineering paper,” and *very* bad. Guess which is which.

B.3 Hints

- Identify parts immediately adjacent to the symbol, forming a distinct group giving symbol, label, and type or value.
- In general, signals go from left to right; don't be dogmatic about this, though, if clarity is sacrificed.
- Put positive supply voltages at the top of the page, negative at the bottom. Thus *npn* transistors will usually have their emitter at the bottom, whereas *pnp*s will have their emitter topmost.
- Don't attempt to bring all wires around to the supply rails, or to a common ground wire. Instead, use the ground symbol(s) and labels like $+V_{CC}$ to indicate those voltages where needed.
- It is helpful to label signals and functional blocks and show waveforms; in logic diagrams it is especially important to label signal lines, e.g., RESET′ or CLK.
- It is helpful to bring leads[1] away from components a short distance before making connections or jogs. For example, draw transistors as in Figure B.2.
- Leave some space around circuit symbols; e.g., don't draw components or wires too close to an op-amp symbol. This keeps the drawing uncluttered and leaves room for labels, pin numbers, etc.
- Label all boxes that aren't obvious: comparator versus op-amp, shift register versus counter, etc. Don't be afraid to invent a new symbol.
- Use small rectangles, ovals, or circles to indicate card-edge connections, connector pins, etc. Be consistent.
- The signal path through switches should be clear. Don't force the reader to follow wires all over the page to find out how a signal is switched.
- Power-supply connections are normally assumed for op-amps and logic devices. However, show any unusual connections (e.g., an op-amp run from a single supply, where $V_- =$ ground), and the disposition of unused inputs.
- It is very helpful to include a small table of integrated circuit (IC) numbers, types, and power-supply connections (pin numbers for V_{CC} and ground, for instance).
- Include a title area near the bottom of the page, with name of circuit, name of instrument, by whom drawn, by whom designed or checked, date, and assembly number. Also include a revision area, with columns for revision number, date, and subject.
- We recommend drawing schematics freehand on coarse graph paper (pale gridlines, five per inch, for example National® Brand "Engineer's Computation Pad" in "Eye-Ease"® green), or on plain paper on top of graph paper. This is fast, and it gives very pleasing results. Use dark pencil (we like HB hardness, 0.5 mm diameter) or ink; avoid ballpoint or felt-tip pen.

[1] *Leads?* Yeah, sure. . . I'll just check with the boys down at the crime lab. They got four more detectives working on the case. They got us working in *shifts*. Hahahaha. . . *LEADS!*

B.4 A humble example

As an illustration, we've drawn a humble example (Figure B.3) showing "awful" and "good" schematics of the same circuit; the former violates nearly every rule and is almost impossible to understand. See how many bad habits you can find illustrated. We've seen all of them in professionally drawn schematics! (We drew the "bad" schematic in an airport while waiting for a flight. It was an occasion of great hilarity; we laughed ourselves silly.)

RESISTOR TYPES

APPENDIX **C**

C.1 Some history

For a half century people used "leaded" (pronounced lēd′-ed) resistors: if you look inside a very old radio (before ∼1950), you'll see colored cylindrical objects with some colored dots painted on them and with a wire wrapped around each end that comes off perpendicularly to the axis ("radial leads"). These carbon-composition resistors evolved into the standard carbon-comp "axial-lead" resistors (still cylindrical, but with colored stripes all the way around, and with the wires now sticking straight out each end) that were dominant through the last half of the 20th century (and that we recommended for noncritical applications in our previous book editions). Axial-lead resistors are still popular for some uses, such as easy breadboarding in the lab. They're also used in applications that require very high resistance ($\geq$100 MΩ), or high voltage or power ratings, or for resistors of very high precision.

However, contemporary electronics has embraced surface-mount packaging, because of its high density (SMT devices are *small*, and you don't have to take up space with holes for the leads). Surface-mount resistors, in common with other two-terminal SMT components (capacitors, inductors), are available in a range of package sizes, characterized by a 4-digit code giving their length and width in units of 0.010″; for example, an "0603" package is 0.06″×0.03″ (1.5 mm × 0.75 mm). We favor that size, or the larger 0805 package, for general prototyping of surface-mount circuits. The smaller packages (0402, 0201, and even "01005") are a major pain – you basically have to work under a microscope (and don't sneeze).

C.2 Available resistance values

You can't get just any old resistance value. Available resistances fall into what's called an EIA Standard Decade, named by the number of values per decade (thus E24 – used for 5% tolerance resistors – has 24 values, spaced approximately 10% apart; see below). Resistors with 1% tolerance are quite inexpensive these days, costing hardly more than an analogous 5% resistor,[1] so you might as well use 1% resistors by default. They come in the E96 set of standard values (96 values per decade, spaced approximately 2% apart; thus 481 values from 10 Ω through 1 MΩ, see below). Resistors of greater precision (e.g., 0.1%) are sometimes available in the **E192** superset,[2] and in convenient round-number values (e.g., 250, 300, 400, or 500) that are not included in the EIA sequences.

Here is the **E24** set of "5%" values (the **E12** subset, used for 10% components, is shown in **bold**):

10	16	**27**	43	**68**
11	**18**	30	**47**	75
12	20	**33**	51	**82**
13	**22**	36	**56**	91
15	24	**39**	62	**100**

And here is the **E96** set of "1%" values (the **E48** set, used for 2% components, or for a reduced set of 1% parts, is in **bold**):

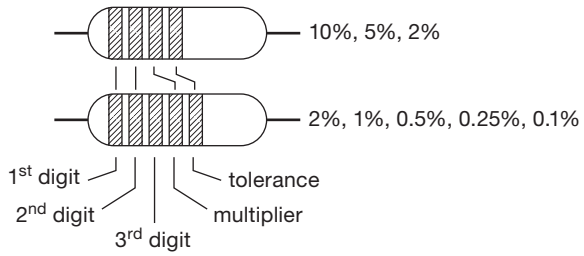
100	137	**187**	255	**348**	475	**649**	887
102	**140**	191	**261**	357	**487**	665	**909**
105	143	**196**	267	**365**	499	**681**	931
107	**147**	200	**274**	374	**511**	698	**953**
110	150	**205**	280	**383**	523	**715**	976
113	**154**	210	**287**	392	**536**	732	
115	158	**215**	294	**402**	549	**750**	
118	**162**	221	**301**	412	**562**	768	
121	165	**226**	309	**422**	576	**787**	
124	**169**	232	**316**	432	**590**	806	
127	174	**237**	324	**442**	604	**825**	
130	**178**	243	**332**	453	**619**	845	
133	182	**249**	340	**464**	634	**866**	

[1] For example, the Digi-Key catalog shows a full selection of Vishay/Dale CRCW-series surface-mount resistors, in sizes from 1210 down to 0201. For the convenient 0603 size, the current prices for 1% and 5% resistors are $0.025 and $0.023 apiece, respectively, in quantity 200. (You'll pay about triple that, in quantity 10, and about a fifth as much, in a full reel of 5000 resistors.)

[2] The full E192 set, along with subsets, is nicely displayed at `http://www.logwell.com/tech/components/resistor_values.html`.

C.3 Resistance marking

Leaded resistors are marked in one of two ways: (a) with a set of four or five color bands, indicating resistance and tolerance; or (b) with a 4-digit resistance code, followed by a letter that indicates the tolerance. Surface-mount resistors use either (a) a 3- or 4-digit resistance code, or, for the smallest package sizes, (b) no marking at all!



color	digit	multiplier	tolerance	(tol. suffix)
black	0	1	–	
brown	1	10	1%	F
red	2	100	2%	G
orange	3	1k	–	
yellow	4	10k	–	
green	5	100k	0.5%	D
blue	6	1M	0.25%	C
violet	7	10M	0.1%	B
gray	8	–	0.05%	A, W
white	9	–	–	
gold	–	0.1	5%	J
silver	–	0.01	10%	K
(none)	–	–	20%	M
			0.02%	N, Q, P
			0.01%	T, L
			0.005%	V
			0.0025%	X
			0.002%	U
			0.001%	S

Figure C.1. The resistor color code, used on some axial-lead resistors (notably carbon-film and carbon-composition types). The resistance is read as a 2- or 3-digit integer (depending on resistor precision) followed by a band indicating the power-of-10 multiplier. For example, yellow-violet-orange-gold is 47 kΩ±5%, and yellow-white-white-black-brown is 499 Ω±1%. The alphabetic tolerance suffix is used on resistors with numerical printed resistance values.

Although it may seem diabolical to the beginner, the practice of color banding makes it easy to recognize resistor values in a circuit or parts bin, without having to search for a printed legend. Each color corresponds to a digit, in a sort of floating-point format (with the final digit indicating the power of ten); a last color band signifies the tolerance. See Figure C.1. Resistors with numerical markings use the same system, but with the digits themselves printed along the body of the resistor (for leaded resistors), or on the top side of a surface-mount package; a final letter signifies the tolerance, as shown in the figure.

C.4 Resistor types

The usual choices for general-purpose use are metal-film (axial-lead) or thick-film (surface-mount) parts. Thin-film surface-mount resistors offer improved characteristics (accuracy, stability, and ability to operate in cryogenic environments). For power applications you usually use wirewound resistors, either in an air-cooled ceramic package or a conduction-cooled ("Dale-type") metal package. High-value resistors ($>10\,M\Omega$, say) are usually of metal-oxide construction (e.g., Ohmite "Mini-Mox" or "Super Mox," or Vishay RNX-series). Film resistors are not tolerant of high peak power; for such applications use something like ceramic or carbon composition, or other styles specified for peak-power use. For the utmost in stability and low temperature coefficient (tempco), you can't beat the excellent metal-foil types from Vishay. They exploit a clever design, in which the positive tempco of the resistive metal element (firmly attached to an insulating substrate) is cancelled by the negative strain-induced tempco caused by differential expansion of the substrate.[3] We've listed some comparative resistor properties in Table C.1; for much more detail see §*1x.2*.

General-purpose resistors are ridiculously inexpensive – thick-film surface-mount resistors cost a few cents apiece in small quantities, and just fractions of a cent apiece in full reel quantities (5000 pieces, for 0603 size). Distributors may be unwilling to sell fewer than 25 to 50 pieces of one value; thus an assortment box (e.g., from Yageo or Vishay/BC) may be a wise purchase. We particularly like the nice packaging and good pricing of the kits from SMT Zone (www.smtzone.com).

C.5 Confusion derby

Component markings should be clear and unambiguous. Sometimes it just ain't so! See Figure 1.130 for some real head-scratchers, both resistive and otherwise.

[3] Check it out: Felix Zandman's 1982 US patent #4,318,072, "Precision resistor with improved temperature characteristics."

Table C.1 Selected Resistor Types

	Resistor Type					
	carbon comp axial	**thick film** SMT-0603	**thin film** SMT-0603	**metal film** axial	**metal foil** SMT	
Parameter	(RC-07)	(Vishay CRCW)	(KOA Speer RN73)	(RN-55D)	(Vishay VSMP)	**Units**
Tolerances	5%, 10%	1%, 5%	0.05%-1%	0.1%-1%	0.01%-1%	Δ*R*/*R*
Temp coef	~1000	100, 200	5, 10, 25, 50, 100	50, 100	0.05 (typ)	ppm/C
Load life	10%	2%	0.25%	0.5%	0.01%	Δ*R*/*R*
Moisture	10%	2%	0.5%	0.5%	0.02%	Δ*R*/*R*
Thermal cycle	2%	2%	0.25%	0.25%	0.01%	Δ*R*/*R*
Low temp	3%	-	-	0.25%	0.01%	Δ*R*/*R*
Overload	2%	0.5%	0.1%	0.25%	0.01%	Δ*R*/*R*
Soldering	3%	0.5%	0.1%	0.25%	0.01%	Δ*R*/*R*
Vibration	2%	-	-	0.25%	-	Δ*R*/*R*
Voltage coef	-	-	-	5	0.1	ppm/V
Self-heating	-	-	-	-	5ppm	Δ*R*/*R*
Price (approx)	$0.35	$0.025	$0.32	$0.05	$10	ea, qty 100
(for tol and TC)	(5%)	(1%, TC=200)	(0.1%, TC=25)	(1%, TC=100)	(0.01%, TC=0.05)	

Properties of selected resistor types. The legendary axial-lead "carbon-composition" resistors have been superseded by inexpensive metal-film (or carbon-film) types, with greatly improved properties (except for peak-power endurance, see Chapter *1x*). We like Vishay's CMF-55 metal-film resistors (industrial version of MIL RN-55D). For most surface-mount applications the "thick-film" (a metal-ceramic composite) types are fine, though thin-film and metal-film resistors have somewhat better properties. The extraordinary Vishay "Z-foil" ultraprecision hermetically sealed resistor is listed to show the best that is currently available (but if you have to ask the price, you probably can't afford it). It's useful to note that a parameter like a voltage coefficient of 5 ppm/V corresponds to a change of 0.1% over a full 200 V operating range.

THÉVENIN'S THEOREM

APPENDIX **D**

In Chapter 1 we stated (but did not "prove") Thévenin's Theorem, namely that any two-terminal network whose internal circuitry consists solely of resistors, batteries, and current sources, interconnected in any manner whatsoever, is equivalent (and indistinguishable) from the two-terminal network consisting of a single battery V_{TH} in series with a single resistor R_{TH};[1] see Figure D.1. We did not prove it, because, in the spirit of this book, we don't *prove* anything; we show you how to design circuits, instead. We make an exception here, because it's nice to see *something* proved, right?

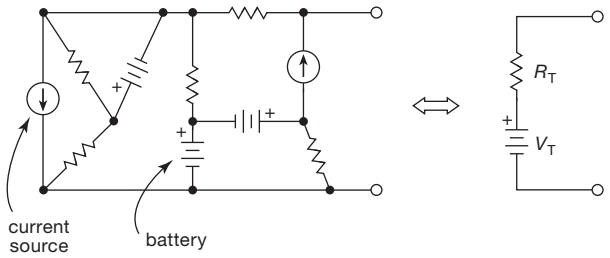


Figure D.1. Thévenin's Theorem: a single resistor in series with a single battery can mimic any mess of a two-terminal network made from resistors, batteries, and current sources.

D.1 The proof

For linear circuit elements (here resistors), the "nodal equations" (Kirchhoff's voltage law, KVL, and Kirchhoff's current law, KCL) are a set of linear equations. So we can find any circuit quantity (a voltage or a current), which depends on all the "independent sources" (batteries, current sources), by turning on each source in turn, and adding the partial contributions. (This is exactly analogous to using superposition to find, say, the electric field from a set of charges.) This technique is often useful in circuit analysis.

Here we wish to mimic the V versus I of the actual circuit with the (simpler) Thévenin equivalent of a single battery in series with a single resistor. Imagine we determine that V versus I function by applying an external current I_{ext} that flows through the two-terminal circuit, and observing the resultant V across those same two terminals. V depends on I_{ext} *and* on all the internal batteries (V_{int}) and current sources (I_{int}).

1. Set all $V_{int} = 0$ and all $I_{int} = 0$; that is, replace all internal batteries with short circuits and all current sources with open circuits. Now, with a given applied I_{ext}, observe V_1.
2. Define $R_T = V_1/I_{ext}$. (They must be proportional, by linearity.)
3. Now set $I_{ext} = 0$, and turn on the internal batteries and current sources. Observe V_2, which we will call V_T.
4. Finally, by superposition it must be the case that

$$V(\text{actual}) = V_1 + V_2 = I_{ext}R_T + V_T.$$

This is true for all I_{ext}, and is exactly what you get with the Thévenin equivalent circuit, when connected to *any* load (which need not be linear); see Figure D.2.

To summarize: (a) you determine R_T and V_T by first finding the open-circuit voltage, which equals V_T; then (b) you find the short-circuit current, I_{SC}, which equals the ratio of V_T to R_T. In other words, $V_T = V_{OC}$ and $R_T = V_{OC}/I_{SC}$. You do this by analysis, if you know the "black-box" circuit; or by measurement, if you don't.

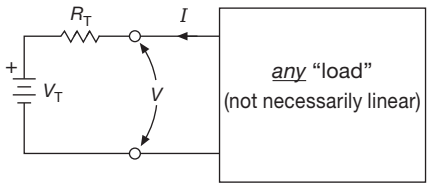


Figure D.2. The Thévenin equivalent circuit behaves exactly like the original network, regardless of the nature of the load.

D.1.1 Two examples – voltage dividers

Figures D.3 and D.4 show two simple examples, variations on the resistive divider. Interestingly, their Thévenin equivalent circuits are different, even though the resistor values and the open-circuit voltages are the same.

[1] A related theorem is Norton's, where the equivalent circuit consists of a resistor R_N in parallel with a current source I_N.

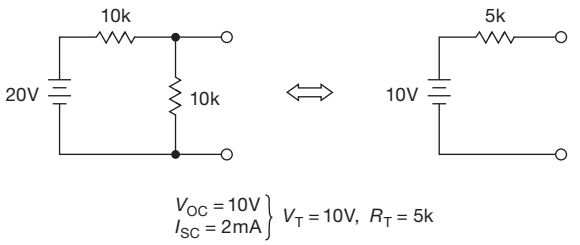


Figure D.3. Thévenin equivalent of a simple resistive divider. Note that R_T is the parallel resistance of the divider (as if the voltage source were replaced with a short circuit).

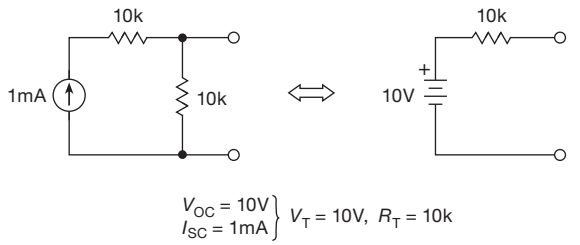


Figure D.4. Note that the Thévenin equivalent resistance is here *not* equal to the parallel resistance of the divider components. Instead it equals the value of the resistor across the output alone (as if the current source were replaced with an open circuit).

D.2 Norton's theorem

You can replace a Thévenin circuit with a Norton circuit, which consists of a current source I_N in parallel with a resistor R_N (Figure D.5). It is easy to show that $I_N = I_{SC}$ and $R_N = R_T$ $(= V_{OC}/I_{SC})$. So, for the two examples above, the Norton equivalents are as shown in Figure D.6.

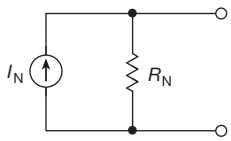


Figure D.5. Norton equivalent circuit: a current source in parallel with a resistor.

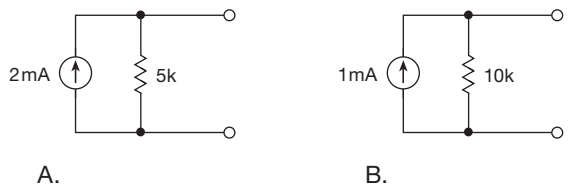


Figure D.6. Norton equivalents of the circuits of Figure D.3 (A) and Figure D.4 (B).

D.3 Another example

Figure D.7 shows a complicated-looking circuit, for which it is pretty easy to see that V_{OC}=25 V (the bottom of the 10k resistor sits at +10 V, and 1.5 mA flows into the top) and that I_{SC}=2.5 mA (10 V across the 10 k, plus the two current sources). From that you get the equivalent circuits shown.

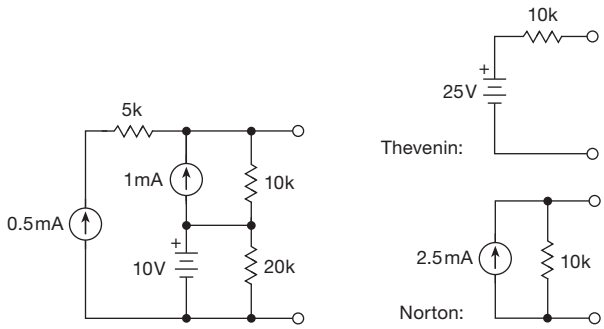


Figure D.7. Thévenin and Norton equivalents of a complicated-looking circuit.

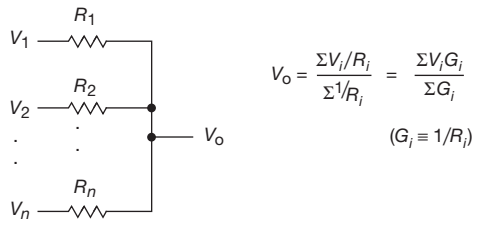


Figure D.8. Millman's theorem for parallel circuits.

D.4 Millman's theorem

A related – and useful – tool is *Millman's Theorem* (also known as the parallel generator theorem), which is helpful when dealing with circuits with several parallel branches. It's shown in Figure D.8, where a set of input voltages V_i are combined via resistors R_i, producing an output voltage V_o. The latter is just $V_o = (\sum V_i G_i)/\sum G_i$, where the G_i are the conductances $G_i \equiv 1/R_i$. The input voltages V_i can of course include ground, forming a voltage divider. Millman's theorem, which comes from the more general class of network theorems, can be generalized to include input *currents* I_k, whose sum is added to the numerator (but whose series resistances, if any, do not appear in the denominator).

LC BUTTERWORTH FILTERS

APPENDIX **E**

Active filters (see Chapter 6) are convenient at low frequencies, but they are impractical at radio frequencies because of the slew-rate and bandwidth requirements they impose on the operational amplifiers. At frequencies of 100 kHz and above (and often at lower frequencies), the best approach is to design a passive filter with inductors and capacitors. (At UHF and microwave frequencies these "lumped-component" filters are replaced by stripline and cavity filters.)

As with active filters, there are many methods and filter characteristics possible with *LC* filters. For example, you can design the classic Butterworth, Chebyshev, and Bessel filters, each in lowpass, bandpass, highpass, and band-reject varieties. It turns out that the Butterworth filter is particularly easy to design, and we can present in just a page or two all the essential design information for lowpass and highpass Butterworth *LC* filters, and even a few examples.

E.1 Lowpass filter

Table E.1 gives the values of normalized inductances and capacitances for low-pass filters of various orders, from which actual circuit values are obtained by the frequency and impedance scaling rules

$$L_n(\text{actual}) = \frac{R_L L_n(\text{table})}{\omega},$$
$$C_n(\text{actual}) = \frac{C_n(\text{table})}{\omega R_L},$$

where R_L is the load impedance and ω is the angular frequency ($\omega = 2\pi f$).

Table E.1 gives normalized values for two-pole through eight-pole lowpass filters for the two most common cases, namely (a) equal source and load impedances and (b) either source or load impedance much larger than the other. To use the table, first decide how many poles you need, based on the Butterworth response (graphs are plotted in Figure 6.30). Then use the preceding equations to determine the filter configuration (T or π; see Figure E.1) and component values. For equal source and load impedances, either configuration is OK; the π configuration may be preferable because it requires fewer inductors. For a load impedance much higher (lower) than the source impedance, use the T (π) configuration.

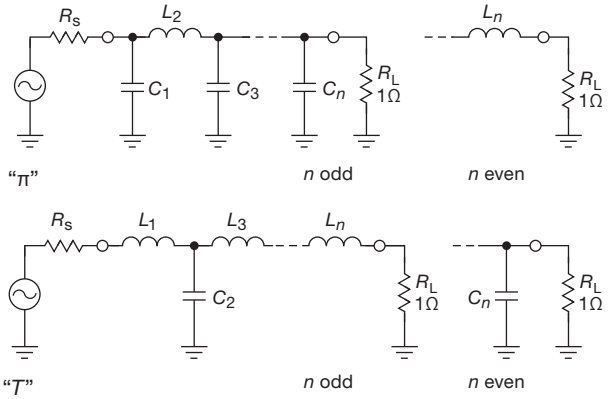


Figure E.1. π and T configurations. See Table E.1 and text.

E.2 Highpass filter

To design a highpass filter, follow the same procedure to determine which filter configuration to use and how many poles are necessary. Then do the universal lowpass to highpass transformation shown in Figure E.2, which consists simply of replacing inductors by capacitors, and vice versa. The actual component values are determined from the normalized values in Table E.1 by the following frequency and impedance scaling rules:

$$L_n(\text{actual}) = \frac{R_L}{\omega C_n(\text{table})},$$
$$C_n(\text{actual}) = \frac{1}{R_L \omega L_n(\text{table})}.$$

E.3 Filter examples

Here are a few examples showing how to design both lowpass and highpass filters.

Example I. Design a five-pole lowpass filter for source

Table E.1 Butterworth Lowpass Filters[a]

	$\pi \longrightarrow R_S$	C_1	L_2	C_3	L_4	C_5	L_6	C_7	L_8
	$T \longrightarrow 1/R_S$	L_1	C_2	L_3	C_4	L_5	C_6	L_7	C_8
$n=2$	1	1.4142	1.4142						
	∞	1.4142	0.7071						
$n=3$	1	1.0000	2.0000	1.0000					
	∞	1.5000	1.3333	0.5000					
$n=4$	1	0.7654	1.8478	1.8478	0.7654				
	∞	1.5307	1.5772	1.0824	0.3827				
$n=5$	1	0.6180	1.6180	2.0000	1.6180	0.6180			
	∞	1.5451	1.6944	1.3820	0.8944	0.3090			
$n=6$	1	0.5176	1.4142	1.9319	1.9319	1.4142	0.5176		
	∞	1.5529	1.7593	1.5529	1.2016	0.7579	0.2588		
$n=7$	1	0.4450	1.2470	1.8019	2.0000	1.8019	1.2470	0.4450	
	∞	1.5576	1.7988	1.6588	1.3972	1.0550	0.6560	0.2225	
$n=8$	1	0.3902	1.1111	1.6629	1.9616	1.9616	1.6629	1.1111	0.3902
	∞	1.5607	1.8246	1.7287	1.5283	1.2588	0.9371	0.5776	0.1951

Notes: (a) Values of L_n, C_n for 1Ω load resistance, and cutoff frequency (–3dB) of 1 rad/s. See text for scaling rules.

and load impedances of 75 Ω, with a cutoff frequency (−3 dB) of 1 MHz.

We use the π configuration to minimize the number of required inductors. The scaling rules give us

$$C_1 = C_5 = \frac{0.618}{2\pi \times 10^6 \times 75} = 1310\,\text{pF},$$

$$L_2 = L_4 = \frac{75 \times 1.618}{2\pi \times 10^6} = 19.3\,\mu\text{H},$$

$$C_3 = \frac{2}{2\pi \times 10^6 \times 75} = 4240\,\text{pF}.$$

The complete filter is shown in Figure E.3. Note that all filters with equal source and load impedances will be symmetrical.

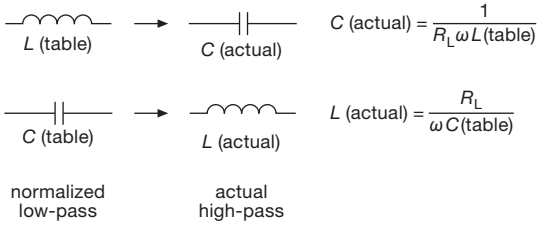


Figure E.2. Lowpass to highpass transformation.

Example II. Design a three-pole lowpass filter for a source impedance of 50 Ω and a load impedance of 10k, with a cutoff frequency of 100 kHz.

We use the T configuration, because $R_S \ll R_L$. For R_L=10k, the scaling rules give

$$L_1 = \frac{10^4 \times 1.5}{2\pi \times 10^5} = 23.9\,\text{mH},$$

$$C_2 = \frac{1.3333}{2\pi \times 10^5 \times 10^4} = 212\,\text{pF},$$

$$L_3 = \frac{10^4 \times 0.5}{2\pi \times 10^5} = 7.96\,\text{mH}.$$

The complete filter is shown in Figure E.4.

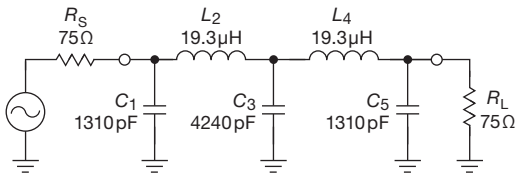


Figure E.3. Circuit for Example I. Five-pole 1 MHz lowpass with equal source and load impedances.

Example III. Design a four-pole lowpass filter for a zero-impedance source (voltage source) and a 75 Ω load, with a cutoff frequency of 10MHz.

We use the T configuration, as in the previous example,

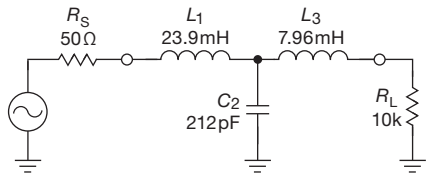


Figure E.4. Circuit for Example II. Three-pole 100 kHz lowpass with 50 Ω source and 10k load.

because $R_S \ll R_L$. The scaling rules give

$$L_1 = \frac{75 \times 1.5307}{2\pi \times 10^7} = 1.83\,\mu\text{H},$$

$$C_2 = \frac{1.5772}{2\pi \times 10^7 \times 75} = 335\,\text{pF},$$

$$L_3 = \frac{75 \times 1.0824}{2\pi \times 10^7} = 1.29\,\mu\text{H},$$

$$C_4 = \frac{0.3827}{2\pi \times 10^7 \times 75} = 81.2\,\text{pF}.$$

The complete filter is shown in Figure E.5.

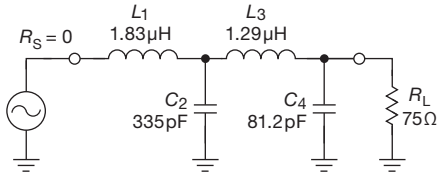


Figure E.5. Circuit for Example III. Four-pole 10 MHz lowpass with voltage source and 75 Ω load.

Example IV. Design a two-pole lowpass filter for current-source drive and 1k load impedance, with a cutoff frequency of 10 kHz.

We use the π configuration because $R_S \gg R_L$. The scaling rules give

$$C_1 = \frac{1.4142}{2\pi \times 10^4 \times 10^3} = 0.0225\,\mu\text{F},$$

$$L_2 = \frac{10^3 \times 0.7071}{2\pi \times 10^4} = 11.3\,\text{mH}.$$

The complete filter is shown in Figure E.6.

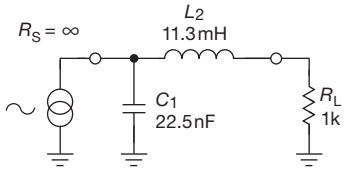


Figure E.6. Circuit for Example IV. Two-pole 10 kHz lowpass with current source drive and 1k load.

Example V. Design a three-pole highpass filter for 52 Ω source and load impedances, with a cutoff frequency of 6 MHz.

We begin with the T configuration, then transform inductors to capacitors, and vice versa, giving

$$C_1 = C_3 = \frac{1}{52 \times 2\pi \times 6 \times 10^6 \times 1.0} = 510\,\text{pF},$$

$$L_2 = \frac{52}{2\pi \times 6 \times 10^6 \times 2.0} = 0.690\,\mu\text{H}.$$

The complete filter is shown in Figure E.7.

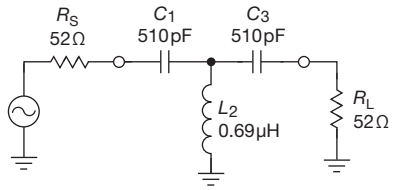


Figure E.7. Circuit for Example V. Three-pole 6 MHz highpass with equal source and load impedances.

We would like to emphasize that the field of passive filter design is rich and varied and that this simple table of Butterworth filters doesn't even begin to scratch the surface.

LOAD LINES

APPENDIX **F**

The graphic method of "load lines" usually makes an early appearance in electronics textbooks. We have avoided it because, well, it just isn't useful in transistor design the way it was in vacuum-tube circuit design. However, it is of use in dealing with some nonlinear devices (tunnel diodes, for example), and in any case it is a useful conceptual tool.

F.1 An example

Let's start with an example. Suppose you want to know the voltage across the diode in Figure F.1. Assume that you know the voltage-versus-current (V–I) curve of the particular diode (of course, it would have a manufacturing "spread," as well as a dependence on ambient temperature); it might look something like the curve drawn. How would you figure out the quiescent[1] point?

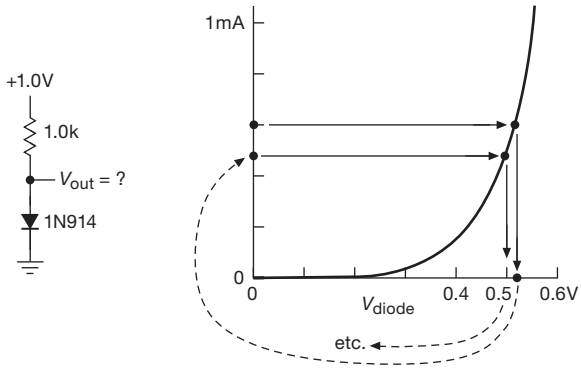


Figure F.1. Finding the operating point by iteration.

One method might be to guess a rough value of current, say 0.6 mA, then use the curve to get the drop across the resistor, from which you get a new guess for the current (in this case, 0.48 mA). This iterative method is suggested in Figure F.1. After a few iterations, this method will get you an answer, but it leaves a lot to be desired.

The method of load lines gets you the answer to this sort of problem immediately. Imagine *any* device connected in place of the diode; the 1.0k resistor is still the load. Now plot, on a V–I graph, the curve of resistor current versus device voltage. This turns out to be easy: at zero volts the current is just V_+/R (full drop across the resistor); at V_+ volts the current is zero; points in between fall on a straight line between the two. Now, on the same graph, plot the V–I curve of the device. The operating point lies on both curves, i.e., at the intersection, as shown in Figure F.2.

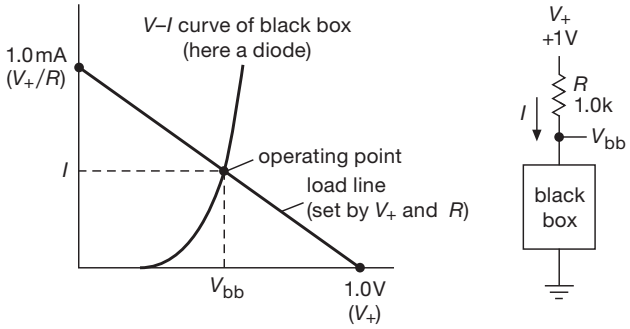


Figure F.2. A "load line" lets you find the operating point directly.

F.2 Three-terminal devices

Load lines can be used with a three-terminal device (tube or transistor, for example) by plotting a family of curves for the device. Figure F.3 shows what such a thing would look like for a depletion-mode field-effect transistor (FET), with the curve family parameterized by the gate-source voltage. You can read off the output for a given input by sliding along the load line between appropriate curves corresponding to the input you've got, then projecting onto the voltage axis. In this example we've done this, showing the drain

[1] The quiescent point, also known as the *operating point*, describes the various dc voltages and currents in a circuit with no ac signals applied.

voltage (output) for a gate swing (input) between ground and −2 V.

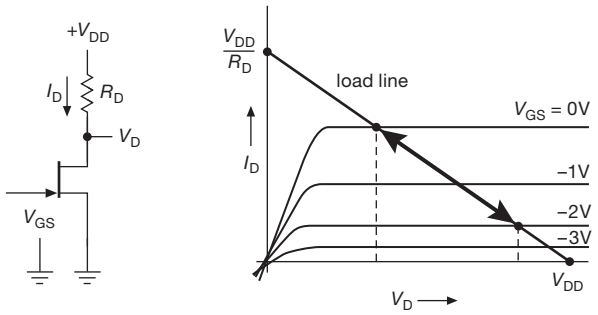


Figure F.3. Load-line solution for a three-terminal device.

As nice as this method seems, it has quite limited use for transistor or FET design, for a couple of reasons. For one thing, the curves published for semiconductor devices are "typical," with manufacturing spread that can be as large as a factor of five. Imagine what would happen to those nice load-line solutions if all the curves shrank to one-fourth their height! Another reason is that for an inherently logarithmic device like a diode junction, a linear load-line graph can be used to give accurate results over only a narrow region. Finally, the nongraphic methods we've used in this book are entirely adequate for handling solid-state design. In particular, these methods emphasize the parameters you can count on (r_e, I_C versus V_{BE} and T, etc.), rather than the ones that are highly variable (β, V_{th}, etc.). If anything, the use of load lines on published curves for transistors only gives you a false sense of security, since the device spread isn't also shown.

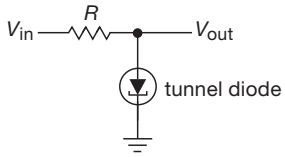


Figure F.4. The tunnel diode: a two-terminal nonlinear device with a region of negative resistance (see Figure F.5).

F.3 Nonlinear devices

Load lines turn out to be useful in understanding the circuit behavior of highly nonlinear devices. The example of tunnel diodes illustrates a couple of interesting points. Let's analyze the circuit in Figure F.4. Note that in this case, V_{in} takes the place of the supply voltage in the previous examples. So a signal swing will generate a family of parallel load lines intersecting with a single device V–I curve (Figure F.5). The values shown are for a 100 Ω load resistor. As can be seen, the output varies most rapidly as the input swing takes the load line across the negative-resistance portion of the tunnel-diode curve. By reading off values of V_{out} (projection on the x axis) for various values of V_{in} (individual load lines), you get the "transfer" characteristics shown. This particular circuit has some voltage gain for input voltages near 0.2 V.

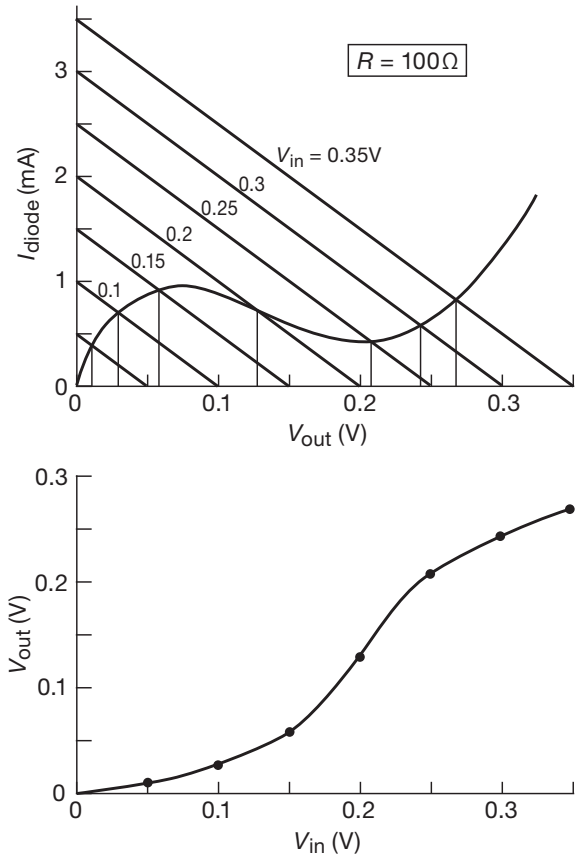


Figure F.5. Load lines and transfer characteristic for the tunnel-diode circuit.

An interesting thing happens if the load lines become flatter than the middle section of the diode curve. That happens when the load resistance exceeds the magnitude of the diode's negative resistance. It is then possible to have *two* intersection points, as in Figure F.6. A rising input signal carries the load lines up until the intersection point has nowhere to go and has to jump across to a higher V_{out} value. On returning, the load lines similarly carry the

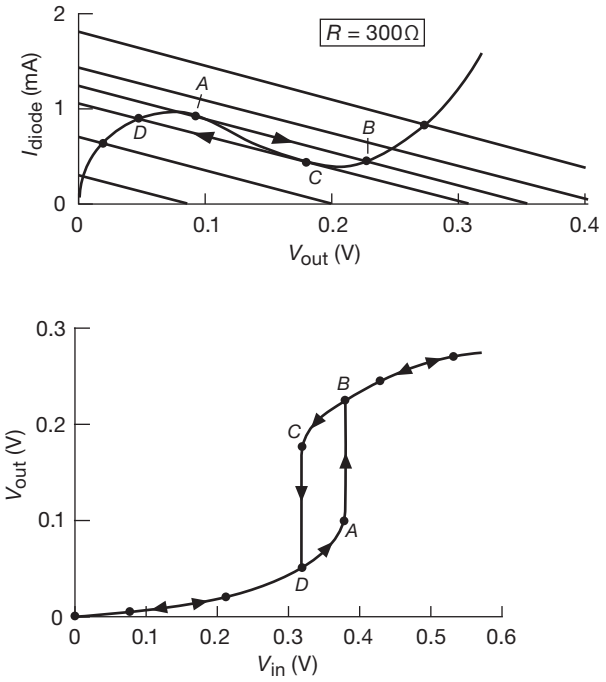


Figure F.6. Having $|R_{load}| > |R_{neg}|$ produces hysteretic switching behavior in the tunnel-diode circuit.

intersection point down until it must again jump back. The overall transfer characteristic has *hysteresis*, as shown. Tunnel diodes have been used in this manner as fast-switching devices (triggers).

THE CURVE TRACER

APPENDIX **G**

A handy instrument for exploring transistor behavior of both BJTs (Chapter 2) and MOSFETs (Chapter 3) is the *curve tracer*. Most simply, it plots collector current versus collector voltage for a family of equally spaced base currents (or, if you want to be an Ebers–Mollian, base *voltages*), and with a selectable current-limiting collector resistor.[1] Figure G.1 shows what you get from a random 2N3904, driven with seven successive base current steps of 5 μA each while sweeping the collector voltage from 0 to 50 V. You can see clearly the rise of beta with collector voltage, and the onset of breakdown somewhat below 50 V (maximum V_{CEO} is specified as 40 V). This particular curve tracer obligingly displays the scale factors, including "β per div," which is about 200 for this specimen (the datasheet specifies $100 \leq \beta \leq 300$ at I_C=10 mA). A curve tracer makes it easy to select closely matched pairs.

Sadly, the traditional curve tracer has disappeared from the product lines of most T&M (test and measurement) manufacturers, including the venerable Tektronix. You can still find them used, for example on eBay, for a thousand dollars or so. Agilent offers some pretty fancy contemporary instruments that will do the job, though it's best to be sitting down when you ask the prices; they go by names like Semiconductor Parameter Analyzer (model 4155C), or Power Device Analyzer/Curve Tracer (model B1505A).

[1] You can run it as common base, if you like; and it has many amusing knobs to play with.

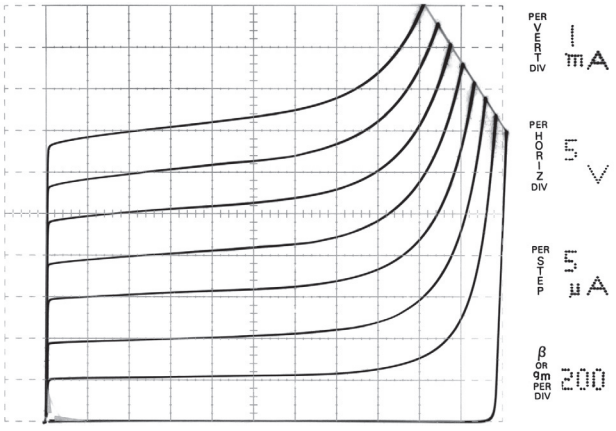


Figure G.1. Tektronix 576 Curve Tracer display of a 2N3904.

A less expensive alternative is to use a "source-measure unit" (SMU), a delightful instrument that lets you source voltages and currents to selected terminals of a device (or subcircuit), simultaneously measuring and logging other voltages or currents. You can program the excitation as dc, or ramps, steps, or pulses, and you can display the logged results via software running on an attached laptop computer; you can also save the logged data as a spreadsheet, to be manipulated to your heart's content. Take a look, for example, at Figure 8.39, or the figures in the "Power Transistors for Linear Amplifiers" section of Chapter *3x*, all of which plot data we collected with a SMU.

TRANSMISSION LINES AND IMPEDANCE MATCHING

APPENDIX **H**

H.1 Some properties of transmission lines

In §12.9 we introduced transmission lines, which most commonly take the form of *coaxial* cable ("coax"), for example the ubiquitous "BNC cables" (RG-58 cables with male BNC connectors at each end) that are used to run all manner of signals between instruments. As we remarked there, for low-frequency applications it is common (and correct) to think of such a cable simply as a well-shielded wire with ∼30 pF/ft of capacitance. However, at high frequencies (say those for which the cable is at least 1/20 of the wavelength) the behavior is fundamentally different: as a bizarre example, an open-ended cable ironically looks like a *short circuit* at a frequency for which the cable's electrical length is $\lambda/4$. For a 5-foot length of coax, that happens at about 32 MHz. An important consequence is that you can't just hook such a BNC cable from a signal generator to some high-impedance circuit under test and assume that it will provide a nice signal source at the circuit's input; instead you will see huge dips and bumps as you tune the frequency, because the generator sees a load impedance that varies from a short circuit (at odd multiples of 32 MHz) to an open circuit (at even multiples of 32 MHz). Perhaps surprisingly, if you were instead to connect a resistor of exactly 50 Ω across the circuit end of the cable, you would find that it now delivered a constant signal amplitude (equal to half the signal generator's open-circuit output amplitude) as you varied the frequency. This nonintuitive behavior is nicely illustrated in the measured data shown in Figure H.1. And even more nonintuitively, at the driving end of such a "terminated" cable the capacitance disappears entirely – you see instead a pure resistive load of 50 Ω!

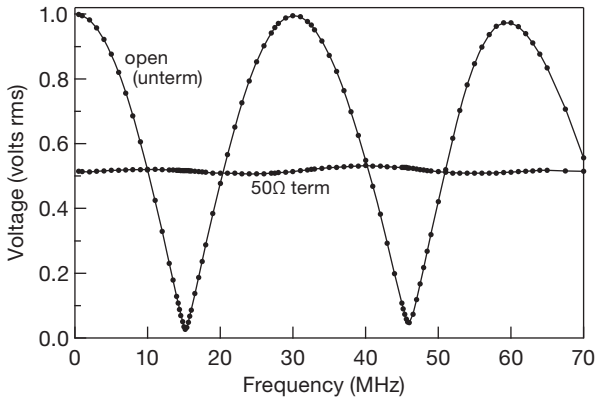


Figure H.1. Measured amplitude at the output connector of a sinewave oscillator of 1 V amplitude (open circuit), under two conditions: driving 10 feet of RG-58 (50 Ω) coaxial cable, open at the far end; and driving the same cable with a 50 Ω resistor connected across the far end.

H.1.1 Characteristic impedance

This simple example illustrates the importance of *termination*: coaxial cable is a form of *transmission line*, with a *characteristic impedance* Z_0 (which is always real: a resistance) that depends on only its physical construction:

$$Z_0 = \sqrt{L/C} = \frac{138}{\sqrt{\varepsilon}} \log_{10} \frac{b}{a} \text{ ohms},$$

where L and C are the inductance and capacitance per unit length, which as indicated depend on only the outer diameter, a, of the inner conductor, the inner diameter, b, of the outer conductor, and the dielectric constant, ε (relative to free space), of the insulating material that separates them. For a wave propagating along a transmission line, Z_0 is the ratio of signal voltage to signal current. The most popular coax line for general purposes is RG-58, with an impedance of 50 Ω (its dimensions are a=0.81 mm, b=2.95 mm, and ε=2.3, for which the above equation gives Z_0=51 Ω). This impedance has become the standard for radiofrequency use, except for video applications where the standard is Z_0=75 Ω; the corresponding popular cable type is called RG-59. In pulse electronics you sometimes see 93 Ω cable (RG-62).

The signal propagates along the cable at a speed

$$v_{\text{wave}} = \frac{c}{\sqrt{\varepsilon}} = \frac{1}{\sqrt{LC}},$$

which is a fraction $1/\sqrt{\varepsilon}$ times c (where c is the speed of light in vacuum, 3×10^8 m/s). The factor $1/\sqrt{\varepsilon}$ is called the *velocity factor*, and ranges from 0.66 (solid polyethylene) to 0.80 (polyethylene foam) for available flexible coaxial cables. In the absence of a dielectric the velocity factor is 1.0, i.e., waves travel at the speed of light in an air-spaced coaxial line. The "electrical length" seen by a propagating signal in a cable of physical length L is larger by the factor $\sqrt{\varepsilon}$, i.e., $L_{\text{elec}}=L_{\text{physical}}\sqrt{\varepsilon}$.

Note that the inductance and capacitance of the cable cannot take on any old values, because they are constrained such that their product LC is related to the speed of light. From this it is easy to show that if you know the characteristic impedance and the velocity factor then you can find the capacitance per unit length (or vice versa) by

$$C = \frac{\sqrt{\varepsilon}}{cZ_0} \text{ Farads/meter.}$$

For example, RG-8 has a specified impedance of 52 Ω, and a velocity factor of 0.66; the above equation then gives C=97.1 pF/m, or 29.6 pF/ft, in good agreement with the specified value of 29.5 pF/ft.

A. Twisted pair and PCB traces

Transmission lines are not *required* to be of coaxial geometry. An extremely popular form of contemporary transmission line is the *twisted pair*, which is just what it sounds like: a pair of insulated wires, gently twisted, and enclosed in an overall insulating jacket (often without any overall shield conductor). These may well be the dominant species of transmission line in our time, because they are the basic stuff of local area networks (LANs). You usually see four twisted pairs bundled into a single unshielded jacket; this is called "UTP" (unshielded twisted pair), and is the most common form of LAN cable. It is available also with a shield (shielded twisted pair, "STP"). Contemporary UTP and STP cables are 100 Ω nominal impedance, and are characterized (in terms of impedance and attenuation) for operation up to 10 Mbps (megabits per second) or 100 Mbps; these are called Category 3 and Category 5, respectively, and appear to differ primarily in the pitch of the twist. Those in the know refer to these casually as cat-3 and cat-5.[1] Ethernet LANs using these data rates are called "10baseT" and "100baseTX," the "T" standing for "twisted"; the corresponding designation for "thinnet" coaxial Ethernet is "10base2."

In high-speed electronics on a printed circuit board it's often necessary to treat connection traces as transmission lines. See the discussion in §*1x.1*, where we described the *microstrip* transmission line, which consists of a thin conducting strip on an outer printed circuit layer, with an underlying ground-plane layer. A popular variant adds a pair of ground-trace shepherds on either side – that's called a *grounded coplanar waveguide* (GCPW). There's also the completely enclosed *stripline* geometry, where the trace(s) are sandwiched between ground-plane layers.

H.1.2 Termination: pulses

A transmission line at low frequencies (wavelength much longer than the cable length) looks simply like a capacitance, typically of order 30 pF/ft. However, at high frequencies, or, equivalently, when dealing with signals with fast rise times, the behavior is different. In order to understand the curious behavior illustrated in Figure H.1, it's helpful to look first at what happens when a simple *pulse* is applied to a length of transmission line. Suppose we connect a fast pulse generator with 50 Ω output impedance (the standard output impedance of signal generators, function generators, and pulse generators) to a length of 50 Ω coax, shorted at the far end. The pulse at first disappears into an impedance Z_0 (thus the signal amplitude is half that of the unloaded generator), but after a round-trip travel time a reflected pulse of opposite polarity returns (Figure H.2). If the signal applied is instead a fast step, the effect of the reflection is to convert the step into a pulse (Figure H.3). An open-ended line produces a reflection of the *same* polarity, with the effects shown in Figure H.4. For arbitrary load resistance R the ratio of reflected to incident wave amplitude (the reflection coefficient) is given by

$$\rho \equiv A_{\text{r}}/A_{\text{i}} = (R-Z_0)/(R+Z_0).$$

Note that a termination resistance of R=Z_0 produces no reflection. A signal applied to such a terminated line is absorbed by the terminating resistor (as heat) and disappears forever. The signal source sees a loading resistance equal to Z_0. (It is for this reason that we did not have to worry earlier about the reflected pulses reflecting again from the pulse

[1] Higher performance standards include Category 5e ("e" for *enhanced*), and Category 6, propelled by the development of Gigabit Ethernet – literally 1 gigabit/sec over unshielded twisted pair – also known as 1000baseT. To achieve this data rate, all four pairs are used simultaneously with 5-level amplitude encoding.

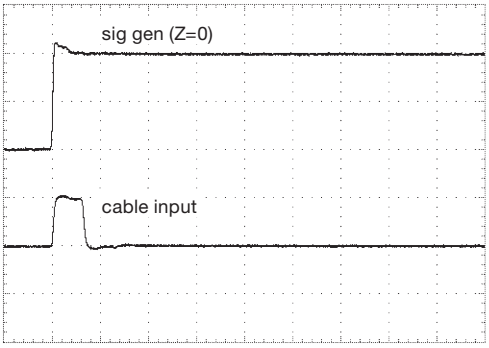


Figure H.3. 'Scope trace of a step waveform applied to an 8-foot length of RG-58A/U (solid polyethylene dielectric, velocity factor of 66%), shorted at the end. Vertical; 1 V/div; horizontal; 40 ns/div.

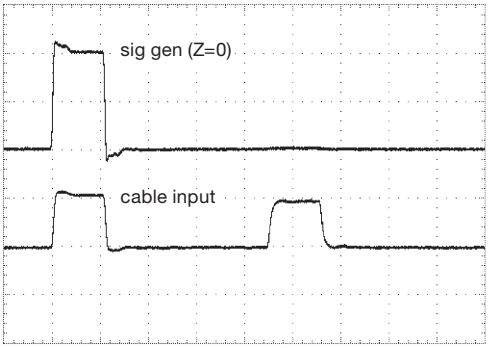


Figure H.4. 'Scope trace showing reflection from open-ended coax line. Same parameters as those in Figure H.2B.

generator – its 50 Ω source impedance swallows any signals returning from an improperly terminated cable, which is the reason most signal sources are standardized at 50 Ω impedance.)

A. Series termination

This last point – that returning (reflected) signals are completely absorbed if the signal source's impedance matches the line – leads to a nice technique called *series termination* (or *back termination*), frequently used for high-speed logic signals (and in other situations where the load has a high input impedance). Look at Figure H.5: a signal source in series with a resistor (equal to the line's impedance) drives a transmission line whose far end is unterminated (i.e., open). Now imagine a step input of amplitude V_{sig} at the signal source; it propagates down the line at half-amplitude, then reflects back from the far end at the full V_{sig} amplitude. Although any point along the line sees a two-step waveform, the surprising fact is that the waveform seen at the far end makes a single step from zero to the full

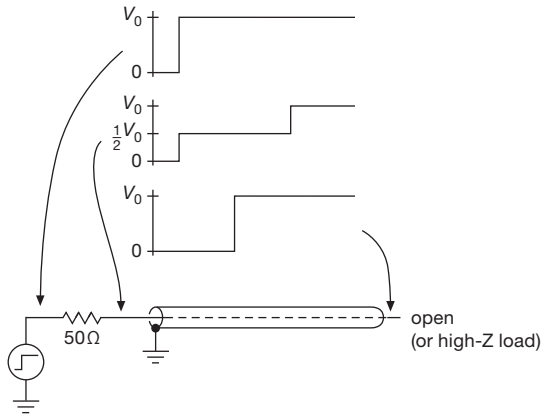


Figure H.5. Series termination of an open-ended line: the line is fed from a signal source of matched impedance; the half-sized step at the input propagates to the far end, from which it reflects in-phase to produce a returning step equal in amplitude to the *unloaded* amplitude of the generator. A high-impedance load at the far end sees only a single full-sized step.

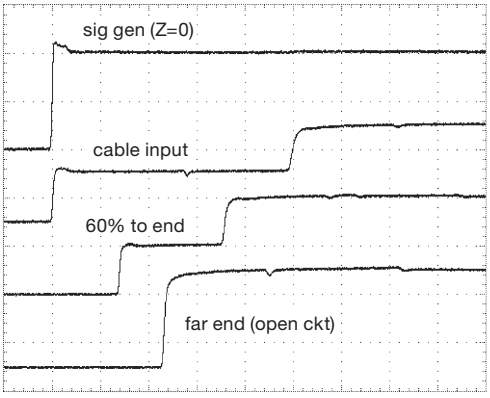


Figure H.6. 'Scope trace showing waveforms at cable input, midpoint, and far end, for series-terminated step input. The generator's zero-impedance signal is also shown; it was set to produce a 2 V step into an open circuit. The cable is 60 ft of RG-58/U (velocity factor of 66%), tapped at 36 ft with a high-impedance voltage probe. Vertical; 1 V/div; horizontal; 40 ns/div.

V_{sig}; at that place the half-sized incident wave arrives at the same time that the half-sized reflected wave departs. This interesting behavior is demonstrated in the 'scope traces in Figure H.6.

You can use this technique for sending CMOS logic signals through a length of coax: three paralleled 74HC buffers (for low source impedance, roughly 15 Ω) in series with a 33 Ω resistor nicely drives lengths of RG-174 or RG-316 (thin 50 Ω coax line), connected to the receiving gate at the far end *without termination*. The receiving gate sees

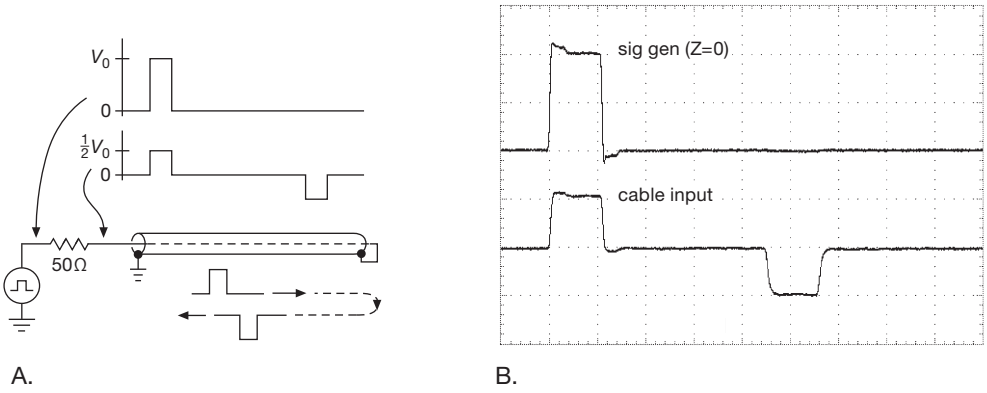


Figure H.2. A. A pulse driving a length of shorted transmission line reflects off the short and returns as a pulse of opposite polarity. B. Scope trace taken with 70 ft of RG-8/U with foam dielectric (velocity factor of 78%), shorted at end. Vertical; 1 V/div; horizontal; 40 ns/div. For this and following figures we used a high-impedance 'scope probe to avoid introducing additional transmission-line effects.

full-swing logic signals. This technique is often preferable to the matched-load alternative – directly driving a line that is terminated in 50 Ω – because with series termination the driver sees a load resistance twice as high (100 Ω in this case), and that only for the round-trip duration of the signal (after which the load becomes an open-circuit).[2] For very fast logic signals (e.g., ECL 100K, or contemporary high-speed CMOS processors, memory, and peripherals), it may be necessary to treat a circuit trace of just a few inches as a transmission line. Typically printed circuit board (PCB) trace impedances are in the range of 50 Ω to 100 Ω, but can be tailored to a specific impedance by proper choice of trace width and spacing above the ground plane; this specialty art is known as *microstrip* technique,[3] useful both for fast digital signals and for signals at frequencies above about 100 MHz (UHF and microwave).

This is all very nice in theory – but in practice you have to contend with sources of fast digital signals that are not matched to the line impedance. This happens often on digital PCBs where the digital output ports of speedy microprocessors and FPGAs are poorly matched to the PCB trace impedances. For example, a line driven by a signal of source impedance $Z_0/2$ produces lots of ringing at both ends of an unterminated line, including 33% overshoot at the far end; such ringing can produce false clocking.

To illustrate what this looks like, we drove a 2.4 m length of unterminated RG-58/U (Z_0=50 Ω) coax with a step input, probing both the input and output signals.

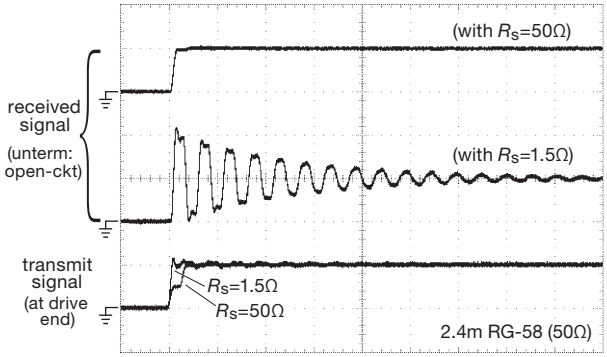


Figure H.7. Signals seen at the far end of an unterminated 8ft length of 50 Ω line when driven with a unit step from a 50 Ω source (top trace) and from a low-impedance source (middle trace). The corresponding signals at the driving end are shown at bottom. Horizontal; 100 ns/div.

We did this under two conditions: (a) when driven with a 50 Ω series termination at the input (i.e., a "back-terminated" line); and (b) when driven with a low-impedance (R_s=1.5 Ω) voltage step.

Figure H.7 shows the results, where the drive signal's open-circuit amplitude (call it V_{OC}) is displayed as one vertical division. The matched series termination generates a clean received step to V_{OC} (and an input waveform with initial step first to $V_{OC}/2$, then to V_{OC} after a round-trip delay, just as seen in expanded scale in Figure H.6). By contrast, the low-Z drive imposes a full V_{OC} step at the input, which is first seen at the far end at $2V_{OC}$ (because the non-inverted reflected wave doubles the amplitude of the arriving wave), subsequently brought nearly back to zero (one round-trip time later) by the inverted wave reflected from

[2] See §12.10, where several methods of driving cables with logic levels are described and illustrated.

[3] If you sandwich the signal-carrying conductors between a pair of ground planes, you've got a *stripline*; see §*1x.1*.

the low-impedance driver. This alternating pattern continues, damped both by cable loss and resistive loss in the 1.5 Ω driver. This signal is a mess! That's why it's worth some effort to match driver impedances to the characteristic impedance of lines whose length (given the signal rise times) qualifies them as transmission lines.[4]

B. A robust logic link

The sort of overshoot behavior seen in Figure H.7 can destroy logic circuits at the receiving end. You see this vulnerability particularly in the laboratory, where a low-impedance logic signal (or pulse-generator output) travels through a length of coax to a logic input on some instrument. The latter is often unterminated, to keep the input impedance relatively high (to prevent heavy loading and attenuation for a signal source unable to drive 50 Ω).

If you want to make your own designs bulletproof against this hazard, you can add a few components as shown in Figure H.8. At the receiving end, series resistor R_2 limits the current, safely clamped by the logic gate's internal protection diodes; the "speed-up capacitor" C_1 prevents loss of speed (1 kΩ into a typical input and wiring capacitance of 10 pF is an RC time constant of 10 ns, a near-eternity in the frenetic world of digital logic). It's always a good idea to include an input pull-down resistor (R_1) to ensure a defined logic level when the input is disconnected. In an abundance of caution we've added resistor R_3, whose job is to prevent a large positive overdrive from forcing the V_+ rail to a positive voltage that can damage other ICs; this could normally be omitted, but it would be a good idea in an instrument with an inviting BNC connector on the front and that contains only a few ICs powered from a small regulator (like a 78L05 – see §9.3.2, Figure 9.6, and Table 9.1) whose dc output can be easily overdriven.

At the driver end the parallel connection of several logic gates generates a drive impedance down in the neighborhood of 5–10 Ω, which the added series resistor R_o brings up to a driving resistance close to the cable's 50 Ω characteristic impedance. That's what you want, of course, and that alone is enough to give you peace of mind as a respectable series-terminated driver. But it never hurts to protect the receiver end, as just discussed: you never know when someone will drive it with a pulse generator, inadvertantly set to deliver *negative* pulses, or 20 V positive pulses (as happened in our laboratory recently).

[4] An analogous effect in long-distance power transmission lines is known as the Ferranti effect; it is said that overvoltages caused by the Ferranti effect, if not properly compensated, can cause damage to power-line switch gear.

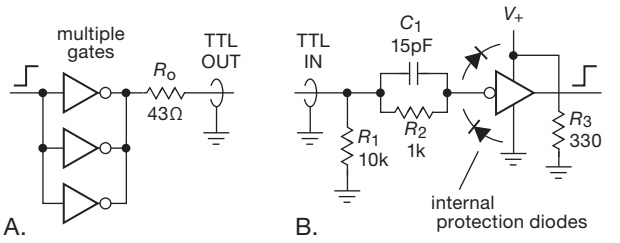


Figure H.8. A. Simple logic-level driver for series-terminated cable; B. Logic-level receiver protected against overdrive. Connect these together to form a complete signal path.

There's further discussion of cable driving and logic interfacing in Chapter 12, beginning at §12.10.

H.1.3 Termination: sinusoidal signals

We have been talking about signals propagating along transmission lines, for clarity using the particular case of pulses or voltage steps. Of course, a *sinewave* applied to a length of cable also produces reflections, unless of course the cable is properly terminated. The effect is to alter the input current, for an applied input voltage, in a way that depends on the (mismatched) load impedance Z_L, and also on the ratio of the signal's wavelength in the cable λ to the physical length of the cable l. The final effect is to produce an input impedance (complex in general) given by

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan(2\pi l/\lambda)}{Z_0 + jZ_L \tan(2\pi l/\lambda)}.$$

From this one can see that:
(a) a matched termination (Z_L=Z_0 = [usually] 50 Ω) results in an input impedance equal to the characteristic impedance of the cable, independent of length or frequency;
(b) a quarter-wave line inverts the load impedance, i.e., Z_{in}=Z_0^2/Z_L;
(c) a half-wave line preserves the load impedance, i.e., Z_{in}=Z_L;
(d) a short length of open-circuited line $l \ll \lambda$ looks capacitive, viz., $Z_{in} \approx -j/\omega C'$, where C' (the effective capacitance) is the constant l/cZ_0;
(e) a short length of short-circuited line $l \ll \lambda$ looks inductive, viz., $Z_{in} \approx j\omega L'$, where L' (the effective inductance) is the constant $Z_0 l/c$.
The impedance-changing properties of transmission lines can be used to match impedances, though any such scheme will be frequency dependent; when you hear words like "stubs," you're dealing with transmission-line impedance matching. Virtuosos in this area make heavy use of network analyzers, and they will try to dazzle you with their

handsome "Smith Charts" (which are well beyond the humble scope of this book[5]).

When you have sinusoidal signals – with reflections – on a transmission line, you generate *standing waves*. That is, you can picture the net result of waves propagating in both directions (at the same frequency) as the sum of a nonpropagating (hence "standing") wave and some additional propagating wave. For example, an open-ended line produces a reflected wave of full amplitude; the result is a pure standing wave of the same frequency and twice the amplitude, with a maximum amplitude of oscillation at the open end (and repeating every half-wavelength), and complete nulls ("nodes" – places with no voltage) midway between. For a shorted-end line a similar thing happens, but the reflected wave is of opposite amplitude, producing a null at the far end (and repeating every half-wavelength), with maxima in between. (You get the same pattern if you tie a length of clothesline to a fence, then wiggle the end up and down at the right rate.) With a smaller termination mismatch you don't get complete cancellation anywhere.

Standing waves are not necessarily bad (though they are never[6] *good*!). But they do increase both the peak voltages and the resistive losses (see next section), relative to a matched line, for the same power transmitted. They are ordinarily seen as the symptom of a mismatched line. So in communications systems people try to minimize the *standing wave ratio* (abbreviated SWR, or sometimes VSWR – for *voltage* standing wave ratio – pronounced "VIZ-wahr"), which is defined as the ratio of the maximum amplitude to minimum amplitude:

$$\mathrm{VSWR} = \frac{V_{\max}}{V_{\min}} = \frac{A_\mathrm{f} + A_\mathrm{r}}{A_\mathrm{f} - A_\mathrm{r}},$$

where V is the ac (signal) voltage, measured at points along the line; and A_f and A_r are the amplitudes of the forward and reflected waves, respectively. Voltage measurements along a cable with no standing waves will give a constant amplitude (hence VSWR=1.0).

The VSWR is a real number, between 1.0 (perfect match, no reflected wave) and ∞ ("perfect mismatch," reflected wave amplitude equal to incident wave amplitude). In terms of the reflection coefficient ρ, the VSWR is just

$$\mathrm{VSWR} = \frac{1+|\rho|}{1-|\rho|}.$$

For a purely resistive mismatch we can use our earlier expression for ρ to find that

$$\mathrm{VSWR} = \begin{cases} R/Z_0 & \text{if } R > Z_0 \\ Z_0/R & \text{if } Z_0 > R. \end{cases}$$

From the VSWR you can find the magnitude (but not the phase) of the reflection coefficient:

$$|\rho| = \frac{\mathrm{VSWR}-1}{\mathrm{VSWR}+1}.$$

The VSWR is measured with a directional power meter. Knowing the values of forward and reflected power, you know from the defining equation above that

$$\mathrm{VSWR} = \frac{1+\sqrt{P_\mathrm{r}/P_\mathrm{f}}}{1-\sqrt{P_\mathrm{r}/P_\mathrm{f}}}.$$

H.1.4 Loss in transmission lines

In the real world of non-ideal transmission lines, things aren't quite as nice as we've led you to believe. Real transmission lines are *lossy*, meaning that signals are attenuated as they travel down the line; they are also slightly *dispersive*, meaning that signals of different frequency travel with slightly different speeds. The loss is *frequency dependent*: its value (often specified as attenuation in "dB per 100 ft") increases proportional to $\sqrt{f}$; i.e., a quadrupling of frequency doubles the loss (in dB) of a given length of line. This happens because the loss is dominated by the *skin effect*: when alternating current flows through a conductor, the current is not uniform throughout the bulk – it is confined to an outer layer (called the *skin depth*) of thickness $\delta = (\pi\sigma f)^{1/2}$, where σ is the conductivity and f is the frequency.[7] Because the skin depth decreases inversely with the square root of frequency, you have to quadruple the frequency to double the resistance (halve the skin depth),

[5] Smith charts are treated in the excellent reference *Fields and Waves in Communication Electronics* by Ramo, Whinnery, and Van Duzer (Wiley, 1994), as well as in the insightful and refreshing *Radio-Frequency Electronics* by Hagen (Cambridge University Press, 2009).

[6] Well, *hardly ever!* For operation over a narrow frequency range you sometimes exploit the impedance-changing properties of mismatched lines, which necessarily have standing waves. Examples are (a) the use of open or shorted lengths of line as high-Q capacitors or inductors, (b) a shorted half-wave or an open quarter-wave line used as an RF bypass capacitor, (c) an open half-wave or a shorted quarter-wave line used as an RF choke, (d) matching two different impedances (cables, signal sources, or loads) by interposing a quarter-wave section of transmission line whose characteristic impedance is the geometric mean of the two impedances being matched (this is analogous to a quarter-wave anti-reflection coating in optics), (e) the use of a slotted line and high-impedance probe for a direct measurement of wavelength, and (f) the use of transmission lines to make "ring hybrids" and "rat-race hybrids." We thank Jon Hagen and Darren Leigh for these applications of "good standing waves."

[7] To be precise, the current density decreases exponentially, falling to $1/e$ (37%) of its surface value at a depth equal to δ.

which is equivalent (in terms of loss) to doubling the length of the line. This explains the approximate slope of the attenuation curves for transmission lines (Figure H.9), where lines of larger diameter have lower losses. Dielectric losses contribute additional attenuation at the highest frequencies. The curves shown are for a matched line; if there are reflections (i.e., if the VSWR is greater than 1.0) then the loss, *for a given net power transmitted down the line*, will be greater.

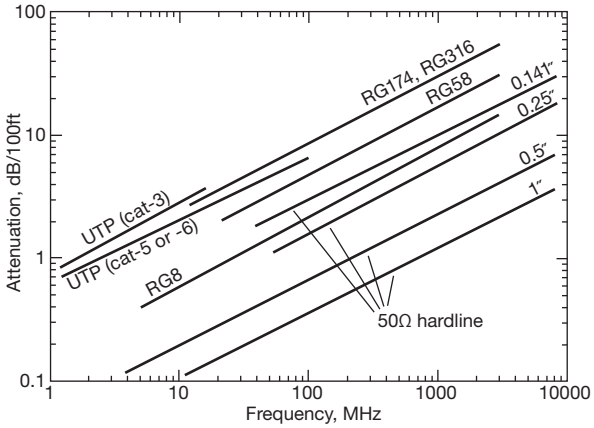


Figure H.9. Attenuation (dB/100 ft) as a function of frequency for several representative cable types.

It's useful to realize that skin-depth effects are significant even at low frequencies – current at the power-line frequency of 60 Hz is confined to a surface layer of roughly 1 cm in copper, for which the skin depth (in centimeters) at room temperature is given by $\delta(\mathrm{Cu})=6.6/\sqrt{f}$; you don't reduce power transmission losses much by using wire thicker than that. At radio frequencies the skin depth is so shallow (e.g., $\delta=10\,\mu$m at 40 MHz) that you can make low-loss connections, inductors, and so on, by silver plating a poor conductor. A common technique for shielding lightweight instruments and computers is to apply a thin metallic plating to a plastic enclosure. Figure H.10 plots the skin depth in copper conductors from 10 Hz to 10 GHz.

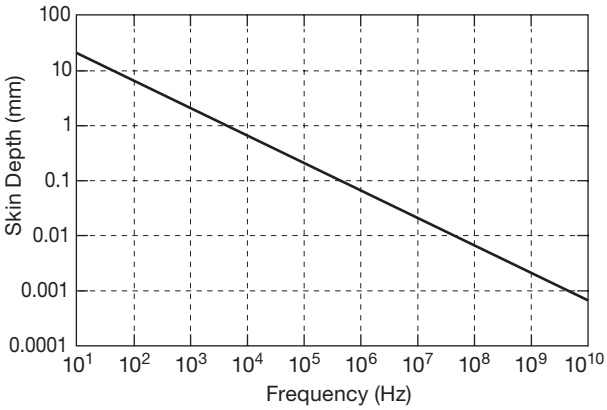


Figure H.10. Skin depth in copper as a function of frequency.

H.2 Impedance matching

Because you get reflections from unterminated (or incorrectly terminated) transmission lines, it's obviously a good idea to make sure you match impedances when you use coax lines whose electrical length is a significant fraction (at least 1/20, say) of the wavelength of the highest frequencies you're using. Stated in terms of time rather than frequency, you have to start worrying about termination when the round-trip propagation time is about 20% of the signal rise time.

We've seen already that a simple way to do this is to terminate the line in its characteristic impedance (resistance), for example 50 Ω for most coax lines. Termination is not required at *both* ends, because a terminated end swallows any incident signals. Thus you can use a mismatched signal-source impedance to drive a line terminated at the far end; or, as we saw above, you can "series terminate" the driven end of a line whose far end is unterminated (i.e., fed to a load of much higher impedance than the line). The usual practice, however, is to terminate *both* ends in the line's characteristic impedance (a conservative instinct that ensures a minimum of reflections). For example, you usually use a 50 Ω cable to pipe the signal from a synthesizer or signal generator to a 50 Ω matched load at the far end; if your load is high impedance, you place a 50 Ω resistor across it (or use a 50 Ω coax feedthrough termination).

Sometimes you need to match a line to a load (or source) of a different impedance; for example, you might want to measure the properties of some 75 Ω video cable (that's the impedance that the video industry has chosen, much to the chagrin of the rest of the electronics community), and all you've got are 50 Ω test instruments. Or, you might want to match the output impedance of a high-frequency amplifier to a length of cable that goes to an antenna.

This brings us to the subject of matching networks. In the following subsections we treat (a) resistive (lossy) networks for broadband impedance matching and attenuation, (b) transformer (lossless) broadband matching, and (c) reactive (lossless) narrowband matching.

H.2.1 Resistive (lossy) broadband matching network

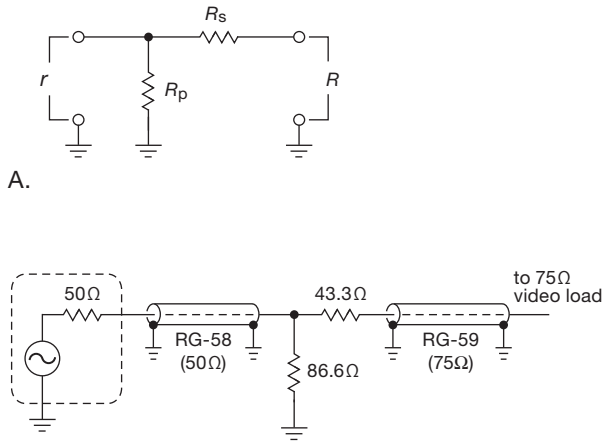


Figure H.11. A. A resistive *L*-network can match any pair of real (resistive) impedances; the parallel resistor R_p goes across the port of lower impedance r. B. Example of matching a 50 Ω signal source and cable to a 75 Ω video cable and load (with a loss of 5.72 dB).

You can easily figure out that two resistors (in the form of an "*L*-network," Figure H.11A) is all it takes to match a pair of impedances r and R (assumed resistive, as all cables are); both sides are happy – each sees a matched load. The values of the matching resistors are

$$R_p = r\sqrt{\frac{X}{X-1}},$$
$$R_s = r\sqrt{X(X-1)},$$

where r is the smaller impedance and X is the ratio of impedances: $X{=}R/r$. Taking the earlier example, you can match a 50 Ω test instrument to a 75 Ω coax line (the common variety is called RG-59) by putting 86.6 Ω across the 50 Ω port and connecting it to the cable through a 43.3 Ω series resistor (Figure H.11B).

The good news is that such a resistive *L*-network is frequency independent; the bad news is that it's lossy. It's easy to show that the loss is

$$\text{loss} = 20\log_{10}\left(\frac{\sqrt{X}}{X+\sqrt{X(X-1)}}\right)\ \text{dB}.$$

For example, the 50Ω-to-75Ω *L*-network above has a transmission loss of 5.72 dB for signals going in either direction. With a resistive match you have to accept this attenuation (this is sometimes called a *minimum loss pad*). We'll see below how to make lossless matching networks with transformers or with networks of *L*'s and *C*'s ("reactive matching networks").

As you might imagine, you can do even *worse*, in terms of loss, with a network containing more resistors! In particular, you can add another resistor, making either a "*T*" network or a "Pi" network, that matches two resistive impedances to each other, with loss greater than the minimum loss we found above. Although this isn't something you ordinarily want to do, there is a variation on that theme that is often useful; namely, a resistive attenuation network between a pair of already-matched impedances.

H.2.2 Resistive attenuator

In radiofrequency circuits you sometimes need to attenuate a signal level – for example, to avoid overdriving a stage of gain. In other situations you need to use a resistive attenuator to provide some isolation between an impedance-sensitive component like an amplifier, mixer, or cable, say, and a component that is not impedance matched; an example of the latter is a filter, which typically is impedance matched in its passband, but reflective (a severe mismatch) in its stopband. Some amplifiers will oscillate if their output directly drives a sharp filter.

The solution to these problems is a resistive impedance-matched attenuator. The two topologies are T and Pi, named for their appearance on a schematic diagram (Figure H.12). It is not difficult to derive the resistor values:

$$R_p = \frac{1+x}{1-x}R_0,$$
$$R_s = \frac{1-x^2}{2x}R_0, \qquad (\text{Pi}-\text{network}),$$

and

$$R_s = \frac{1-x}{1+x}R_0,$$
$$R_p = \frac{2x}{1-x^2}R_0, \qquad (\text{T}-\text{network}),$$

where x is given by the attenuation: $\text{atten(dB)} = -20\log_{10}x$ (or, equivalently, $x{=}10^{-\text{atten(dB)}/20}$), and Z_0 (assumed resistive) is the impedance at both input and output. Tabulated values for 50 Ω source and load impedances are given in Table H.1.

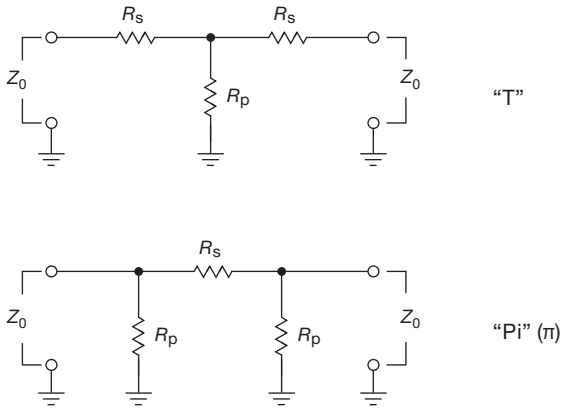


Figure H.12. Resistive T and Pi attenuators for equal input and output impedances.

H.2.3 Transformer (lossless) broadband matching network

If the unavoidable loss of a resistive matching network is unimportant in your application, that certainly is the simplest method. However, in many applications it is essential to minimize loss – for example, in a communications transmitter or in low-level circuits whose performance is limited by amplifier or thermal noise.

In that case you can use either a transformer or a reactive matching network; neither method provides coupling all the way down to dc, however. Transformer coupling is relatively broadband, but it is limited in impedance ratios; reactive matching, by contrast, flexibly matches impedances (including reactive impedances), but only around some chosen center frequency. We treat reactive matching in the next subsection.

Transformers for use at signal frequencies are similar in principle to ordinary ac power transformers, that is, they use a pair of windings that are coupled magnetically, and whose turns ratio is the desired voltage ratio. The impedance ratio is then the square of the turns ratio. (For example, a transformer with a 1:4 primary:secondary turns ratio, driven with a 50 Ω signal source, would present an output impedance of 800 Ω, and should be loaded with that resistance.) However, because of the higher signal frequencies, it is necessary to use magnetic cores that do not have significant conductive paths for eddy currents. At audio frequencies the solution is to use the same sort of laminated metal stacks used in power transformers, but with much thinner laminations. At still higher frequencies, laminated cores are replaced either by powdered iron cores or by completely nonconducting magnetic "ferrite" materials. Because of the devastating effects of parasitic capacitance and inductance, transformers for use at high radio frequencies (say above 10 MHz) generally are constructed from transmission lines (coaxial or parallel) wound around a magnetic core.

Table H.1 50Ω T and Pi Attenuators

	Pi		T	
Attenuation (dB)	R_P (Ω)	R_S (Ω)	R_P (Ω)	R_S (Ω)
0	∞	0	∞	0
0.25	3.47k	1.44	1.74k	0.72
0.50	1.74k	2.88	868	1.44
0.75	1.16k	4.32	578	2.16
1.00	870	5.77	433	2.88
1.25	696	7.22	346	3.59
1.50	581	8.68	288	4.31
1.75	498	10.1	247	5.02
2.0	436	11.6	215	5.73
2.5	350	14.6	171	7.15
3	292	17.6	142	8.55
4	221	23.9	105	11.3
5	178	30.4	82.2	14.0
6	150	37.4	66.9	16.6
7	131	44.8	55.8	19.1
8	116	52.8	47.3	21.5
9	105	61.6	40.6	23.8
10	96.3	71.1	35.1	26.0
15	71.6	136	18.4	34.9
20	61.1	248	10.1	40.9
25	56.0	443	5.64	44.7
30	53.3	790	3.17	46.9
35	51.8	1.41k	1.78	48.3
40	51.0	2.50k	1.00	49.0
45	50.6	4.45k	0.56	49.4
50	50.3	7.91k	0.32	49.7
55	50.2	14.1k	0.18	49.8
60	50.1	25.0k	0.10	49.9

Resistor values for T and Pi attenuators for use with 50 Ω source and load. The values shown can be scaled for use at some other impedance, assuming equal input and output impedances.

Impedance-matching transformers are widely available commercially, though for special applications you may need to design and wind your own. At audio frequencies many manufacturers offer miniature impedance matching transformers with "telephone" bandwidths (200 Hz–4 kHz), or full audio bandwidth (20 Hz–20 kHz); impedances go from loudspeaker and microphone impedances (8–600 Ω) up to "hi-*Z*" values of 10k–50kΩ. There's further discussion in §8.10.

A nice series of radiofrequency transformers is made by North Hills, including models that transform 50 Ω or 75 Ω to impedances up to 1200 Ω; these cover the frequency range between 20 Hz and 100 MHz, with a typical

frequency range of 1000:1 or more for a given transformer. For higher frequencies you can get broadband matching transformers from Mini-Circuits, covering the range of 4 kHz to 2 GHz with impedance ratios from 1:1 to 16:1, and with frequency ranges of 1000:1 or more for a given transformer. These are constructed with transmission-line techniques.

It is worth noting that transformer coupling provides *galvanic isolation*: input and output circuits need not share the same ground. This is particularly useful when you need to send a signal (or distribute a "house clock") between instruments whose individual cases are each grounded through their power cords or enclosure rails. We have seen several instances in which instrument "ground," in the same laboratory, differed by as much as *several volts* of 60 Hz ac. Here an isolated 50 Ω: 50 Ω broadband transformer is ideal, for example the Mini-Circuits FTB1-6 (10 kHz–125 MHz) or the North Hills 0016PA (20 Hz–20 MHz).

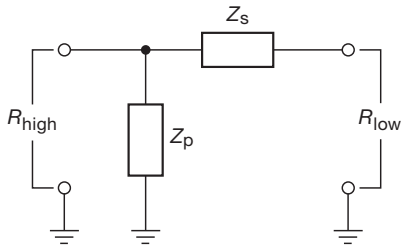


Figure H.13. Lossless reactive impedance-matching L-network.

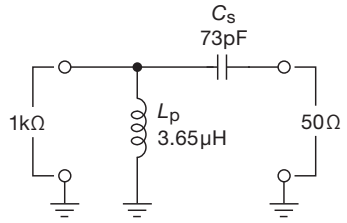


Figure H.14. Example of lossless network to match a 1 kΩ source impedance to a 50 Ω load, at a center frequency of 10 MHz.

H.2.4 Reactive (lossless) narrowband matching networks

You can match *any* pair of impedances, real or complex, with just two reactive components. The resulting match is perfect only at a single frequency, but "good enough" over some modest band of frequencies. This can be considered an alternative to (broadband) transformer matching, with considerably greater flexibility in target impedances. It is worth noting that a lossless match between impedances that are not purely real (i.e., that have a reactive component) will always be narrowband.

The simplest reactive matching network is an L-network with one inductor and one capacitor (Figure H.13). You can choose either the inductor or the capacitor as the parallel element, but the network must have the parallel reactance located across the port with the larger impedance. The design procedure is straightforward, and is nicely motivated and explained in Hagen (see Appendix N). It amounts to choosing the parallel reactance to produce (in combination with the higher port's impedance R_{high}) the correct lower resistance R_{low} as its real part, then using the series reactance to cancel the resulting reactance.

The procedure goes like this.

1. Calculate the quantity

$$Q_{EL} = \sqrt{\frac{R_{high}}{R_{low}} - 1}$$

(this will be twice the actual Q value, or frequency selectivity, of the matching network).

2. Now select the form of parallel reactance (i.e., inductor or capacitor), and set the magnitude of its reactance equal to R_{high}/Q_{EL} at the center frequency. In other words, $L_{parallel}=R_{high}/2\pi f Q_{EL}$ or $C_{parallel}=Q_{EL}/2\pi f R_{high}$, respectively.
3. Finally, set the magnitude of the series reactance (i.e., capacitor or inductor, respectively) equal to $Q_{EL}R_{low}$ at the center frequency. In other words, $C_{series}=1/2\pi f Q_{EL}R_{low}$ or $L_{series}=Q_{EL}R_{low}/2\pi f$, respectively.

As an example, let us match a 1000 Ω source (an amplifier output) to a 50 Ω load (an antenna) at a frequency of 10 MHz. We find $Q_{EL}=4.36$, and, choosing a parallel inductor at the input, $L_{parallel}=3.65\,\mu$H and $C_{series}=73$ pF (Figure H.14). The Q of the resultant coupled network is equal to $Q_{EL}/2$, roughly $Q \approx 2$; its bandwidth is thus about 50% between half-power points, though the match is perfect only at the center frequency. Note that the Q rises with increasing impedance ratio, and that you have no control over it. The network becomes a narrow bandpass filter for very large impedance-transformation ratios.

If you want a higher Q, you can get it by adding another reactive component, to form a Pi (or T) network. You can think of this as a pair of L-networks, going to an intermediate impedance that is much lower (or higher) than either port impedance. Each L-section then has an impedance

ratio greater than the final transformation, hence the higher Q value. You might think narrow bandwidth is bad, but in fact it is often desirable in communications circuits where you want to suppress out-of-band signal energy.

Alternatively, you can get lower Q than the simple L-network gives you by cascading a pair of Ls – a "double L." Here the impedance transformation is taken in two half-steps – each section's ratio is smaller than the final ratio, hence a lower Q.

H.3 Lumped-element delay lines and pulse-forming networks

The continuous transmission line with inductance and capacitance per unit length of L/l and C/l, respectively, can be approximated by an array of N discrete series inductors L and shunt capacitors C (Figure H.15). It is easy to show that the resultant circuit approximates a transmission line whose propagation time per element is $\tau_i = \sqrt{LC}$, and whose impedance is $Z_0 = \sqrt{L/C}$; the total propagation time is $t_p = N\tau_i = N\sqrt{LC}$.

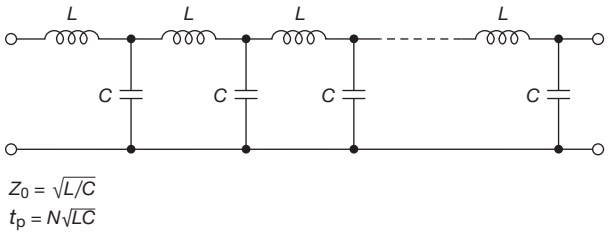


Figure H.15. Lumped-element delay line formed with an array of equal-value capacitors and inductors.

You can make a delay line this way, approximating a long transmission line. Roughly speaking, such a discrete approximation to a continuous transmission line will preserve details of the waveform only down to time scales of τ_i, or $1/N$th of the total propagation time. For example, a $1\,\mu$s lumped delay line with 20 LC sections will swallow details shorter than about 50 ns. It is a lowpass filter that attenuates frequencies above $f = 1/2\pi\sqrt{LC}$.

Lumped-element delay lines were used in early analog oscilloscopes to allow time for the sweep to begin before the (delayed) signal reached the deflection circuitry; this let you see the triggering event (and a bit before). Later analog scopes used lengths of a helical-conductor coaxial line for the same purpose. Take a look at §H.4.3 for a bit more on this fascinating application.

Lumped delay lines are useful as *pulse-forming networks*, as shown in Figure H.16. Here the parallel capacitance of a set of LC delay sections is charged to a high positive voltage; then the "center conductor" of the coax equivalent is switched to ground with a high-voltage switching element such as a thyratron. The common terminal (analogous to the coax "shield") then produces a negative voltage pulse, of duration equal to *twice* the delay-line propagation time; its source impedance is just that of the delay line. This can drive a load directly; often it is converted to a different amplitude (and perhaps opposite polarity) with a pulse transformer, as shown. Pulse-forming networks find use in radar and other applications in which the pulse voltage and/or duration are inconvenient to produce with the analogous transmission-line circuit.

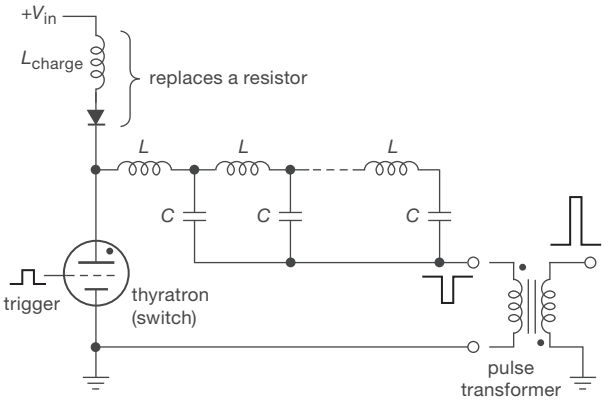


Figure H.16. Pulse-forming network for producing a high-voltage pulse of high energy. The thyratron is a special type of vacuum tube, containing a small amount of hydrogen or other gas, designed for switching really high voltages and high currents (10s of kV, 1000s of amperes: 10s of megawatts!). The inductor–diode fragment shown is a way to implement efficient "resonant charging" of a capacitor from a fixed dc voltage.

Shown also in this diagram is the method of "resonant charging," in which an inductor L_{charge} plus diode replaces the conventional charging resistor for the purpose of charging a capacitor C_{total} (the N capacitors in parallel). This has several benefits: (a) the charging wastes no energy, whereas resistive charging wastes exactly 50%; (b) it is complete after a time equal to half the period of the resonant circuit formed by L_{charge} and C_{total}; and (c) it charges the capacitor(s) to twice the supply voltage. Resonant charging is a clever technique, and it is used also in switching converters and power supplies, flashlamp circuits, and elsewhere.

H.4 Epilogue: ladder derivation of characteristic impedance

In a formal course on "waves" you are usually subjected to an analysis that uses Maxwell's equations to derive the relationship between the traveling $\vec{E}$ field and $\vec{B}$ field, from which follows the relation between voltage and current, and hence impedance. As a bonus you get the capacitance and inductance per unit length, and the speed of propagation.

But there's a nice "circuit" way to convince yourself that a properly terminated transmission line looks like a pure resistance (equal in value to its "characteristic impedance," for example 50 Ω), namely to model it as a discrete *LC* ladder (Figure H.17) consisting of little increments of length δx, each having a series inductance $\mathscr{L}\,\delta x$ and shunt capacitance $\mathscr{C}\,\delta x$, where $\mathscr{L}$ and $\mathscr{C}$ are the inductance and capacitance per unit length of the coaxial line. There are $l/\delta x$ of these in the whole length l of the line.

H.4.1 First method: terminated line

We start[8] by writing down the impedance looking into the last section of a terminated line ($\mathbf{Z}_1$ in Figure H.17), which will give us a condition on the ratio $\mathscr{L}/\mathscr{C}$ in order for $\mathbf{Z}_1$ to equal, approximately, R_0. Then we'll see that the impedance $\mathbf{Z}_{\text{in}}$ looking into the whole ladder converges exactly to R_0 as we convert the discrete ladder approximation to a continuous transmission line by taking the limit as δx goes to zero.

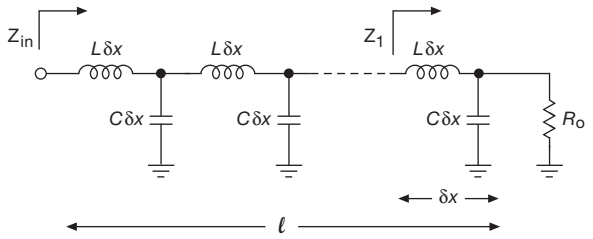


Figure H.17. *LC* ladder model of a transmission line of length l. We're ultimately interested in the limit $\delta x \to 0$, where the number of sections $N{=}l/\delta x \to \infty$.

Let's do it. $\mathbf{Z}_1$ is just the impedance of $\mathscr{L}\,\delta x$ in series with the parallel impedance of R_0 and $\mathscr{C}\,\delta x$:

$$\mathbf{Z}_1 = j\omega\mathscr{L}\delta x + \frac{R_0\cdot(-j/\omega\mathscr{C}\delta x)}{R_0 - j/\omega\mathscr{C}\delta x}$$

$$= j\omega\mathscr{L}\delta x + \frac{R_0}{1 + j\omega\mathscr{C}R_0\delta x}$$

$$\approx R_0 + j\omega\delta x(\mathscr{L} - R_0^2\mathscr{C}),$$

where in the last step we've kept only the first term of the binomial expansion, i.e., $1/(1+\varepsilon) \approx 1-\varepsilon$.

The second term vanishes when $\sqrt{\mathscr{L}/\mathscr{C}}{=}R_0$, which is the formula for the characteristic impedance of a transmission line. But... not so fast – that term would have vanished anyway as $\delta x \to 0$; however, our transmission line would have vanished as well! What we need to do is to include the next-order binomial terms, and see what happens as we let δx go to zero, while holding the total line length l constant; the number of sections then increases, as $N{=}l/\delta x$.

You can do the math. You'll find that the next two terms add contributions[9] of order δx^2 and δx^3, making $\mathbf{Z}_1$ look like

$$\mathbf{Z}_1 \approx \left\{R_0 + \mathscr{O}(\delta x^2)\right\} + j\left\{\omega\delta x(\mathscr{L} - R_0^2\mathscr{C}) + \mathscr{O}(\delta x^3)\right\}$$

and so, cascading N such sections (where $N{=}l/\delta x$), with the condition $\sqrt{\mathscr{L}/\mathscr{C}}{=}R_0$, the higher order terms vanish as $\mathscr{O}(\delta x)$ and $\mathscr{O}(\delta x^2)$ (for the real and imaginary parts, respectively) as $\delta x \to 0$. Thus the input impedance of a transmission line of length l, terminated in its characteristic impedance (resistance) R_0, is pure resistive and equals R_0.

H.4.2 Second method: semi-infinite line

Here's a clever method[10] that doesn't require approximation or worries about convergence. The idea is to look into one end of a lumped-element transmission line that extends to infinity (Figure H.18), noticing that it looks just the same if we step one notch to the right. So, calling the (complex) input impedance $\mathbf{Z}_0$, we have, simply,

$$\mathbf{Z}_0 = j\omega L + \mathbf{Z}_0 \parallel \mathbf{Z}_{\mathbf{C}} = j\omega L + \frac{\mathbf{Z}_0\cdot(-j/\omega C)}{\mathbf{Z}_0 - j/\omega C}.$$

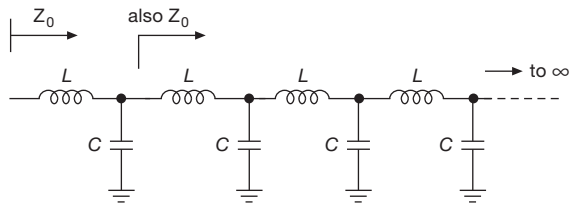


Figure H.18. Semi-infinite *LC* ladder model makes for an easy calculation.

[8] This treatment was inspired by Hagen's *Radio Frequency Electronics*; see Appendix N.

[9] They are $-R_0^3\omega^2C^2\delta x^2 + jR_0^4\omega^3C^3\delta x^3$, if you want to check your math (or ours!).

[10] Suggested to us by Jene Golovchenko.

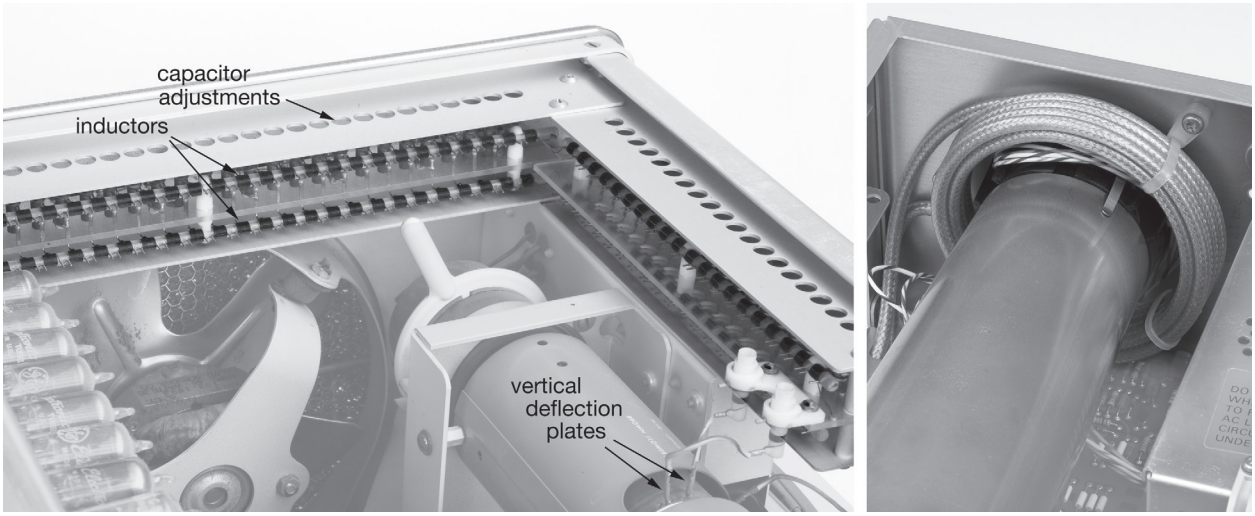


Figure H.19. Analog oscilloscopes used delay lines in the signal path, so you could see the triggering event. The 1959-vintage vacuum-tube Tektronix 545A (left) had a 30 MHz bandwidth, and used a two-channel lumped-element 200 ns delay line consisting of 50 inductor pairs and 50 (adjustable!) trimmer capacitors; (Figure H.20); their 1982 vintage solid-state 2213 'scope (right) had 60 MHz bandwidth, and used a 2.5 m length of spiral-conductor coaxial cable for its 100 ns delay line.

Multiplying through by the denominator of the last term and rearranging terms, we get a quadratic equation for $\mathbf{Z}_0$:

$$\mathbf{Z}_0^2 - j\omega L\mathbf{Z}_0 - L/C = 0,$$

with the solution

$$\mathbf{Z}_0 = j\omega L \pm \frac{\sqrt{4L/C - \omega^2 L^2}}{2}.$$

Now for the *coup de grâce*: we let the individual segments shrink toward zero while keeping the full line length. The individual L's and C's go to zero, but their ratio remains constant. Only the $4L/C$ term survives, giving us the (real) impedance $\mathbf{Z}_0 = \sqrt{L/C}$. No approximations!

H.4.3 Postscript: lumped-element delay lines

The clever designers of analog oscilloscopes, back in the dark ages of electronics, found a way to get the horizontal trace going *before* the triggering event, namely by delaying the displayed signal by ∼100 ns, using a delay line.[11] The designers of early vacuum-tube 'scopes (like the legendary Tektronix 545A) used a lumped-element transmission line like the one in Figure H.19 to achieve the delay (200 ns in this case) that otherwise would have required more than 100 feet of cable; presumably they also felt some satisfaction in exploiting the theory they learned in an electronics course they had taken years earlier.[12] Later analog 'scopes replaced the lumped-element delay line with a cleverly engineered coax delay line with a pair of spiral inner conductors; its larger inductance per unit length increased the signal delay,[13] and, happily, increased the characteristic impedance as well. The photograph shows an example, where the cable has been stuffed into an empty space at the rear, conveniently wrapped about the CRT.[14] Figure H.21 reveals the inner secrets of this elegant differential-pair delay line, whose crisscrossing counterwound helical "center conductor" produces a delay of 12 ns/ft. And Figure H.22

[11] Digital 'scopes finesse this problem by using digital memory to store some pretrigger digitized samples.

[12] The capacitors in Figure H.19's lumped-element delay line are in fact connected to the *midpoint* of each inductor, as can be seen in the official schematic of Figure H.20. It turns out that this is a more efficient implementation of a finite lumped line, as explained to us over a dinner of fine conversation and Persian cuisine by Larry Baxter ("Mr. Capacitive Sensors"; see the book by the same name).

[13] By a factor of πnD, approximately, where n is the number of turns per unit length and D is the spiral diameter. Because the coarse-pitch spiral is closely surrounded by the shield, you can think of the signal as propagating, corkscrew fashion, *along* the spiral; that approximation then gives you this simple expression, without the need to calculate the inductance and capacitance per unit length.

[14] That's *cathode ray tube*, for those born in this millennium, and thus deprived of the opportunity to admire one.

Figure H.20. Working hard to achieve perfection: Tektronix used fifty inductor pairs and fifty trimmers for the delay line in their type 545 'scope (pictured in Figure H.19) from the 1950s. Reproduced with permission of Tektronix Inc.

shows the observed delay when a differential triangular pulse-pair waveform, launched into one end, is received at the other end.

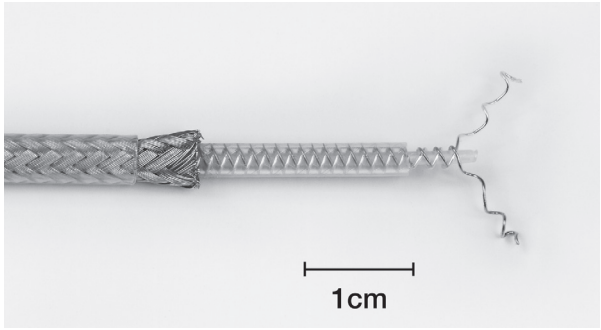


Figure H.21. We borrowed a Tektronix 2213 carcass from Brian Shaban, and, after a bit of surgery, look what we found inside! We measured a 100 ns delay, and a differential impedance of 155 Ω, for the 8.5 ft cable that used to live in this 'scope. The counterwound helix consists of two insulated 30 ga wires, with a pitch of 1.125 mm and an average diameter of 2mm, insulated from the surrounding braided shield of 3.25 mm inside diameter.

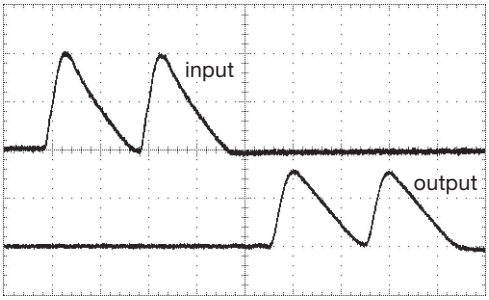


Figure H.22. We launched this analog waveform down the cable of Figure H.21 (well, what remained of it, anyway) to see the signal delay of ∼95 ns. The delayed signal is of good fidelity, with a test signal bandwidth somewhat greater than that of the 60 MHz 'scope. Vertical: 1 V/div; horizontal: 20 ns/div.

TELEVISION: A COMPACT TUTORIAL

APPENDIX I

This appendix evolved from a tutorial, originally written for a nontechnical audience.[1] It is organized as follows:

Television: video plus audio
Combining and sending the audio + video: Modulation
Recording analog-format broadcast or cable television
Digital Television: what Is it?
Digital Television: broadcast and cable delivery
Direct satellite television
Digital video streaming over internet
Digital cable: premium services and conditional access
Digital cable: video-on-demand
Digital cable: switched broadcast
Recording digital television
Display devices (CRT; LCD; plasma; OLED)
Video connections (analog; DVI/HDMI; DisplayPort)

I.1 Television: video plus audio

¶ 1. Television involves the remote delivery of a moving picture plus sound. It is accurate to think of the *sound* as continuous; however, the *picture* is captured, and then delivered, as a succession of still images, at a rate fast enough that the viewer perceives a scene of continuous motion.[2]

¶ 2. Television is distinguished further, of course, by the transmission of this movie-like content to the remote viewer. Originally this was carried out exclusively by terrestrial transmission, via radio waves, to the viewer's antenna and television set. Over time, other methods of transmission have been added – electrical cable,[3] optical fiber, direct satellite transmission via microwaves, and, of course, the Internet – along with recording methods such as magnetic videotape (Betamax, VHS, D-VHS), and optical discs (Laserdisc, VideoCD, DVD,[4] HD-DVD, Blu-Ray, and others).

I.1.1 The audio

¶ 3. The *audio* portion of television is perhaps more easily understood, as it differs little from ordinary sound-recording techniques. A microphone converts the instantaneous sound pressure variations into an electrical signal; that is, it creates as its output an electrical voltage that at each moment is proportional to the pressure of the sound wave to which it is exposed. Contemporary audio recording and delivery usually employs two or more microphones, creating "stereo" sound (i.e., two channels), or multichannel sound (e.g., "5.1 channel sound").

¶ 4. Traditionally these signals were processed, stored, and delivered by "analog" methods, which means simply that the signals were treated as smoothly varying voltages as they passed through the electronic innards of the amplifiers, recorders, modulators, and so on.[5] Contemporary "digital" technology does it differently: almost as soon as possible, the microphone's signal (the varying voltage that represents the sound) is converted to a succession of numbers (it is *digitized*), and everything that follows is some form of arithmetic on this torrent of numbers that come tumbling out. Only at the final stage – recreating the recorded sound for the listener – is the digital representation converted back to an analog voltage, and then, in the loudspeaker, to a reproduction of the original sound pressure wave.

¶ 5. Just to give a sense of the quantity of numbers involved, in the standardized recording technology of the

[1] Following the stylistic conventions of that audience, the paragraphs are numbered sequentially.

[2] For conventional cinema-style movies, the rate is 24 frames/second; television in the United States uses a rate of approximately 30 frames/second.

[3] Known technically as *coaxial transmission line*.

[4] "Digital Versatile Disc."

[5] Common analog recording technologies, now nearly obsolete, include the vinyl record (where the audio signal waveform is embossed as small displacements of a fine groove), and the audio cassette tape (where the audio signal waveform is recorded as patterns of magnetization on a thin layer of magnetic oxide coating on a flexible plastic tape).

compact disc (CD), the instantaneous sound is *sampled* at a rate of 44,100 times per second (in both stereo channels simultaneously), and each such sample pair is converted ("digitized") to a 16-bit binary number.[6] So the bits are tumbling out at a rate of $2\times44{,}100\times16 = 1{,}411{,}200$ bits per second, or nearly 100 million bits per minute.[7]

¶ 6. One might ask why any sane person would want to deal with such a quantity of numbers, when the original analog representation of the sound was so much simpler – just a pair of voltages that were varying at most 20,000 times per second.[8] The reasons are several, but they boil down to the contemporary ease and economy of digital processing, combined with the higher efficiency and quality of storage and transmission of audio (and video) that has been properly digitized. To get a sense of those advantages, one need only marvel at the gorgeous images transmitted daily from planetary probes visiting Mars and Saturn – images that are free of "snow" and other artifacts irreparably added to analog transmission by the effects of unavoidable electrical interference – to appreciate the benefits of error-free digital transmission. And, to get a sense of the density of digital storage, we note that a contemporary 5″ optical disc (dual-layer Blu-ray disc) holds 80 hours of CD-quality audio, or ten times that amount if modestly "compressed," compared with a mere hour's storage of analog audio on the 12″ vinyl LPs of yesteryear.[9]

I.1.2 The video

¶ 7. The *video* is by far the more complicated part of television. The challenge is to reproduce a scene with motion, in color, while preserving adequate fidelity and absence of artifacts. And this must be done within the resources of the storage and delivery channels – that is, with finite disk storage and speed, and with finite transmission (via broadcast tower, cable, Internet, or satellite) *bandwidth*.[10]

¶ 8. Video systems begin with a camera that has an electronic sensor (analogous to a digital camera) and that converts the two-dimensional color scene on that sensor into a succession of *frames*, each of which represents the image at those successive times (for US TV, the rate is approximately 30 frames per second). In traditional *analog* television, the two-dimensional image is converted into an electrical signal by the following method: imagine a single frame, that is, a still picture. To keep it simple, imagine further that it is monochrome; that is, "black and white."[11] We begin at the upper left, and move horizontally across the picture, generating an electrical voltage proportional to the brightness at each point as we pass by. When we reach the right-hand border, we jump back to the left edge, continuing with another horizontal path, slightly below the first. See Figure I.1. We continue in this way until we reach the bottom right-hand corner, at which time we have scanned the entire frame once, in what is known as a *raster* ("grid" in German) pattern.[12]

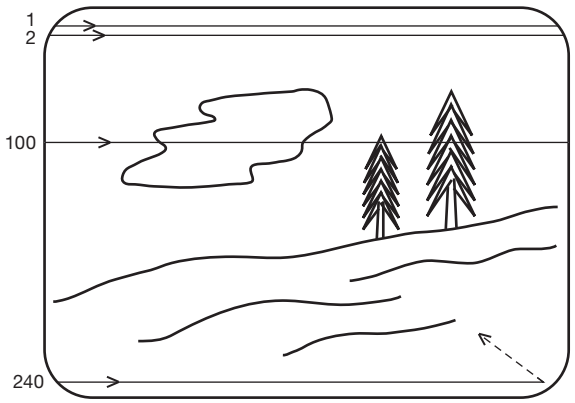


Figure I.1. A static two-dimensional image is "raster scanned" to create a video waveform (Figure I.2) representing intensity along the scan lines.

¶ 9. What we have done, then, is to generate an elec-

[6] That is, a number ranging from −32768 to +32767, those bracketing the "full-scale" range of the recorded sound.

[7] The *recorded* bitrate is roughly triple this figure because of coding, error-correction redundancy, and the like.

[8] Or 20 kHz, the upper limit of human hearing; and that only for one of relative youth, such youth further possessed of sufficient wisdom to avoid deafening rock concerts.

[9] And a contemporary 3 terabyte 3″ magnetic hard drive that you can hold in your hand holds yet another factor of 60, or 50,000 hours of excellent quality (128 kbps AAC) compressed stereo audio; that's 15 years of 40-hour per week music!

[10] *Bandwidth* refers to the range of frequencies that can be carried on the cable or other transmission medium. It is technically accurate to think of this, for example, as the range of stations on the radio dial that could be carried with fidelity by a single electrical cable (or other medium). The term is sometimes used loosely to refer to rate of data transfer.

[11] Or, more accurately, *grayscale*.

[12] Traditional standard definition television (SDTV, usually called "NTSC," for National Television System Committee, and going back to the 1940s) in the United States divides the whole picture into 480 horizontal lines, along each of which roughly 640 features (picture elements, or *pixels*) could be resolved; a computer user would not be terribly impressed – he or she would say that standard NTSC television has only "VGA" resolution (i.e., 640×480).

trical representation, in time (a varying voltage proportional to brightness at each point in the image) of a single two-dimensional image; that is, we've converted a two-dimensional image into a one-dimensional output voltage. See Figure I.2.

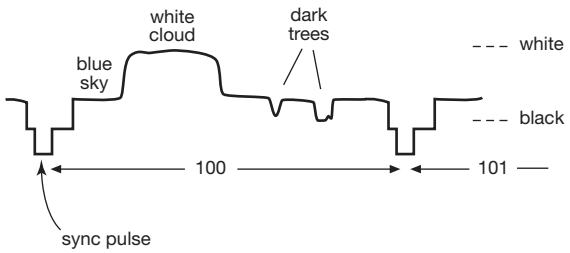


Figure I.2. A portion of the video waveform from Figure I.1, representing one of the 240 horizontal scan lines.

¶ 10. This time-varying voltage is called the *video* signal, and it is the first step in creating a television image. In traditional NTSC analog television, this signal was transmitted by analog methods, after a process called *modulation* (more later), and was recovered and used by the television set to paint the picture on the screen, performing the same raster scan (left to right, top to bottom). Each frame follows in sequence, presenting a succession of 30 pictures per second on the television set's viewing screen.[13]

¶ 11. To complete the video signal, it is necessary to add some synchronizing information, so the television set knows when to begin painting a frame, and also when each horizontal line begins. In traditional NTSC television this is done by adding a horizontal *sync pulse* at the beginning of each horizontal line, which is just a short[14] voltage pulse that, if it were in the middle of a picture, would be interpreted as "blacker than black." See Figure I.3 The television set detects these pulses and uses them to synchronize its scanning across each line. Likewise, a unique *vertical sync pulse* is transmitted for each field which informs the

[13] To complicate things, NTSC uses a method known as *interlacing*, in which a coarser raster – omitting every other horizontal line – is performed at twice the rate. Thus, in standard NTSC US television, the viewer sees 60 pictures ("fields") per second, each of which has only 240 horizontal lines; two such fields, with their interlaced lines, form one complete 480-line frame. This is sometimes called a "480i" format, to distinguish it from formats with higher resolution (e.g., high-definition TV, HDTV, with 720 lines or 1080 lines, or "4K Ultra HDTV," with 2160 lines), or from those without interlacing (which are known as *progressive*; e.g., *720p*).

[14] About 4.5 millionths of a second.

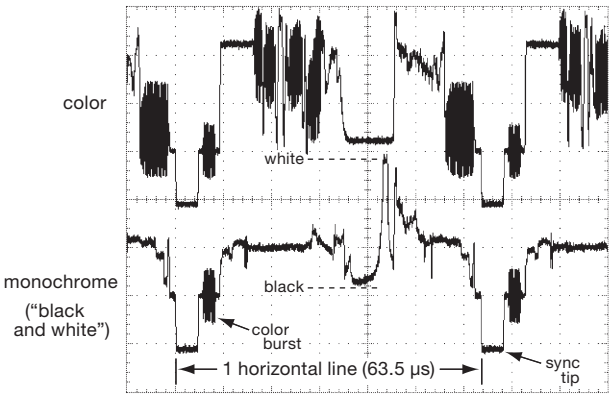


Figure I.3. Composite analog video signal of one horizontal line, framed by horizontal sync pulses. The brightness ("luminance") is represented by its amplitude. Color is accommodated by adding a modulated 3.58 MHz "chrominance" subcarrier, whose amplitude represents degree of saturation and whose phase encodes the colors.

television set when to return to the top to begin painting the next field or frame (see Figure I.4. The complete video picture signal, with its added sync pulses, is called *composite video*.

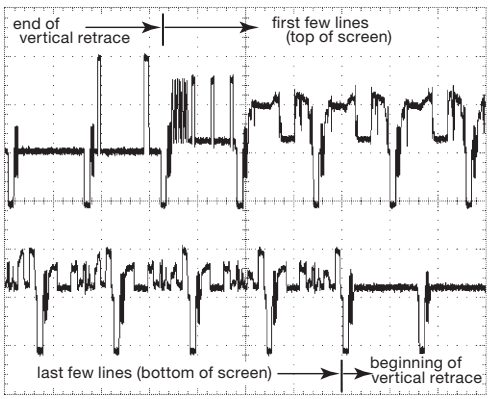


Figure I.4. The vertical retrace (beginning of a new field) is signaled by a set of tailored sync pulses, the first and last of which are shown here.

I.2 Combining and sending the audio + video: modulation

¶ 12. Continuing for the moment with traditional NTSC television (as opposed to *digital* television, whose standards are known as *ATSC*, set by the Advanced Television Systems Committee, and which will be explained later), the composite video, along with the audio, must now be

sent, via transmitting tower or cable, to the home viewer. Naively one might think of simply transmitting these signals "as is." This is not done, however, for at least two reasons: first, if the composite video signals were transmitted directly, then any two television signals would overlap and jam each other (because they would all share the same frequency band, namely that of the raw video signal itself); second, some wavelengths are more conveniently generated and propagated than others. For these reasons, the audio and video content of television signals (and, indeed, all communications and broadcast signals) are instead used to vary some aspect of a "carrier" wave, chosen at some specified wavelength. That carrier wavelength (or, equivalently, frequency) defines the "channel"; and the process of impressing the information (video and audio) onto the carrier wave is known as *modulation*.

¶ 13. Radio stations use the same technique: AM stations vary the strength (amplitude) of the carrier (hence "amplitude modulation"), whereas FM stations vary the frequency ("frequency modulation"). The carrier frequency itself defines the channel: in the US, AM stations are assigned to carrier frequencies between 520 and 1710 kilohertz (kHz, thousands of cycles per second), while FM is assigned to the band of carriers from 88 to 108 megahertz (MHz, millions of cycles per second). In the US, broadcast television begins at 54 MHz (Channel 2), and ends at 698 MHz (Channel 51), with gaps for FM, aeronautical, and other services.[15]

¶ 14. When information (video, for example) is modulated upon a carrier, the resultant signal spreads out and occupies a small band of frequencies. For example, when an FM station varies the frequency of its assigned carrier to carry its audio signal, the resulting signal occupies about 150 kHz. So FM stations are assigned channels separated by 200 kHz (to allow a "guard band" of 50 kHz in addition to their 150 kHz signal) – and that is why the frequencies of FM stations always end in an odd number after the decimal point (for example New York City's WNYC is at 93.9 MHz), ensuring a minimum spacing of 0.2 MHz (= 200 kHz).

¶ 15. Traditional analog NTSC broadcast television used a variant of AM for the picture signal (composite video) and, separately, FM for the sound signal.[16] The assigned TV channels are spaced apart by 6 MHz, each station being permitted to occupy nearly that amount, after allowing for a small guard band. Television sets "know" the frequencies allocated for each channel and tune to the correct frequency when the user chooses the channel number. For example, if (during television's analog era) one tuned to Channel 13 in the New York City area, the television set's electronics selected the station transmitting on 210 MHz (assigned by the FCC as Channel 13), namely WNET. The electronics in the set *demodulates* the received signal, recovering composite video and audio. The video, with its embedded sync signals, is used to paint the picture, frame after frame, while the audio is sent to the loudspeakers.[17]

¶ 16. Broadcast television (and radio) takes place on what is often called the "public airwaves." One needs only a television set (or radio) and an antenna to receive these over-the-air (OTA) public transmissions. Although some countries require licensing of receiving devices (radios and television sets), in the United States the broadcast services are freely available to anyone within range of a transmitting tower.

¶ 17. Depending on the distance and path from the broadcast station to the viewer, the "antenna" can be as simple as an indoor "bowtie" or pair of "rabbit ears," or as elaborate as a roof-mounted multi-element structure. Whatever its form, the antenna's function is to create an electrical signal on the feedline, induced by the speed-of-light broadcast signal passing by the antenna's site. Receiving antennas intended for broadcast television are designed to work over the range of frequencies used by broadcasters (see ¶13); thus the electrical signal delivered to the television set includes multiple stations, and it is the job of the TV tuner to select and process the channel to be viewed.

¶ 18. *Cable* television sends traditional analog TV channel signals in almost exactly the same way as broadcast. An evident difference, however, is that the channelized signals are received at the viewer's end from a coaxial cable (rather than being received by the viewer's television antenna), and then connected to the television set directly (i.e., to its normal antenna connector on the rear). Alternatively, for additional cable services (such as premium channels) the incoming cable connects to a "set-top box" (STB) provided by the cable company, the output of which is connected to the viewer's television set (or flat-screen monitor,

[15] You can download a gorgeous multicolor wall-sized spectrum allocation figure from `http://www.ntia.doc.gov/files/ntia/publications/spectrum_wall_chart_aug2011.pdf`.

[16] That is, the picture and sound signals are carried simultaneously on a pair of designated carrier frequencies within the single assigned television channel.

[17] In this primer we have ignored details associated with reproduction of *color* (versus black and white).

projector, etc.; see ¶56ff).[18] The channel *frequencies* are also somewhat different, with Channels 2–13 chosen the same as for terrestrial broadcast, but with the remaining channels reassigned to eliminate gaps; this can be done because the cable is a private world of its own, isolated so as not to interfere with, or be interfered by, other broadcast services.

¶ 19. A third difference is that some of the cable content is delivered as a subscription, for which the viewer pays additional fees; examples are premium services such as Home Box Office (HBO). These require some method for permitting or denying viewable delivery of selected channels or programs. Continuing for the moment with analog cable (whose days are numbered!), this can be done in several ways: the simplest is by installing filters (to block unsubscribed channel frequencies) at the utility pole, where the subscriber's cable splits off from the trunk running along the street;[19] A more sophisticated method involves scrambling the cable-borne analog signals of subscription programs,[20] and then instructing the set-top box (via digital communication from the cable provider to the individual STB) as to which programs may be unscrambled.

¶ 20. It is worth noting that cable companies have been required to carry the broadcast stations in their area, normally as analog cable channels.[21] Each such program occupies a cable channel (frequency). However, they may distribute additional services via digital methods ("digital cable") on additional channels (frequencies), which they much prefer; that is because, with digital methods, it is possible to carry up to *ten* NTSC-quality programs (i.e., SDTV, for Standard-definition TV) on a single channel. This is called *multicast*: the ability to carry multiple *programs* on a single channel (i.e., frequency). And, note that a cable can carry more than 100 such carriers – permitting more than 1000 simultaneous programs.

¶ 21. Previewing some additional characteristic of digital television: digital cable permits flexible subscriptions, with a program being authorized on-the-fly (e.g., pay-per-view, or video-on-demand). It also provides a natural mechanism for content protection via encryption. It allows for interactive participation, via a reverse channel back to the cable provider. It permits the delivery of high-definition content, with more than the 480 lines of NTSC (up to 1080 lines, at the highest quality currently supported). Finally, it provides a natural way to time-shift, pause, or replay live programs, via computer-type hard-disk storage.

¶ 22. Analog broadcast was sent into retirement in the United States in June of 2009, and all television broadcast delivery is now done by digital methods (more to follow). This conversion-to-digital process is taking place worldwide and will likely be complete by 2020 or earlier.

I.3 Recording analog-format broadcast or cable television

¶ 23. Video recording was complex and costly (and therefore confined to the broadcast studios) until 1975, when home video-recording devices were introduced in the US by Sony ("Betamax") and its competitors ("VHS," for video home system). These devices replicate the "front end" of a television set, to recover video and audio from the incoming signal (broadcast or cable), and use a clever spinning tape head arrangement[22] to capture on magnetic tape a reasonable replica of an NTSC analog television program. The technique is entirely analog (no digitization, no numbers) and records only onto special video-tape media (no computer media, no hard disks, etc.), as a magnetic recording (analogous to an analog audio tape recording; see the footnote at ¶4).

¶ 24. Videotape technology has been upstaged by digital alternatives such as optical disc recording (most famously in the form of DVDs and Blu-Ray discs – whether sold with prerecorded content, or recorded with a disk recorder), which creates a permanent copy of the video material; or by recording to a computer-type hard-disk drive (hdd), where the video copy is stored as a computer file. These digital methods require that the program material be converted from analog to digital form, if it is not already. (This is done internally and automatically in devices like TiVo®

[18] For better picture quality, the latter connection is usually made not to the set's antenna input (called "RF," for Radio frequency, meaning the modulated channels discussed above, in ¶¶12–15), but to special audio–video inputs, with names like *s-video*, *component video*, *composite video*, *DVI*, or *HDMI*; see ¶64ff. The latter pair are *digital* connections, discussed below in connection with digital TV.

[19] Vintage cable subscribers will remember calling the cable company to add a movie channel, whereupon a cable truck appeared, the cable guy went up the pole to fiddle with something (changing the filter), and, voilà, movies on your television!

[20] For example, by suppressing the horizontal sync pulses or inverting the video (interchanging black and white).

[21] Unless all subscribers are provided with STBs that can receive digital delivery.

[22] This is known as a "helical" tape head, which creates successive narrow slanted tracks across the slowly moving tape, each one holding one field of video. The use of a rapidly moving tape *head* eliminates the need for rapidly moving *tape*.

and other personal video recorders.) Digital television and digital video are discussed next.

I.4 Digital television: what is it?

¶ 25. Just as an *audio* signal can be digitized (i.e., its instantaneous amplitude is measured, at rapid intervals, and converted to a succession of numbers), and subsequently transmitted, stored, or processed (¶¶4–6, above), so it is possible to digitize the *video* signal that represents successive frames of picture. Although one could imagine simply sending the digitized version of traditional NTSC as "digital TV," in practice one takes advantage of the enormous processing finesse of contemporary digital electronics to economize by *compressing* the raw video signal to a small fraction of its native size before it is delivered. The use of compression, along with the fact that a digital signal is "just numbers," permits the delivery of multiple programs on what would otherwise carry just a single video signal (program), typically by a factor of five to ten.

¶ 26. There are several reasons for this improvement. One is the ability to detect and correct transmission errors by numerical techniques, allowing one to operate with received signal levels that are close to the "noise" (from interference, or signal loss owing to range or obstructions); with purely analog transmission a large received signal/noise ratio is necessary to reduce the visible effects of noise ("snow") to acceptable levels.

¶ 27. A second reason is the spectral efficiency of digital transmission – or, more accurately, its improvement compared with the *inefficiency* of analog signaling. This can be seen in Figures I.5 and I.6, a pair of spectra taken directly off the home antenna of one of the authors in March of 2009, a time during the switchover to digital when both analog and digital broadcasts were taking place (see also Figure I.7).

¶ 28. Compression aims to reduce by a large factor (tenfold or more) the quantity of numbers needed to describe the succession of picture frames, without significantly degrading the image quality. This seemingly impossible task takes advantage of redundancies in a moving picture, and of limitations in human visual perception.

¶ 29. Contemporary digital-video compression is a rich and mathematically complex subject, the result of enormous effort in the applied mathematics and electrical engineering communities over the last several decades. But the basic tricks are easy enough to understand. The process begins by exploiting the fact that portions of an image near each other tend to be similar; so one can encode and send the (smaller) *differences* of brightness and color from a set of reference points, rather than the full description of brightness and color at each point. Likewise, successive frames tend to be similar, so one can define a sparse collection of index frames and send only the differences for intervening frames.[23] A further trick exploits the fact that the image often contains moving objects or a panning camera; so it is efficient to calculate "motion vectors" predicting

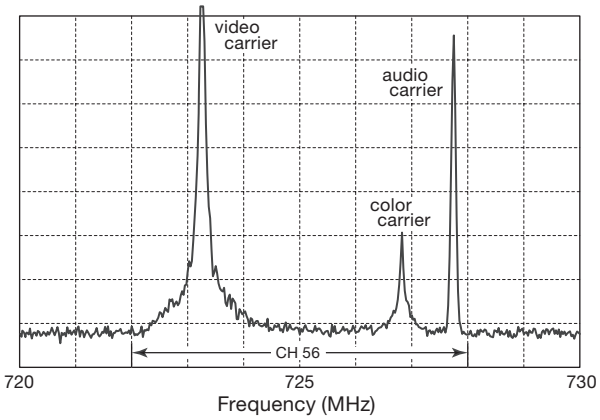


Figure I.5. The spectrum of 6 MHz-wide analog Channel 56 in Boston, as seen in May 2009. The video information resides in the sideband tails, while most of the transmitted power is wasted in the non-informational video carriers.

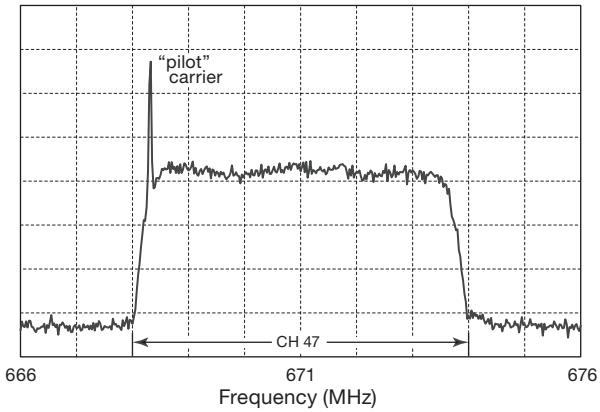


Figure I.6. Digital Channel 47, seen also in May 2009, fills its 6 MHz spectrum allocation with digitized video. It carries five times as many programs, with comparable (or better) picture quality.

[23] More precisely, it is the corrections from an *interpolated guess* between index frames (or reference points within a frame) that are sent.

the approximate motions, and then send only the (smaller) corrections from the predicted values.

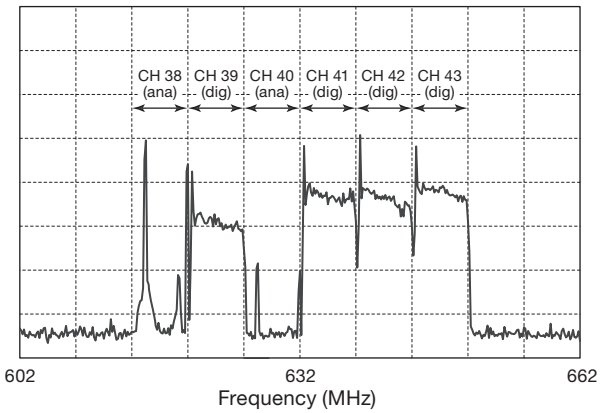


Figure I.7. A country in transition: RF spectrum of Channels 36–45 (each permitted 6 MHz of spectrum) as seen on our antenna feedline in Cambridge, Massachusetts, on May 6, 2009. Analog Channels 38 and 40 each carry one standard-definition (SDTV) program, whereas Digital Channels 39 and 41–43 can carry up to five SDTV programs each (though it's more common to see one HDTV and one SDTV program).

¶ 30. These methods greatly reduce the needed bitrate (number of numbers per second), and they do so *without any loss of picture quality whatsoever* – they are "lossless." That is because the original digital image can be fully and exactly recovered by applying the numerical differences in the reverse order. However, further bitrate reduction is desirable (and often necessary), and this is accomplished by *lossy* compression. This consists essentially of discarding the less-important image information (from a psychovisual standpoint); the tradeoff is a somewhat degraded image (the degree of degradation depending on the degree of compression), which, however, can differ from the pristine original in ways that are hardly perceptible to the viewer.[24] The mathematics involves methods with names like discrete cosine transform, variable quantization, and Huffman coding; but the bottom line is that these methods permit a large reduction in bitrate with a relatively small reduction in perceived image quality.[25]

¶ 31. The tradeoff of image quality with bitrate is gradual, and somewhere in the process a decision is made as to the desired final bitrate.[26] A major constraint is imposed by the fact that both digital broadcast and digital cable television in the US is sent on channels that conform to the same 6 MHz channel spacing that has been used for television since the 1940s. In practice (see ¶35, below) it is possible to send about 20 million bits per second (Mbps) on an over-the-air digital-broadcast channel, and nearly double that on a digital cable or satellite channel. A typical compressed bitrate for over-the-air NTSC-quality (SDTV) digital video is about 4 Mbps; thus digital-broadcast television stations are able to combine up to 5 or so NTSC-quality programs on a channel. (Recall that a frequency "channel" is no longer a single "program," because of multiplexing. More on this beginning at ¶33, below.) High-definition (HDTV) content requires nearly the full broadcast bitrate, so only one HDTV program can be broadcast on a channel. By contrast, cable or satellite systems, which are not constrained to MPEG-2 compression, are able to combine as many as eight HDTV programs onto one channel when using efficient H.264/MPEG-4 encoding. The second revision of the broadcast standard (ATSC 2.0) incorporates these more efficient codecs, as well as a host of transport and delivery enhancements that are aimed at mobile and interactive viewing, thus allowing over-the-air broadcasting to compete with services available on the internet.

¶ 32. It is worth admiring the impressive bitrate reductions that these methods are achieving: a simple calculation[27] shows that digitizing an HDTV program without any compression would produce a bitrate of roughly 1000 Mbps, whereas contemporary compression methods reduce this to a modest (and deliverable) 20 Mbps, a 50-fold reduction! And comparable reductions are routinely achieved with SDTV.

[24] If such effects are noticeable, they are called *compression artifacts*; these are sometimes seen in over-compressed "jpeg" still photographs, as the patchy blocks or the "mosquito noise" around edges. Similar considerations apply to lossy audio compression, for example highly compressed "MP3" music files.

[25] The video-compression recipe currently used for all digital TV broadcasting in the US is named "MPEG-2" and described in the Advanced Television Systems Committee documents A/53 and A/54 (see www.atsc.org). An improved set of compression methods is incorporated in the set of standards known as MPEG-4; these are widely used by the cable and direct broadcast satellite services, as well as for video streaming over the internet.

[26] Which is permitted to vary, as program content changes. This is known as *variable bitrate*, or VBR, as distinguished from *constant bitrate*, or CBR.

[27] Bitrate $\approx$ 1080 lines $\times$ 1920 pixels/line $\times$ 30 frames/second $\times$ 16 bits/pixel = 995,328,000 bits/second.

I.5 Digital television: broadcast and cable delivery

¶ 33. Over-the-air digital television broadcast and digital cable television both use traditional frequency "channels," upon which they put a stream of numbers (the compressed video described in ¶¶25–32, plus the associated digitized audio[28]), instead of the continuous analog waveform that was used in traditional NTSC television. Because of the economical bitrates produced by compression, there is adequate capacity on a single cable (or broadcast) channel frequency to accommodate several simultaneous programs. This is called multicasting, and permits up to four or five SDTV programs (a "multiplex") to be carried on a single broadcast channel frequency.[29] One can think of these as subchannels.[30]

¶ 34. For either OTA digital broadcast or digital cable, the set-top box (STB) or equivalent hardware within the television set receives the multiple channel frequencies, each with its multiplex of programs. The STB or television knows the program assignments within each channel and is able to pull out the subchannel that the viewer selects, which it identifies by assigning a "virtual channel number." That is what the viewer chooses – it is displayed on the STB and on the screen during selection. For example, the viewer might select HBO, which is assigned a virtual channel (e.g. 82) and which might actually be just one of ten subchannel programs carried on one digital cable channel frequency. The STB then captures the HBO stream, decrypts and decodes its MPEG-2 encoding, and converts it to displayable video for a television monitor (or flat screen, etc.).

¶ 35. In more detail, and in the language of digital television engineering, the delivery channel (digital broadcast or digital cable) is called the "transport stream," which can be thought of as a data pipe carrying some 20 Mbps (broadcast) or 38 Mbps (cable) in each frequency channel.[31] The ATSC specification dictates that the data put onto the transport stream must be broken into little *packets* of data, each of length 188 bytes and each belonging to an individual program. When multiple programs are sent on one transport stream, it is called a "multiprogram transport stream," or MPTS; if a single program, it's a "single-program transport stream" (SPTS). Repeating what was said earlier: a broadcaster can can put five standard-definition programs (or one HD and one SD program) onto a single broadcast channel's MPTS. The individual packets are identified by program, and they are interleaved in time (see Figure I.8).

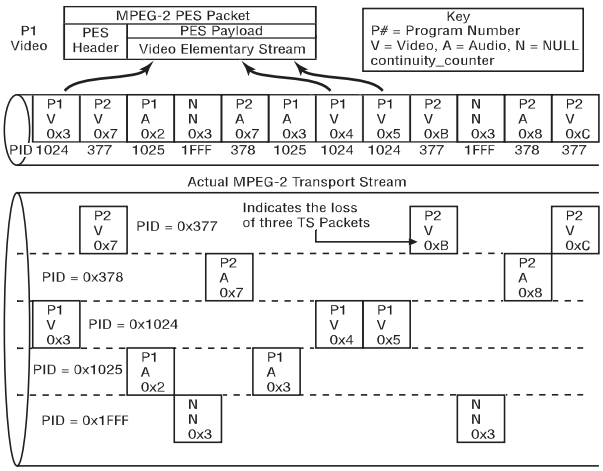


Figure I.8. Multiple programs can be interleaved into one digital transport stream, as a "multiprogram transport stream." Their individual video and audio packets are tagged with program identifiers (PIDs), by which they can be selected and reassembled (adapted from Figure 7.1 of ATSC Doc. A/54A, courtesy ATSC; readers are recommended to reference the current version of the standard or recommended practice available on the ATSC website).

¶ 36. To put it another way, the several programs that will be put into a multiprogram transport stream (a "program multiplex") are cut into short pieces (packets, about 40 μs in length for digital cable), tagged with unique identifiers (called PIDs, for "program identifiers"), and then interleaved with the pieces (packets) from the other programs

[28] And also some "conditional access" (CA) information that enables legitimate subscribers to decrypt and view protected content; see ¶47.

[29] *Cable* delivery is more efficient, and permits as many as ten SDTV programs on a single channel.

[30] Because the cable (or broadcast channel) has a fixed total bitrate, the bitrates of the individual programs that are being multiplexed must be adjusted such that their total combined bitrate matches the channel capacity. This is called *bitrate grooming* and involves *null padding* (adding nulls, to increase a program's bitrate), on-the-fly compression (to reduce a program's bitrate), or even time shifting of program content (to prevent unlucky alignment of peak bitrates of the various programs). Digital television packets include "presentation time stamps," so it's OK to move things around a bit as they flow through the various digital pipes on their way to the television screen.

[31] The disparity has to do with the particular modulation schemes used: for broadcast it is called "8-VSB," whereas cable uses the more efficient "256-QAM" (pronounced "quahm"), thereby exploiting cable's better signal-carrying properties to carry roughly twice the information content.

that share the same transport stream. At the STB or television set the packets belonging to the selected program are identified ("filtered") by looking for their PIDs, and then reassembled into a single program transport stream to be decoded and displayed. In Figure I.8 there are two programs (P1 and P2), each with video (V) and audio (A), with their corresponding PIDs (1024, 1025, 377, 378); they are shown as the interleaved multiprogram stream at the top, and as filtered into their respective single-program streams at bottom.

I.6 Direct satellite television

¶ 37. Satellites provide an alternative to over-the-air or cable/fiber delivery of television programming, and satellite delivery is particularly welcome in areas not served by wired broadband connections. This is known variously as direct-to-home (DTH), direct broadcast satellite (DBS), or broadcast satellite services (BSS), and exploits (usually) satellites in the geostationary constellation, i.e., in the equatorial "Clarke" orbit[32] of radius 42,200 km, where a satellite's period matches Earth's rotational period of 23h56m4s.[33] As surprising as it may seem, a single satellite with a ∼100 W transmitter can deliver a half-dozen high-definition programs (or 30 "standard-definition" programs) simultaneously to small dish antennas on houses everywhere in the continental US. Typical direct-broadcast satellites are equipped with a dozen or more such "transponders," and the receiving dish antennas use multiple "feeds" (as many as four, for DirecTV or DISH Network) to point at several satellites, making available many hundreds of television programs.

¶ 38. Early DBS systems used a 4 GHz (C-band) downlink, and required large dishes (>3m diameter) and expensive RF electronics. Contemporary systems operate around 12 GHz (Ku-band), with mass-produced oval receiving dishes (typically 0.6m×0.8m) that incorporate several low-noise RF amplifiers, each with local oscillator, downconverting mixer, and IF amplifier in an integrated LNBF (low-noise block-downconverter plus feed) unit that sits at the focus of the parabolic dish. That's an impressive amount of hardware for $100.

¶ 39. One may be puzzled by the peculiar orientation of home dish antennas – why are they sometimes pointed so low that they seem to be aimed at or below the horizon? There are two parts to the answer: first, the constellation of geostationary satellites spans an arc across the southern sky, populated worldwide with some 200 satellites; over the longitudes of the US alone there are some 35 satellites parked[34] in geostationary orbit. The line of satellites across the southern sky dips down to the horizon at its eastern and western ends.[35] Second (and technologically more interesting), the geometrical arrangement of the receiving dish is what's called an "offset-feed paraboloid." That is, the little conical feeds are offset below so they do not block the incoming signal. This makes the dish appear to be pointing some 25° lower than it is, thus the explanation for the apparently "subterranean" satellites. This peculiar arrangement is used to eliminate blockage of incoming signal by the feedhorns, and also to reduce the encroachment of thermal radio noise that is emitted by the surrounding environment – another bit of thoughtful design that enables successful direct satellite broadcasting.

¶ 40. The individual transponder RF channels are 27 MHz wide, with as many as 32 such transponders on a satellite, which (with guard bands) adds up to a total downlink bandwidth of ∼1000 MHz. Satellite transponders typically use phase-shift keying (QPSK, 8PSK), with downlink bitrates of 40 Mbps per channel, adequate to deliver a half-dozen or more 1080i HD programs with efficient H.264/MPEG-4 encoding. The transponder channels are divided into two sets, one of each circular polarization at each transponder frequency, so that the total downlink spectrum occupied by a given satellite is about 500 MHz. This is downconverted in the LNBF to a pair of 500 MHz-wide IF bands, centered at 1.2 GHz and 1.9 GHz.

¶ 41. Rather more hardware resources are expended on the *up*link, with steerable dishes of ∼10 m-diameter class (Figure I.9) illuminating the uplink receiving dishes at the satellite, with transmitted power of the order of hundreds of watts per transponder; dishes of that size produce diffraction-limited beam diameters of about 0.15°

[32] See "Extra-terrestrial relays – can rocket stations give world-wide radio coverage?" [Sir] Arthur C. Clarke, *Wireless World*, October 1945.

[33] Ha! You thought it was 24 hours. Not so – we're in orbit around the sun, which adds an extra 4 minutes (1/365th part of a day) to Earth's true rotational period (the *sidereal* day) to arrive at the 24-hour *solar* day (the average time from solar noon to solar noon).

[34] Hardly "parked," of course – they are in equatorial orbits, whizzing around the Earth at nearly 7000 miles per hour, to keep up with Earth's rotation.

[35] You can get a good sense of the satellite arc with with mobile apps like DishPointer or Satellite Locator: when you point your smartphone into the sky it shows the satellites as bright red circles (with their locations) on a red arc, superimposed on the camera view (with obstructing trees, etc.) seen by the mobile device.

($\theta \approx \lambda / D$), small enough to prevent illumination of a satellite in an adjacent orbital slot, but requiring some active steering to maintain alignment on the desired satellite. Direct-broadcast providers such as DirecTV and DISH Network like to position their satellites close enough along the Clarke belt so that a single dish with several feeds can capture the downlinked signals from several of them; for example, both DirecTV and DISH Network offer an oval dish with three feeds, targeting satellites separated by 10° (currently at 110°W, 119°W, and 129°W longitudes).



Figure I.9. A few of the several dozen transmitting antennas at EchoStar's Network's uplink facility in Cheyenne, Wyoming.

¶ 42. You hook the coax cables from the receiving dish to a set-top box (usually with DVR), similar to what's used for cable TV, but which is designed to power the LNBF electronics up there on the roof, to select from the dish's several feeds and polarizations, and to receive the intermediate-frequency (IF) bands coming down from the dish on standard 75 Ω video coax (quad-shielded RG-6, usually).[36] As with cable TV, the content providers control your available programming, via subscription-enabled decryption. Satellite systems are not bound by over-the-air terrestrial standards (e.g., MPEG-2 encoding), and they generally use more efficient schemes such as MPEG-4. When compared with cable or fiber television service, the inability to target millions of subscribers individually (along with the absence of a reverse uplink channel to the satellite) limits the possibilities for interactive services such as video-on-demand.

[36] There may be an intervening module or two, with names like "multi-switch" or "node," to deal with selection of polarizations and IF-band stacking.

I.7 Digital video streaming over internet

¶ 43. With steady improvements in internet bandwidth (i.e., speed), it has become practical to deliver video (and associated audio) through the same internet infrastructure that serves personal computers and mobile devices (cellphones, tablets) with their email, web browsing, and so on. Some familiar examples of internet video streaming are news services such as CNN, government services such as NASA TV, movie streaming services such as Netflix and Hulu, and peer-to-peer services such as Skype. Just as with broadcast or cable delivery, the ultimate payload is a stream of numbers that constitute the video and audio content, in some efficient compressed format that takes advantage of sophisticated encoding schemes with names like "H.264" (also known as AVC, for advanced video coding), one of the current favorites. At the viewer's end the digital content is decoded[37] to recover the video and audio. For some services (e.g., Skype) a dedicated client program must be installed, whereas for others (e.g., NASA TV or CNN) a standard internet browser (such as Internet Explorer, Safari, or Firefox) suffices.

¶ 44. Compared with delivery via broadcast or cable, however, internet delivery of the time-critical video data presents some unique challenges. That is because data sent over the internet (via "Internet Protocol," IP) travels as independent packets of data, each typically some thousand bytes in length, and each including headers that specify its destination (IP address). Packets make their way through the multiply-connected nodes of the internet, and (usually) reach their destination. But there is no reserved pathway for a stream of packets (as there is for the "circuit-switched" architecture of the telephone system), and no guarantee of speedy delivery, sequential delivery, error-free delivery, or, indeed, of delivery of any sort. Various schemes are used to circumvent these evident deficiencies, for example by requesting retransmission of missing packets (they are numbered) or of corrupted packets (they include error-revealing "checksums"). These work well, and a data file that is downloaded via "TCP/IP"[38] is effectively guaranteed to be bit perfect.

[37] Hence "codec," a contraction for coder–decoder, usually appended to the name of the compression scheme, e.g., "the H.264 codec."

[38] Transmission Control Protocol/Internet Protocol, a universally used standard for transfer of data that needs to be transported and reconstituted without error.

¶ 45. Because of this disorganized "packet switching" of internet data, streaming video may suffer interruptions or intervals during which the average delivery rate is reduced. For this reason it is common for the receiving end to store ("buffer") a few seconds of video beyond what is currently being displayed; and if the internet delivery speed is inadequate (as evidenced by buffer underruns), the sender will reduce the data rate (thus delivering lower-quality video). Contemporary broadcast-quality high-definition TV (known as 1080i, meaning that the picture consists of 1080×1920 pixels, delivered 30 frames/sec) requires upward of 2 megabits/sec of download speed, available now in most homes with a broadband internet connection (cable, optical fiber, or telephone-line DSL[39]). That speed is well within the capability of wireless (Wi-Fi) connections, so high-definition video can be delivered to mobile devices such as laptop computers and tablets.

I.8 Digital cable: premium services and conditional access

¶ 46. Cable television providers offer premium services, such as Showtime, HBO, and pay-per-view, for which the subscriber pays additional monthly fees. A subscriber whose subscription includes Showtime, for example, is able to view (and record, if the STB includes a DVR) programming delivered on Showtime channels. The cable provider needs a method to control each subscriber's access to the full channel lineup. Although each customer has a cable coming into the home (and therefore one could imagine that different content is sent to each home), in fact the same signal is sent to all the homes within a neighborhood group, known as a "service group."[40]

¶ 47. To limit access within the full suite of distributed programming, the cable provider includes *conditional access* information along with the video and audio. This is done by including "CA" packets, along with the usual V (video) and A (audio) packets that comprise one program within the multiprogram transport stream. The STB includes decryption hardware, and uses the CA packets to provide the key information to unlock the encryption that is imposed on the video and audio packets by the cable company.

I.8.1 Digital cable: video-on-demand

¶ 48. How is it possible to provide program material specifically for an individual subscriber, for example "on-demand" delivery of a previously broadcast program, or of a movie, on a cable network that serves an entire city?[41] Such services go by names like "video-on-demand" (VOD), and are made possible by the fact that the cable provider is able to tailor the actual signals carried on its cables that go to different groups of cable subscribers in an area.

¶ 49. A cable network in a metropolitan area is more complex than one might at first imagine: rather than a city-wide distribution of common program material, the network is organized into smaller groupings of *nodes* and *service groups*. A service group consists of segments of cable, typically running past no more than 500 homes, carrying identical material; those signals are fed into the cable at a "node" in the neighborhood, which in turn is fed via optical fiber from a more distant "hub" at which the provider inserts the group of channels that is to go to that particular service group.

¶ 50. The trick to providing individual on-demand material, then, is first to ensure that there are at least a few extra channels (in addition to the standard lineup) available to carry such content; and, second, to divide up the city into many service groups, so that those extra channels can carry a different suite of on-demand content to the different service groups. For example, suppose there are five channels available for on-demand material and that each can carry ten programs (as described earlier); if a service group includes 500 houses, of which 200 are cable subscribers, then the cable provider can satisfy 25% of those subscribers with simultaneous on-demand programming (because its five extra channels, each sending ten custom programs, deliver 50 simultaneous programs).

¶ 51. On-demand programming requires also a reverse channel for each subscriber, so that the subscriber can select programming, and also pause (or fast-forward or rewind) the material. However, these reverse channels need carry only a few simple commands up to the provider (as opposed to the high-bitrate video coming down in response), and are easily accommodated in an "interactive" uplink signaling band of the cable network.

[39] Digital subscriber line, technology for bidirectional digital transport over analog telephone lines.

[40] See ¶49, below, for more details.

[41] The term *unicast* is sometimes used to distinguish such individual delivery from *broadcast*.

I.8.2 Digital cable: switched broadcast

¶ 52. Some cable providers use another service that exploits the flexibility inherent in the cable network's use of separate service groups, namely "switched broadcast." Switched broadcast delivers programs only when requested by viewers, on a service-group basis, as compared with delivering all programming to all customers. This allows the cable provider to offer more programming choices than could be carried simultaneously on the available number of channels.[42]

¶ 53. Although switched broadcast and VOD use similar methods to deliver their content to a subscriber, it is worth noting a difference: switched broadcast content is delivered whenever that program material is normally scheduled, and not at the whim of the subscriber; what distinguishes it from normal program material is that it is not put onto the service group's cable at all, if no one in that group has tuned to the program. Once a subscriber has tuned that program (causing it to be sent to that service group, on a particular subchannel), that same program is present on the cable serving any additional subscribers in that service group, on the same subchannel. By contrast, VOD is delivered on-demand, at the time requested by the subscriber; likewise, it can be paused, or fast-forwarded, etc. (tasks that are performed on-the-fly by the cable provider, not by the set-top box). That is possible because VOD content is being sent on a particular subchannel on the subscriber's service group only, and is available for the requesting subscriber only (via encryption and user-specific "entitlement control," in the case of protected content). The terms *narrowcast* and *unicast* are sometimes used to refer to service-group-specific cable delivery of these two varieties of content, namely scheduled ("linear") material, and interactive user-specific on-demand material, respectively.

I.9 Recording digital television

¶ 54. The conversion of analog material (audio and video of the real world) into digital form is hard work – but it makes the task of *recording* straightforward. That is because a single program received at the STB is, in essence, just a stream of numbers, which can be filtered from the multiprogram stream, assembled in a temporary memory "buffer," and written to a hard-disk file just like any computer file. In that sense, a set-top box with DVR is simply a special-purpose computer, with the usual processor, hard drive, etc., and having additional hardware to do the special STB tasks – receive the cable signal, generate the displayable output, take control commands from the infrared remote "clicker," and so on. A typical contemporary set-top box with DVR contains a dual-processor chip, 64 MB program memory, a 160 GB (minimum) hard drive, and various video-related additional hardware (input tuner, video memory, display and audio drivers, etc.). Along with buffering and storing the video and audio content, the STB controls the access, decryption, and re-encryption of the (protected) video content.

¶ 55. It is worth noting that any digital storage medium of adequate speed and capacity can be used to store digital video content; at the consumer level there are many "personal video recorders" (PVRs) that store programs on recordable DVDs or onto solid-state "flash" memory chips. There are also digital video tape recorders that can store both SDTV and HDTV onto a digital variant of VHS tape.

I.10 Display technology

¶ 56. For over half a century television images were displayed with a cathode-ray tube (CRT), in which electrons emitted from a hot cathode and accelerated to potentials of kilovolts were deflected (usually magnetically) to paint a raster, at the video frame rate, on a phosphor surface coated on the interior of the viewing face of the evacuated tube. The intensity was modulated with a grid electrode near the cathode. Early CRTs were monochrome (black and white); color tubes used arrays (triads or stripes) of red, green, and blue phosphors, aligned with a metallic mask so that electrons from each of three electron-emitting cathodes (or, in Sony's Trinitron, steered from a single electron gun) struck only one color of phosphor.

¶ 57. Cathode-ray tube displays worked; but they were heavy (over 100 lbs for a TV with 32″ display), bulky, and required fussy "convergence" adjustment to get the colors to track. Shortening the tube to reduce cabinet depth exacerbated the convergence and geometry problems. CRTs are now obsolete, replaced by several technologies, among them liquid-crystal displays (LCDs), plasma displays, and organic light-emitting diodes (OLEDs).

¶ 58. In an LCD a liquid-crystal layer is sandwiched between a pair of crossed optical polarizers; an applied electric field alters the polarizing properties of the layer, thus varying the optical transmission. In the classic display there's a uniform white rear illumination (from white LEDs, or from one or more cold-cathode fluorescent lamps – CCFLs – combined with diffusers and light pipe

[42] This is sometimes called *narrowcast*, versus broadcast or unicast.

material); an array of electrodes applies local electric fields to the image pixels, which are overlaid with red, green, and blue color filters. The array of liquid-crystal pixels act as video-rate dimmable shutters; all the light originates in the rear illuminator.

¶ 59. LCDs are dominant in computer displays and popular in televisions. They are thin (a centimeter or so) and bright. But they have somewhat limited dynamic range (or contrast ratio: ratio of maximum brightness to "maximum darkness") and speed, and their color balance and black level degrades when viewed off-axis. There have been great improvements in speed and in off-axis performance, owing to methods with names like in-plane switching (IPS), fringe-field switching (FFS), and the like. And the dynamic range can be improved by using LED-array backlighting, which can be dimmed locally and rapidly, adapting to the light and dark areas of the displayed image.[43]

¶ 60. For the most realistic rendering of cinematic material, however, the plasma display is superior to the LCD. It consists of an array of tiny cells, in each of which a switchable gas discharge generates ultraviolet light that causes a spot of phosphor to glow. A high-definition display of 1080×1920 pixels has three such cells for each pixel (one each with red, green, and blue phosphors) to generate the pixel's overall emitted color. Unlike an LCD (which filters an underlying light source), the cells of a plasma display generate the emitted light directly. You can think of it as an array of 6 million little CRTs (1080×1920×3), each one time-switched to achieve the required light intensity.

¶ 61. Plasma displays retain their color fidelity and contrast ratio regardless of viewing angle, and they have fast response. In larger sizes they are currently somewhat less expensive than LCDs. They are not as bright as LCDs, however, and a static pattern that is displayed for a long time can cause some image retention, or (in extreme cases) phosphor "burn." Their contrast ratio is very good, but not infinite, because the gas discharge in every pixel must be sustained at a low level (i.e., it cannot be switched off entirely) so that it can be rapidly modulated.

¶ 62. LCD and Plasma have dominated display technology, but the future is likely to be ruled by OLED (organic LED), a direct-emitting array of tiny LEDs (either in three colors, or white LEDs with filters). These emerged first in small displays (e.g., cellphones and camera viewfinders), but by 2014 they had made the big time, with full 4K "Ultra HD" resolution (3840×2160), and screen sizes to 65″ and beyond. Unlike earlier display technologies, OLEDs can be made flexible, and the current fad is *curved* TV screens (capable of 3D, if you're interested in that). OLED is likely to be the ultimate winner, because of their very high dynamic range (1,000,000:1), wide viewing angle, low power consumption, elegant form factor (3 mm thick, lightweight, and nearly borderless), and potential manufacture by an inkjet-like process. These have been winning Best of Show awards, and they deserve it.[44]

¶ 63. Two other technologies that looked good, but have fallen onto hard times, are field-emission display (FED) and surface-conduction electron-emitter display (SED). Both involve an array of phosphor cells (like the plasma display), but with electron (rather than ultraviolet) excitation of the phosphor. An SED prototype from Canon generated great enthusiasm[45] at the 2006 CES, but subsequent patent disputes and economic realities took their toll. FED and SED may rise again – but don't hold your breath.

I.11 Video connections: analog (composite, component) and digital (HDMI/DVI, DisplayPort)

¶ 64. What are all those incompatible cables, anyway? To add to the analog–digital confusion, the *consumer* market (of large-screen TVs and flat panels, etc.) and the *computer* market (of LCD monitors) have gone their (mostly) separate ways. Here's a quick rundown of the most-used connections; their connectors are pictured in Figure I.10.

¶ 65. In the *consumer television* world there are four types of connections (and connectors), the first two of which are nearly extinct.

Composite video. Low quality standard-definition (SDTV: 480i) analog video, recognizable by the yellow RCA-type connector ("phono jack"), usually bundled with an audio pair (red = right, white = left). The single yellow line carries bandwidth-limited luminance

[43] For marketing purposes these LCDs are sometimes called "LED TVs." Don't be fooled: it's an LCD, but with LED rear illumination replacing the CCFL. And it may or may not have local dimming – read the fine print.

[44] The reviewers are *gushing*: "The best picture I've seen on any TV, ever." (CNET), "The best-looking TV I've ever seen. Ever." (Digital Trends), and "The best direct-view display – of any size, at any price – we've ever laid eyes on." (HDTVtext, UK).

[45] A breathless review in SlashGear (19Oct2006) proclaimed "SED-TV is something that no amount of words can describe. It is something that must be SEEN to be believed; literally." and "SED-TV is the future of digital image displays; it's as simple as that."

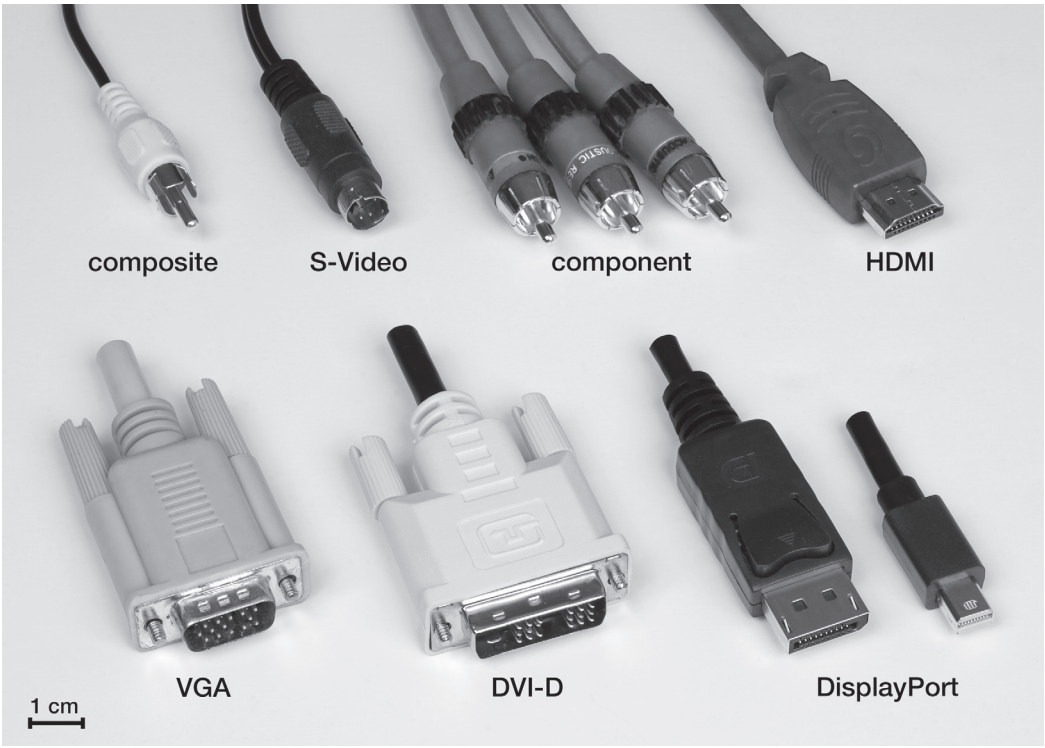


Figure I.10. Cable connectors for computer monitor and video signals. Top row: TV video connectors, from the legacy analog composite to contemporary HDMI high-definition digital video. Bottom row: computer monitor connectors, from the legacy analog VGA to the popular digital DVI and newer DisplayPort (standard and mini connectors).

("luma," abbreviated Y, essentially the grayscale image) and chrominance ("chroma," abbreviated C, the color difference signal pair that is modulated on a 3.58 MHz color subcarrier), along with line and frame synchronization pulses; it is sometimes called CVBS (composite video, blanking, and sync). Don't use such a connection unless there's nothing else available!

S-video. Somewhat better SDTV analog video, recognizable by the 4-pin miniDIN connector with fragile pins. It separates the luma (+ sync) and chroma signals, retaining more bandwidth. Avoid this one, too, unless you like fuzzy pictures.

Component video. Now we're talking! This analog format uses three 75 Ω coax lines, with (usually) RCA-type connectors (or, occasionally, BNCs) that are colored red, green, and blue, and that can carry full-bandwidth HDTV. The colors are misleading: the green line carries luminance (+ sync), while blue and red carry color *difference* signals. That is, the GBR-colored connectors denote "YPbPr," where Y is luminance (red+blue+green), Pb is blue-minus-Y, and Pr is red-minus-Y. As with composite and S-video, component video carries only video; the audio needs it own cables. Component video does not know, or care, about things like content protection; for that reason it is not embraced by content providers, who may prevent full-resolution HDTV (1080×1920) output on the component jacks (e.g., on a Blu-Ray player).

HDMI. A purely digital format, whose initials stand for high-definition multimedia interface. HDMI is the digital alternative to component video. It is recognizable by the flat 19-pin USB-like connector (sadly without any required latching mechanism), and electrically it is equivalent to the DVI-I computer-monitor format (see below). It carries both audio (up to eight channels, digitized to 24 bits, up to 192 ksps) and video (digitized at 8–16 bits per component, at rates adequate for full HDTV ("1080p," 1080×1920 progressive at 60 Hz; HDMI versions 1.4 and later support full "4K," 4096×2160, with 60 fps progressive from version 2.0). The video data is a digitized version of analog video: uncompressed numerical data representing the amplitude of the color

components, sent over four twisted pairs (R, G, B, clock). HDMI supports digital content protection (HDCP, high-bandwidth digital content protection), a protocol by which an HD video source authenticates a display device before sending (encrypted) data, so you are allowed to view the full-resolution video. HDCP seems to work, most of the time anyway (though you may get annoying messages and glitches).

¶ 66. In the *computer monitor* world there are three types of connections (and connectors) in wide use.

Analog VGA. Legacy analog link, recognizable by the 15-pin D-type connector with locking screws. VGA (for video graphics array) carries separate RGB analog signals, plus Hsync and Vsync (thus "RGBHV"). In contemporary implementations it also has an I^2C channel for monitor identification and control. VGA will work up to resolutions of 1600×1200 or so (there's no specified limit, but you'll see smearing going rightward of sharp features when pushing the resolution), but many monitors have abandoned VGA altogether in favor of the digital formats: DVI and DisplayPort.

DVI. Currently the standard digital interface, recognizable by the 29-pin (maximum) connector with locking screws, looking somewhat like a "VGA on steroids." It is electrically similar to the more compact and inexpensive HDMI, above, that evolved from it for the consumer television market, and that includes audio (thus a single connection from a cable box or Blu-Ray player to the television monitor). It comes in several variants, all using the same connector (in which some pins may not be loaded): DVI-D is video-only, and comes in single-link and dual-link varieties (the latter needed for resolutions greater than 1920×1200 at 60 Hz, for example the 2560×1600 at 60 Hz used in $30''$ displays); DVI-A is analog video only (for compatibility with analog monitors); and DVI-I ("integrated") has both digital (single- or dual-link) and analog video. DVI, like VGA, carries no audio.

DisplayPort. Newer standard, intended to supersede DVI; it uses a 20-pin USB-like connector with latching mechanism, and supports very high data rates (up to 4.3 Gbps on each of four differential pairs, thus 17.3 Gbps). It departs from the "digitized-raster" scheme of DVI/HDMI, using instead a packetized data protocol; but it's got enough bandwidth to handle the full video bandwidth of dual-link DVI (which can be converted to DisplayPort protocol). It supports up to 16 bits per color, and 8-channel audio at 24 bits and 192 ksps. It includes provision for fiber optics (instead of copper) for long cable runs (to 50 m or more), and it supports both existing 56-bit HDCP and its own DPCP (DisplayPort content protection, with stronger 128-bit AES encryption). The current revision of DisplayPort can handle 4K 60 fps progressive with ease.

SPICE PRIMER: GETTING STARTED WITH FREE ICAP/4 DEMO

APPENDIX **J**

Free SPICE is easy, and fun. It's available for several platforms (Macs, Linux, PCs). Here's how to get started in Windows.

J.1 Setting up ICAP SPICE

1. Download "ICAP/4Windows Demo" zip file (ICAP4Demo.zip) from intusoft.com.
2. Extract zip file to some temporary folder.
3. In that folder, double click (abbreviated hereafter as "**cc**") on setup.exe.
4. Launch ICAP/4 (you may have to be admin mode): (Programs → ICAP_4 Demo → Start ICAPS).

J.2 Entering a Diagram

- Add standard components with typing
  R resistor
  C capacitor
  L inductor
  o ground (the letter "oh," not the number)
  v signal or voltage source
  Y voltage test point.
  (These are case-insensitive: r = R, etc.)
- For a library component, Parts → Parts Browser,
  – browse by type, and subtype, or use Find,
  – then Place.
- Moving and orienting:
  – highlight component by click-hold, then can drag;
  – when highlighted, type + to rotate (90°CW), or − to flip 180°.
- Editing a component label:
  – highlight label by click-hold, drag to relocate;
  – when label highlighted, **cc** on label to edit values.
    * Note units multipliers: f, p, n, u, m, k, meg (not M).
    * Note that SPICE only needs the units *multiplier*, not the units themselves. For example, for a capacitor "1u" is OK for SPICE; but you can type "1.0uF" for clarity on the schematic, which SPICE will treat the same.
    * You can **cc** on the component, instead, to bring up the value dialog.
  – You can add parameters by entering the value and clicking Apply; this is necessary, e.g., for a voltage source (add dc or ac, give amplitude). Or you can click ≫Add≫, which also adds the parameter to the schematic drawing.
- Wires:
  – type w, then drag a wire between points;
  – you can drag multiple separate wires; ESC when done.
- Settings:
  – set "Rubberbanding" in Options, to maintain connections when dragging components.

J.3 Running a simulation

Can a passive circuit consisting only of resistors and capacitors have *voltage gain*? The surprising answer is yes. Let's use this simple example to illustrate an ICAP/4 SPICE entry and simulation.

J.3.1 Schematic entry

We launch ICAPS, and place two resistors (type the letter "r"), two capacitors (type the letter "c"), and a voltage source (type the letter "v") with the bottom terminal grounded (type the letter "o" right after typing "v"). If the pieces are too small to see on your screen, you can resize them by using Options→Zoom (or the function keys F6, F7, and F8). We drag them around to connect them as shown in Figure J.1, and add an output-voltage test point (type the letter "y" and change the label to Vout). We assign values by double clicking on each component, then entering the value in the value field (where "???" appears initially); note that pressing the Enter key after each value moves you along to another field in the dialog – you have to click OK to complete the operation.

For the capacitors we could have used microfarads (e.g., 0.01uF instead of 10nF); note that the "F" can be omitted, but including it makes the schematic marking clearer. For the voltage source we enter the value 1V (i.e., 1 volt) in the

"AC" field, and then we click on the ADD button so that the value shows on the schematic (SPICE uses the value, even if we do not ADD it; but it's nice to show it on the circuit diagram). As before, the unit symbol "V" can be omitted.[1]

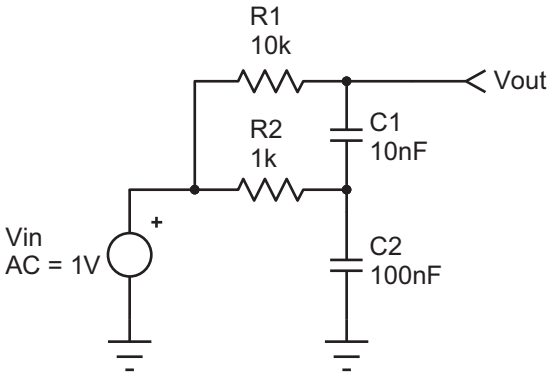


Figure J.1. An *RC* circuit with voltage gain! Schematic entry in Intusoft ICAP/4 is swift, taking no more than a couple of minutes. Printing the captured schematic to "Adobe PDF" produced this figure.

J.3.2 Simulation: frequency sweep

Now for the fun! We begin by saving the project to a folder of our choosing, using the File→Save As menu. Then we set up the simulation by Actions→Simulation setup→Edit (or by clicking on the button with a pencil over a wavy line). Choose and click "AC Analysis," then enter 20 points per octave, start at 100 Hz, end at 10 kHz (type 10k in the box), and click OK. Then click DONE. Now run the simulation by clicking on the little running person (or Actions→Simulate). This launches the Spice engine, with a small window ("IsSpice4") showing status and errors, and (if successful) a little graph of the result. Running the simulation also launches the "IntuScope" display program (which you can launch manually by clicking on the 'scope icon next to the running person, or by Actions→Scope). At this point there are multiple windows piled up, which you should resize and drag to convenient locations on the screen and then do Options→Save Preferences to stick them down. Now click the "Test Pts Only" box in the "Add Waveform" window that belongs to IntuScope, then highlight Vout[2] and click "Add." Voilà – this produces a nice plot of V_{out} versus frequency (see Figure J.2), with default axis labels, grids, and scaling. You can change the axis labels by double clicking on the label; and you can fiddle with the scaling in the "Scaling" window.

To add a plot of phase shift versus frequency, highlight "ph_Vout" in the Add Waveform window, click on the box labeled "With Like Traces" (unless you want a separate graph), and click "Add." We've done that to make Figure J.2, where we've also used the cursor tools on the IntuScope window to mark the point of maximum voltage gain.

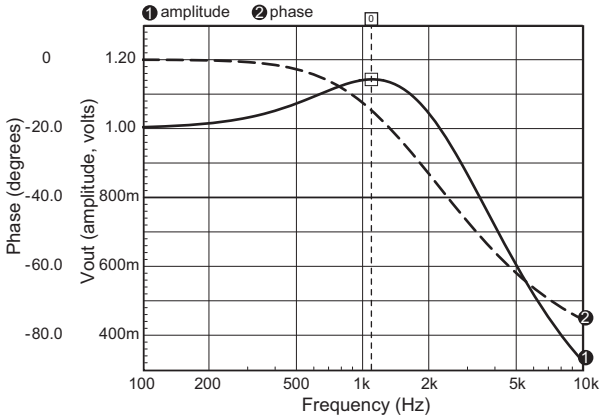


Figure J.2. "'Scope" output of the *RC* gain circuit, showing amplitude and phase ("AC Analysis") for the circuit of Figure J.1. The cursor indicates the peak of the voltage gain, V_{out}/V_{in}=1.142 at f=1.096 kHz. The original IntuScope output plotted phase in blue, which we've converted to a dashed black line; we enlarged the text, also, for readability.

J.3.3 Simulation: input and output waveforms

Do we really believe this gain-versus-frequency plot? Everyone knows that a simple *RC* network just mushes things out – shifted phase and reduced amplitude. We won't believe this thing has voltage gain unless SPICE shows us the actual input and output signal waveforms for a sinewave near the frequency of maximum voltage gain (and maybe not even then!).

Here's how you do it: first, add another voltage test point at the input ("y" again), and change the label to Vin. Then double click on the input voltage source and click on the field labeled "Tran Generators" (SPICE calls any old waveform, including a sinewave, a "transient"). Select SIN, set the peak amplitude to 1, the offset to 0, and the frequency to 1.096 kHz. Click OK, and OK. Then go to Simulation Setup (pencil over wavy line), deselect AC Analysis, and select Transient. In the Transient Analysis

[1] For SPICE's ac analysis, the frequency-sweep gain computation is performed with infinitesimal small-signal amplitudes, which are then normalized to the signal amplitude you specified. Here, for example, SPICE uses a signal amplitude much smaller than 1 V.

[2] Or, if you have not renamed the output to Vout, select from among the signals with names like v2, etc., in the Y Axis output list.

window that pops up, set Data Step Time to 1us, Total Analysis Time to 105ms, and Time to Start Recording Data at 100ms (this gives the circuit time to settle into its steady state). Leave the Maximum Time Step field blank (i.e., accept default value). Click the runner icon to run the simulation. At this point the IsSpice4 window should show a wiggly trace. Now go to the IntuScope window, clear the old graph by pressing the DEL key for each selected trace (you can save it, if you want, or start a new plot with File→New Graph), and Add "vin" and "vout" in the Add Waveform subwindow of IntuScope, with the "With Like Traces" box checked as before.

Not bad... but the waveforms look a bit jagged. If you zoom in (drag the box boundaries in the "Scaling" window) you'll see that there are only about 20 points plotted per cycle. We asked for 1 μs steps, but we're getting output at $\sim$50 μs intervals! That's because we didn't constrain the Maximum Time Step in the Transient Analysis window. Go back now (Simulation Setup→Transient) and set Maximum Time Step to 1us. Then click on the running person (the simulation will take slightly longer, maybe a full second this time); then clear the old graph and Add the two waveforms in IntuScope (output first, to set scale, then input). You should see silky-smooth plots, as in Figure J.3. If you still have doubts about an *RC* circuit with voltage gain, you can hook up the circuit and check it out on a real-live oscilloscope, as we did (Figure J.4). Evidently this *RC* voltage-gain stuff is real!

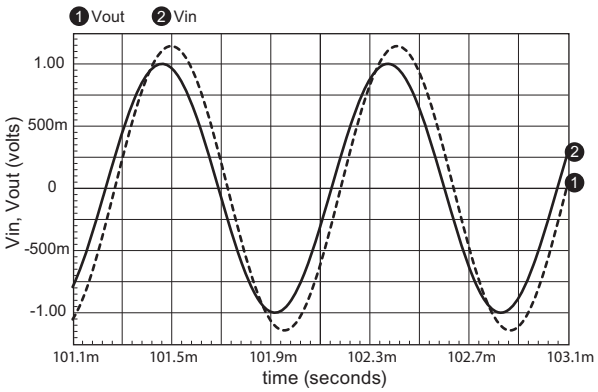


Figure J.3. "'Scope" output of the *RC* gain circuit, showing sinewave response ("Transient Analysis") at the frequency of maximum gain for the circuit of Figure J.1. The original IntuScope output plotted V_{out} in blue, which we've converted to a dashed black line; we enlarged the text, also, for readability, and adjusted the axis scales to match the scope trace.

J.4 Some final points

The ICAP/4 demo program includes ten "tutorial movies" (in the Help menu), which are mercifully short (about a minute each) and quite helpful for the beginner. Take ten minutes to view them! You'll learn, for example, that you can create many signal sources by going directly to Parts→Parts Browser→!Generators. You'll also learn some tricks to simplify schematic entry (copying, zooming), and ways to make the parts labels more readable ("tall," "wide," and "split" options). And by experimenting a bit you can discover that both the schematic and the simulation graphs can be copied ("Print") to the clipboard and then pasted into Word or Wordpad, where they exist as scalable graphics. If you Print to Adobe PDF you'll get a scalable graphic that can be cleaned up in Adobe Illustrator.

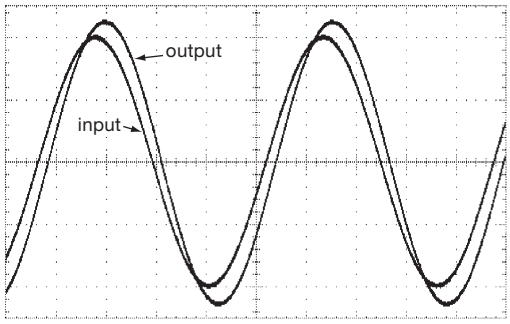


Figure J.4. A *real* 'scope (Tek TDS3044B) validates that crazy SPICE stuff! Horizontal; 200 μs/div; Vertical; 0.5 V/div.

We have been using the Intusoft "ICAP/4" full version of SPICE now for many years; it provides a convenient Windows-based schematic entry and simulation environment, which we use to explore circuit configurations. Typically we begin with a starting circuit design, making explanatory notations in a Word document as we go. We copy and paste the schematics and simulation graphics into the running text, making a "lab notebook" of sorts; see next.

J.5 A detailed example: exploring amplifier distortion

In Chapter *2x* ("BJT Amplifier Distortion: a SPICE exploration") we do some serious work with SPICE, in the form of an exploration of distortion in discrete BJT amplifier designs.[3] The amplifier stuff is lots of fun; and there are some nice SPICE tricks there revealed. Take a look!

[3] When we do this in real life, we like to paste circuit diagrams and screen shots of simulated performance into a Word document, annotating as we go. We've kept that spirit alive in the discussion in Chapter *2x*, albeit in the form of a LaTeX conversion of the original Word document.

J.6 Expanding the parts database

You can freely add SPICE models to the ICAP/4 parts library. An easy way to start the process is to download the folder AoE-PR from our website (www.artofelectronics.com), and place it under the default PR folder. Then do File→Update Part Database.

"WHERE DO I GO TO BUY ELECTRONIC GOODIES?"

APPENDIX **K**

Good question! Here are some hints, from our experiences.

I. Mail order and online

Digi-Key Corp (Thief River Falls, MN: 1–800–digikey). We used to say "*Get their catalog!!*" – but, sadly, they've abandoned paper, replacing it with an impressive search capability. You can get everything here, even in small quantities, with fast delivery. It's often worth designing with their webpage open in front of you. Online ordering and stock/price checking: www.digikey.com.

Mouser Electronics (www.mouser.com). Broad stocking distributor, with service comparable to Digi-Key's, and willingness to ship small quantities. Good selection of precision passives; and they are still printing a comprehensive paper catalog.

Newark Electronics + Farnell (1–800–2–newark; www.newark.com). Broadest stocking distributor, with service comparable to Digi-Key's and good selection of tools; paper catalog still in print.

"Stocking Distributors". These are the standard distribution channel for quantity buying; names like Allied (still publishing a paper catalog), Arrow, Avnet, FAI, Heilind, Insight, Pioneer, Wyle. Substantial minimum quantities – not generally useful for prototyping or small production.

Manufacturers' Direct. Many semiconductor manufacturers (Analog Devices, TI, Maxim,. . .) will not only send free samples with the slightest provocation, they will also sell in small quantities via credit card; check out Mini-Circuits for RF components, and Coilcraft for inductors, transformers, and RF filters.

Oddballs. Marlin P Jones, Jameco, B&D, Herbach & Rademan, Omnitron, ABRA, All Electronics; ephemeral collection of "surplus" stuff, some real bargains.

eBay (www.ebay.com). If you haven't been here, you've probably just arrived from Mars. LOTS of stuff, literally millions of items, an online auction. You can get plenty of electronic stuff, but CAVEAT BIGTIME EMPTOR. Feedback Forum helps.

Alibaba Small Pacific-rim companies that have stocks of obsolete components are easily found on Alibaba. You can place a quote request for a part number, and you'll get dozens of useful reasonably-priced suppliers.

II. Indexes and Locators

Octopart, FindChips, NetComponents (octopart.com, findchips.com, netcomponents.com). Give it a part number and it searches dozens of distributors, returning (sometimes unreliable) information on availability and pricing.

WhoMakesIt (www.whomakesit.com). A bit like the EEM catalog, helpful if you know the category of stuff you want, but not the manufacturer, etc.

Google (www.google.com). Our standard "portal," reads your mind and vectors you to the goodstuff. Helpful, sometimes, in finding parts and equipment manufacturers and vendors.

III. Local

Sometimes it's nice to shop in person; here are the sorts of places to go.

Radio Shack (www.radioshack.com). They call themselves "America's Electronic Supermarket"; we'd call them "America's Electronic Convenience Store." Their stores are everywhere, and they stock (pretty reliably) an idiosyncratic collection of parts and supplies, of uncertain quality or duration. However, in the changing marketplace of consumer electronics, their future path is unclear.

Electronics Flea Markets. Also known as "swap meets," perhaps somewhat in decline; two legendary meets are on opposite coasts: De Anza College (Cupertino, every second Saturday, March–October), and MIT (Cambridge, every third Sunday, April–October). What meets

are three cultures (electronics, computers, hams); haggling is mandatory; caveat very emptor.

Electronics Surplus Supply Stores. These are incredibly cool! Several well-known haunts are Halted (www.halted.com; officially "HSC Electronics Supply"), in Santa Clara, Sacramento, and Santa Rosa; and Murphy's Surplus Warehouse (www.murphyjunk.bizland.com) in El Cajon.

IV. Miscellaneous

Obsolete ICs. Your best place to start is Rochester Electronics (www.rocelec.com), a wonderful place that apparently buys up inventories of ICs being discontinued. Jameco also has a lot of obsolete parts. Freetradezone has broker lists for obsolete parts. Also try Interfet (www.interfet.com), a manufacturer of small-signal FETs, including ones that the big guys have abandoned.

PC Board Manufacture. We like a place called Advanced Circuits, www.4pcb.com; you can get online quotes, and they do a good job and deliver pronto. Another inexpensive and fast PC house is Alberta Printed Circuits in Canada (www.apcircuits.com).

WORKBENCH INSTRUMENTS AND TOOLS

APPENDIX **L**

Here are some electronic favorites, stuff that we really like using when we design and build electronic circuits. It's best to check current catalogs and websites – much of this sort of equipment becomes obsolete with terrifying speed.

Soldering iron
Metcal MX-500 (not cheap; it will change your life!)
Weller WSL (less expensive, variable temp)

Desoldering station
Pace MBT201-SD

Surface-mount prototype and rework
Zephyrtronics Airbath and Airpencil
Metcal offers SMT "tweezers" and other goodies

Bench DMM
Keithley 2100 ($6\frac{1}{2}$-digit, with USB control & readout)
Agilent 34410A ($6\frac{1}{2}$-digit, with LAN & USB control & readout)

Pocket DMM
Amprobe 37XR-A (cheap, good enough)
Fluke 289 (not cheap, very good)
Agilent U1252A/53A (everyone's favorite lately)
"Smart Tweezers" (auto-ranging SMT tweezer-style meter, neat!)

Triple bench power supply
HP E3630A (excellent performance, reasonable price)

High-voltage bench power supply
SRS PS300-series (single and split, to 20 kV)

Device programmer (if needed; JTAG in-ckt is taking over, pods from mfgrs)
BP Microsystems 1610 (universal and reliable; lifetime free algorithm updates)

LCR meter
HP 4263B (a cheaper one is the SRS model SR720)

Analog oscilloscope
B&K Precision and Hameg still offer some models, to 200 MHz bandwidth

Digital oscilloscope
Tek DPO2024B (cheap "lunchbox"); DPO/MSO 3k-, 4k-, 5k-series
Agilent DSO/MSO 5k-, 6k-, and 7k-series (Agilent's answer to the "lunchbox")
Lecroy WaveRunner, WaveSurfer, WaveJet series (lot of models, do your homework)

Arbitrary function generator
Tek AFG3000-series (single and dual channel, to 240MHz and 2Gs/s)

Low-distortion function generator
SRS DS360 (0.01 Hz–200 kHz, 0.001% distortion)

RF and microwave synthesizer
Agilent N9310A
SRS SG380-series (2, 4, and 6 GHz models, low phase noise, see §13.13.6B)

Low frequency spectrum analyzer
SRS model SR785

RF spectrum analyzer
Agilent ESA series (depends on frequency range, and $$)

Source measure unit
Agilent B2900-series (single and dual channel)
Keithley 2600-series (single and dual channel)

Precision time and frequency standard
Symmetricom 4411A (uses GPS constellation)

Engineering software
Altium System Designer, OrCad, or Eagle (for schematic capture and layout; includes simulation)
Xilinx WebPack (for PLD and FPGA design), and analogous tools from Altera, Lattice, Actel, etc.
ICAP/4, LTspice ($0!), MicroCap 9, MMICAD, PSpice (for simulation)
FilterCAD ($0!, from LTC), FilterPro ($0!, from TI) (simple analog filter design)
MathCAD, MATLAB, Mathematica (engineering/math worksheets)
LabVIEW™ (virtual instruments; control of real instruments)

CATALOGS, MAGAZINES, DATABOOKS

APPENDIX **M**

Here are some recommendations for data books, catalogs, and magazines; some have become on-line services only (e.g., IC Master). You will need some of these if you want to practice electronic design.

"Master" catalogs

EEM (Electronic Engineer's Master) *lists all categories of electronic stuff*
IC Master *ditto, for ICs*
Octopart *excellent on-line part (and datasheet) finder*

Parts and equipment catalogs

Digi-Key Corp (a distributor), *wide range, quick delivery*
Mouser Electronics (a distributor), *wide range, quick delivery*
Newark/Farnell Electronics (a distributor), *best broad-line catalog*
Allied Electronics (a distributor), *similar to Newark*
TechniTool (a distributor), *measurement, assembly, and test tools*
Stanley Supply & Services, (a distributor) *equipment and supplies for assembly*
Keysight Technologies (Agilent, HP), *broadest manufacturer of test and measurement equipment*
Tektronix, *test and measurement equipment, especially scopes*
Fluke, *test and measurement equipment*
Stanford Research Systems (SRS) *test and measurement equipment; great documentation and app notes*

Magazines and tabloids

EDN, *keeps you up to date on new products and methods*
Electronic Design, *ditto; a bit thin, lately*
Electronic Products, *ditto, emphasis on products only*
EE Times, *tabloid format*
Computer Applications Journal, *microcontroller project mania*
Circuit Cellar Ink, *emphasis on microcontrollers*
Nuts and Volts, *quirky do-it-yourself, tabloid format*
Make Magazine, *cool stuff you can build*

Data books; short-form/design guides A starter selection, with product emphasis.
Altera: cPLD, FPGA
Analog Devices: all linear functions; converters; DSP
Atmel: PLDs and microcontrollers
Avago (←Agilent←HP←Avantek): opto, rf
Cirrus/Apex: power op-amps etc.
Cypress: memory, processors
Diodes Inc/Zetex: discretes, etc.
Fairchild: discretes, linear, digital
Freescale: processors, DSP, automotive
IDT, Micron, Samsung: memory
Infineon: discrete, power, processors, RF
Intel: microprocessors and microcontrollers
Linear Technology Corp: all linear functions
M/ACom: RF and microwave
Maxim/Dallas: linear, digital, μC
MiniCircuits: RF – broad line, inexpensive
NXP (Philips): logic, microcontroller
ON Semi/Sanyo: linear, logic, discrete
Renesas: discrete, memory, processors
TI/National/Burr–Brown: linear, logic, opto, power, processors, DSP
Xilinx: cPLD, FPGA

Datasheets (in *.pdf format) are available online from nearly every semiconductor manufacturer's website; try, e.g., www.analog.com, www.maxim-ic.com, www.linear.com, www.ti.com, etc.

Another way to find datasheets, prices, and availability is via a search website like octopart.com or findchips.com; these display price and inventory (and sometimes datasheets), and link you through to the stocking distributor.

FURTHER READING AND REFERENCES

APPENDIX **N**

General

Ashby, D., ed., *Circuit Design: Know It All*. Newnes (2008). A collection of fascinating electronics engineering stuff from 14 acclaimed authors.

Camenzind, H., *Much Ado About Almost Nothing, Man's Encounter with the Electron.* Booklocker.com (2007). Fascinating stories about electronics, by the famed designer of the 555.

Dobkin, B. and Williams, J., eds., *Analog Circuit Design: A Tutorial Guide to Applications and Solutions*. Newnes (2011). Excellent selection of informative and well-written application notes from Linear Technology. Lively and entertaining, too.

Dunn, P. C., *Gateways into Electronics*. Wiley (2000). Fascinating physics-based approach to electronics; deep coverage of critical areas.

Jones, R. V., *Instruments and Experiences: Papers on Measurement and Instrument Design.* Wiley (1988). Classic on instrument design, based on Jones' papers.

Lee, T. H., *The Design of CMOS Radio-Frequency Integrated Circuits*. Cambridge University Press (2nd ed., 2003). From the originator of gigahertz CMOS comes this delightful volume, covering much more than its humble title suggests. Terrific introductory chapter on the history of radio.

Pease, R. A., *Troubleshooting Analog Circuits*. Butterworth–Heinemann (1991). The curmudgeon-in-chief reveals his tricks.

Purcell, E. M., and Morin, D. J., *Electricity and Magnetism*. Cambridge University Press (2013). Excellent textbook on electromagnetic theory. Relevant sections on electrical conduction and analysis of ac circuits with complex numbers.

Scherz, P. and Monk, S., *Practical Electronics for Inventors*. McGraw-Hill (3rd ed., 2013). The title says it all.

Sedra, A. S. and Smith, K. C., *Microelectronic Circuits*. Oxford University Press (6th edition, 2009). Popular classic engineering text.

Senturia, S. D., and Wedlock, B. D., *Electronic Circuits and Applications*. Wiley (1975). Good introductory engineering textbook.

Sheingold, D. H., ed., *Nonlinear Circuits Handbook*. Analog Devices (1976). Highly recommended.

Sheingold, D. H., ed., *The Best of Analog Dialog, 1967 to 1991*. Analog Devices (1991). Outstanding collection of analog-engineering techniques.

Terman, F. E., *Radio Engineers' Handbook*. McGraw-Hill (1943). Three score and ten years later it continues to amaze, with excellent sections on passive circuit elements and other basic engineering.

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THE OSCILLOSCOPE

APPENDIX **O**

The oscilloscope ("scope" for short) is, by far, the most useful and versatile electronic circuit test instrument.[1] As usually used, it lets you "see" voltages in a circuit as a function of time, triggering on a particular point of the waveform so that a stationary display results. Contemporary scopes are almost invariably *digital* (input signals are digitized, processed, and displayed), and they do (and usually *better*) what their analog ancestors did. To understand how to use an oscilloscope, we think it best to start with the traditional (and nearly extinct) 2-channel *analog* scope, for which we've drawn a block diagram (Figure O.1) and typical front panel (Figure O.2). Digital scopes carry forward nearly all of its features, to which they add an impressive array of capabilities (and a few hazards).

O.1 The analog oscilloscope

O.1.1 Vertical

Beginning with the signal inputs, most analog scopes have two channels; that's very useful, because you often need to see the relationship between signals. Each channel has a calibrated gain switch, which sets the scale of **VOLTS/DIVISION** *on the screen*.[2] There's also a **VARIABLE** gain knob (concentric with the gain switch) in case you want to set a given signal to a certain number of divisions. Warning: be sure the variable gain knob is in the "calibrated" position when making voltage measurements! It's easy to forget. The better scopes have indicator lights to warn you if the variable gain knob is out of the calibrated position.

The scope is dc-coupled, an essential feature: what you see on the screen is the signal voltage, dc value and all. Sometimes you may want to see a small signal riding on a large dc voltage, though; in that case you can switch the input to **AC COUPLING**, which capacitively couples the input with a time constant of about 0.1 second. Most scopes also have a grounded input position, which lets you see where zero volts is on the screen. (In **GND** position the signal isn't shorted to ground, just disconnected from the scope, whose input is grounded.) Scope inputs are usually high-impedance (1MΩ in parallel with about 20 pF), as any good voltage-measuring instrument should be.[3] The input resistance of 1 MΩ is an accurate and universal value, so that high-impedance attenuating probes can be used (as will be described later); unfortunately, the parallel capacitance is not standardized, which is a bit of a nuisance when changing probes.

The vertical amplifiers include a vertical **POSITION** control, an **INVERT** control on at least one of the channels, and an **INPUT MODE** switch. The latter lets you look at either channel, their sum (their difference, when one channel is inverted), or both. There are two ways to see both: **ALTERNATE**, in which alternate inputs are displayed on successive sweeps of the trace, and **CHOPPED**, in which the trace jumps back and forth rapidly (0.1–1 MHz) between the two signals. **ALTERNATE** mode is generally better, except for slow signals. It is often useful to view signals both ways, to make sure you're not being deceived.

O.1.2 Horizontal

The vertical signal is applied to the vertical deflection electronics, moving the dot up and down on the screen. The horizontal sweep signal is generated by an internal ramp generator, giving deflection proportional to time. As with the vertical amplifiers, there's a calibrated **TIME/DIVISION** switch and a **VARIABLE** concentric knob; the same warning stated earlier applies here. Most scopes have a **10× MAGNIFIER** and also allow you to use one of the input channels for horizontal deflection (this lets you generate those beloved but generally useless

[1] It is sometimes said that practitioners of other engineering disciplines are especially envious of EEs, because we are blessed with such a splendid instrument with which to visualize the happenings in our circuits.

[2] Note that the two channels can be set for different scale factors, offsets, and coupling. This goes also for digital scopes, which commonly have four channels.

[3] Scopes intended for high-frequency measurements, going beyond 100 MHz, say, offer also a 50 Ω input impedance option.

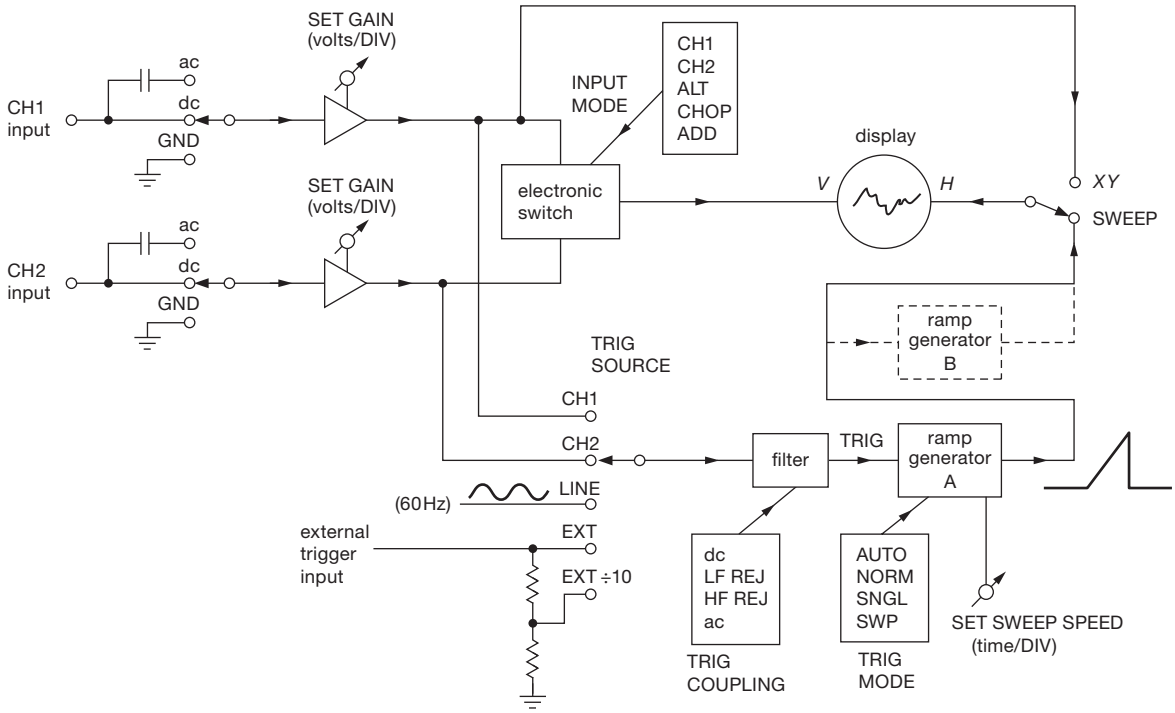


Figure O.1. Block diagram of a 2-channel analog oscilloscope.

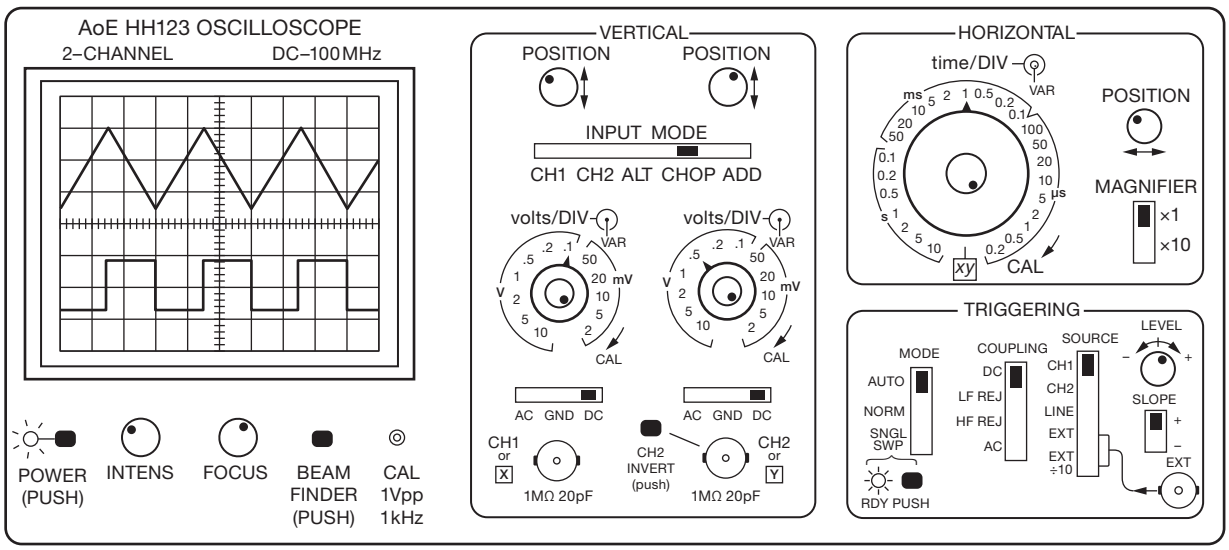


Figure O.2. Portrait of a 2-channel analog scope.

"Lissajous figures" featured in elementary books and science fiction movies).

O.1.3 Triggering

Now comes the trickiest part: *triggering*. We've got vertical signals and horizontal sweep; that's what's needed for a graph of voltage versus time. But if the horizontal sweep doesn't catch the input signal at the same point in its wave-

form each time (assuming the signal is repetitive), the display will be a mess – a picture of the input waveform superimposed over itself at different times. The trigger circuitry lets you select a **LEVEL** and **SLOPE** (+ or −) on the waveform at which to begin the sweep. You can see from the front panel that you have a number of choices about trigger sources and mode. **NORMAL** mode produces a sweep only when the source selected crosses through the trigger point you have set, moving in the direction (**SLOPE**) you have selected. In practice, you adjust the level control for a stable display. In **AUTO** the sweep will "free run" if no signal is present; this is good if the signal sometimes drops to small values, since the display won't disappear and make you think the signal has gone away. It's the best mode to use if you are looking at a bunch of different signals and don't want to bother setting the trigger each time. **SINGLE SWEEP** is used for nonrepetitive signals. **LINE** causes the sweep to trigger on the ac power line, handy if you're looking at hum or ripple in a circuit. The **EXTERNAL** trigger inputs are used if you have a clean signal available at the same rate as some "dirty" signal you're trying to see; it's often used in situations where you are driving some circuit with a test signal, or in digital circuits where some "clock" signal synchronizes circuit operations. The various coupling modes are useful when viewing composite signals; for instance, you may want to look at an audio signal of a few kilohertz that has some spikes on it. The **HF REJ** position (high-frequency reject) puts a low-pass filter in front of the trigger circuitry, preventing false triggering on the spikes. If the spikes happen to be of interest, you can trigger on them instead in **LF REJ** position.

Many scopes now have **BEAM FINDER** and **TRIGGER VIEW** controls. The beam finder is handy if you're lost and can't find the trace; it's a favorite of beginners. Trigger view displays the trigger signal; it's especially handy when triggering from external sources.

O.1.4 Hints for beginners

Sometimes it's hard to get *anything* to show on the scope. Begin by turning the scope on; set triggering for **AUTO**, **DC COUPLING**, **CH 1**. Set sweep speed at 1 ms/div, cal, and the magnifier off (×1). Ground the vertical inputs, turn up the intensity, and wiggle the vertical position control until a horizontal line appears (if you have trouble at this point, try the beam finder).[4] Now you can apply a signal, unground the input, and fiddle with the trigger. Become familiar with the way things look when the vertical gain is far too high, when the sweep speed is too fast or slow, and when the trigger is adjusted incorrectly.

O.1.5 Probes

The oscilloscope input capacitance seen by a circuit under test can be undesirably high, especially when the necessary shielded connecting cable is included. The resulting input impedance (1MΩ in parallel with 100 pF or so) is often too low for sensitive circuits and loads it by the usual voltage divider action; for example, at 10 MHz a 100 pF load looks like 160 Ω – ouch! Worse yet, the capacitance may cause some circuits to misbehave, even to the point of going into oscillation. In such cases the scope obviously is not acting like the "low-profile" measurement instrument we expect; it's more like a bull in a china shop.

The usual solution is the use of high-impedance "probes." In simplified form,[5] the popular 10× probe works as shown in Figure O.3. At dc it's just a 10× voltage divider. By adjusting C_1 to be $\frac{1}{9}$th the parallel capacitance of C_2 and C_3, the circuit becomes a 10× divider at all frequencies, with input impedance of 10MΩ in parallel with a few picofarads. In practice, you adjust the probe by looking at a square wave of about 1 kHz, available on all scopes as **CALIB**, or **PROBE ADJ**, setting the capacitor on the probe for a clean square wave without overshoot. Sometimes the adjustment is cleverly hidden; on some probes you twist the body of the probe and lock it by tightening a second threaded part. One drawback: a 10× probe makes it difficult to look at signals of only a few millivolts; for these situations use a "1× probe," which is simply a length of low-capacitance shielded cable with the usual probe hardware (wire "grabber," ground clip, handsome knurled handle, etc.). The 10× probe should be the standard probe, left connected to the scope, with the 1× probe used when necessary. Some probes feature a convenient choice of 1× or 10× attenuation, switchable at the probe tip.

Even with a 10× probe, the circuit loading may be unacceptable; after all, its improvement is just the same factor of ten by which it attenuates the input signal. You *can* get 100× probes, with correspondingly higher input

[4] Curiously, some scopes (for example the once-popular Tektronix 400 series) don't sweep on **AUTO** unless the trigger level is adjusted correctly.

[5] In practice the cable itself is made from resistance wire, to damp transmission-line effects (frequency peaking and transient reflections, see Appendix H), an elegant 1959 invention by Kobbe and Polits (US Patent 2,883,619); you also see tricks such as a series *RC* across the scope terminals (e.g., 500 Ω and a trimmer capacitor), to provide a transmission-line match at high frequencies.

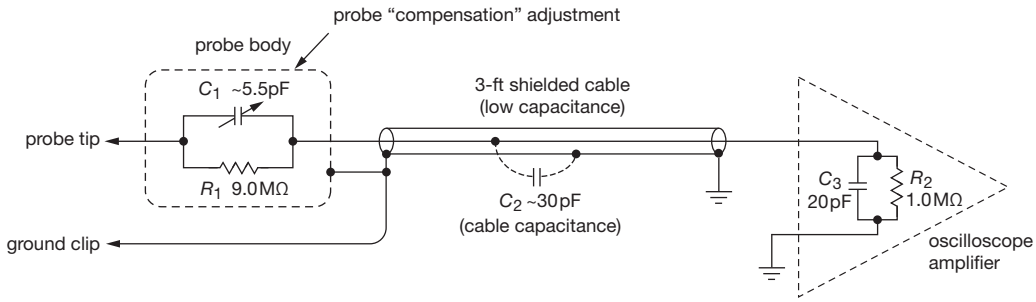


Figure O.3. A 10× passive scope probe attenuates signals by a factor of ten at all frequencies, conveniently raising the input impedance by the same factor. (In practice additional tricks are used to suppress transmission-line effects, see text.)

impedance (e.g., the Tektronix P5100 series), but these are intended primarily for viewing high-voltage signals (the scope itself is usually limited to a maximum of ±400 V at the input connector), and they do not excel in important features such as small physical size. What you do, instead, is to use an *active probe*, which uses a FET follower at the tip to achieve an input capacitance <1 pF.[6] Active probes, being intended for wideband use, are intended to drive a 50 Ω input (available on most high-speed scopes; if not, attach a 50 Ω pass-through terminator); they require a source of power, available at the scope's input connector (on digital scopes), or provided by a stand-alone box like the Tektronix 1103.

Any discussion of probes would be incomplete without a mention of *current* probes: these handy devices, when clipped around a wire in some circuit, convert the circuit's current to a voltage waveform that's displayed on the scope. The simplest current probes are inherently ac-coupled (they wrap a secondary winding around a magnetic split core that surrounds the one-turn wire "primary") and thus do not sense dc current; the fancier types use a combination of Hall effect and transformer coupling to achieve response down to dc. Examples of the latter are the Tektronix A622 (dc to 100 kHz) and the TCP312A (dc to 100 MHz); the latter requires the matching TCPA300 amplifier.

O.1.6 Grounds

As with most test instruments, the oscilloscope input is referred to the instrument ground (the outer connection of the input BNC connectors), which is usually tied electrically to the case. That, in turn, connects to the ground lead of the ac power line, via the 3-wire power cord. This means that you cannot measure voltages between the two arbitrary points in a circuit, but are forced to measure signals relative to this universal ground.

An important caution is in order here: if you try to connect the ground clip of an oscilloscope probe to a point in the circuit that is at some voltage relative to ground, you will end up shorting it to ground. This can have disastrous consequences to the circuit under test; in addition, it can be downright dangerous with circuits that are "hot to ground" (for example line-powered switching power supplies). If it is imperative to look at the signal between two points, you can make a differential measurement by inverting one input channel and switching to ADD, or you can use an external differential preamp (e.g., the LeCroy DA1855A). In desperate situations we have been known to "float" the scope by lifting the ground lead at the power cord, but this is *not recommended*, unless you really know what you're doing (and agree to waive any liability on our part).

Another caution about grounds when you're measuring weak signals or high frequencies: be sure the oscilloscope ground is the same as the circuit ground where you're measuring. The best way to do this is by connecting the short ground wire on the probe body directly to the circuit ground,[7] then checking by measuring the voltage of "ground" with the probe, observing no signal. One problem with this scheme is that those short ground clips are usually missing, lost! Keep your probe accessories in a drawer somewhere.

O.1.7 Other analog scope features

Many scopes have a **DELAYED SWEEP** that lets you see a segment of a waveform occurring some time after the trigger point. You can dial the delay accurately with a multiturn adjustment and a second sweep-speed switch. A delay

[6] One of our favorites is the Tektronix P6243, <1 pF and 1 GHz bandwidth.

[7] See the illustrations in Figure 12.32

mode known as **A INTENSIFIED BY B** lets you display the whole waveform at the first sweep speed, with the delayed segment brightened; this is handy during setup. Scopes with delayed sweep sometimes have "mixed sweep," in which the trace begins at one sweep speed, then switches to a second (usually faster) speed after the selected delay. Another option is to begin the delayed sweep either immediately after the selected delay or at the next trigger point after the delay; there are two sets of trigger controls, so the two trigger points can be set individually. (Don't confuse delayed sweep with "signal delay." All good analog scopes have a delay in the signal channel, so you can display the event that caused the trigger; it lets you look a little bit backward in time! See the photographs of the analog delay lines in Figures H.19 and H.21).

A common feature of analog scopes is a **TRIGGER HOLDOFF** control; it inhibits triggering for an adjustable interval after each sweep, and it is very useful when viewing complicated waveforms without the simple periodicity of, say, a sinewave. The usual case is a digital waveform with a complicated sequence of 1s and 0s, which won't generate a stable display otherwise (except by adjustment of the sweep-speed vernier, which means you don't get a calibrated sweep).

All scopes (analog and digital) include some **BANDWIDTH LIMIT** vertical amplifier options (for simplicity, not shown in Figures O.1 and O.2), useful for reducing the amount of wideband "fuzz" on the displayed trace when you're working with relatively slow signals.

During the height of the analog scope era, you could get scopes with on-screen "storage" (for single-shot capture) and scopes with an impressive array of plug-in modules that let you do lots of interesting stuff, including display of eight traces, or spectrum analysis, or accurate (digital) measurements of voltage and time on waveforms, and so on. Happily, these functions and many others (e.g., looking far backward in time from the triggering event) are now embodied in the dominant oscilloscope species, the *digital* oscilloscope. Let's take a look.

O.2 The digital oscilloscope

Analog scopes are easy to use, but they are seriously limiting in what you can do. For example, (a) it's hard to see a "single-shot" event; (b) you can't store a trace, or compare a live trace with an earlier trace; (c) you can't extract a trace for measurement or illustration; and (d) you can't look back in time to see what happened before the triggering event.

Digital scopes effortlessly provide these and many other capabilities; and, because of the stunning capability and low cost of digital conversion and processing, they are, ironically, less expensive than an analog scope (if you can find one) of comparable bandwidth. The transition to usable and friendly digital scopes was rocky at first, but they are now ubiquitous and universal.

The basic scheme (Figure O.4) is to digitize the incoming signal after the frontend stages of programmable gain and bandwidth limiting, capture the samples in a fast circular buffer memory, and then use a processor (or multiple processors) to do all the signal processing, measurements, conversion to a meaningful display, user interface, and I/O. We'll keep this section brief, and merely run through some capabilities of digital scopes.

O.2.1 What's different?

In no particular order:

Front-end: The signal emerging from the (variable-gain) input amplifiers is digitized at some sampling rate f_{samp} (typically 1 Gsps or more, but always above the minimum Nyquist rate of $2f_{\text{max}}$ when the scope is set at fast enough sweep rates to resolve the scope's bandwidth f_{max}. But – **important** – see "aliasing," below). The digital samples, typically of 8 bits resolution, are stored at full speed into a *sample memory* (or "capture memory"), often of length 1 Mpt or more per channel (and reaching 1 Gpt at the high end). Note that, even though digital scopes let you zoom in after a trace is captured, the resolution depends on the vertical scale factor, because of the fixed bit depth of conversion.

Simultaneous on all channels: Digital scopes digitize all channels simultaneously; there's no "alternate" or "chop." Most digital scopes come in 2- or 4-channel flavors, augmented in "mixed-signal" scopes by sixteen or more 1-bit (i.e., logic-level) channels.

Pre-trigger: Because digitized input signals are pouring into memory, you can set a trigger condition (most simply, level and slope; but see "Smart trigger," below) and, when it is satisfied, you've got substantial pre-history in the sample memory. From the user's point of view, you can simply set the displayed trigger pointer to the right portion of the screen to reveal what came before. And you can walk backward or forward to your heart's content through a saved single-shot capture (see "Single-shot capture," below).

Display: The time interval between points on the *displayed* waveform (the "waveform interval") is typically longer (often much longer) than the sampling interval

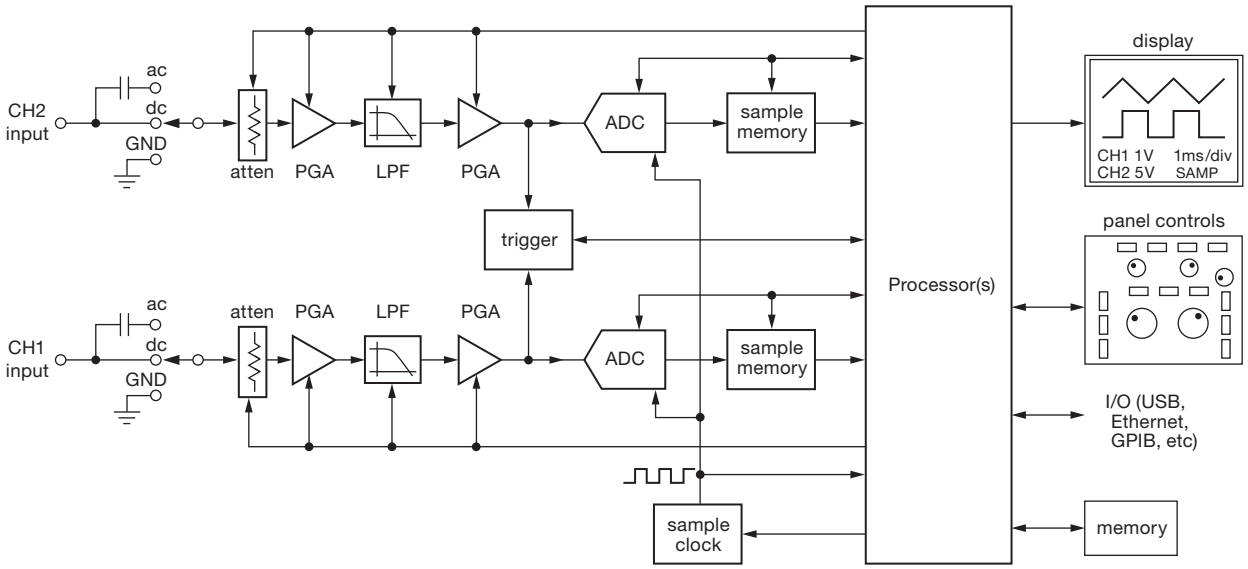


Figure O.4. Block diagram of a 2-channel digital oscilloscope.

$1/f_{\text{samp}}$. This allows for various modes of processing the sampled points to produce the displayed waveform, in particular:

sample one sample point per waveform interval is displayed; the rest are discarded. Simple, but susceptible to aliasing, see below.

peak detect the highest and lowest sampled points in two successive waveform intervals are displayed. Produces a thicker waveform, but no short spikes are lost.

envelope similar to peak detect, but combines min/max from multiple triggered acquisitions. Useful for seeing excursions from an ideal repetitive waveform.

average each point in the displayed waveform is the calculated average of single samples (as in sample mode) over many (a settable number, e.g, 2, 4, 8, ...512) triggered acquisitions. Greatly reduces noise, without reducing bandwidth, but requires a repetitive signal.

high resolution each displayed point is calculated as the average value of the multiple samples captured within one waveform interval. Provides higher resolution, and does not require a repetitive signal, but reduces bandwidth.[8]

Persistence: Engineers with gray beards wax poetic about the beautiful gradations of intensity with which analog scopes display waveforms. Digital scopes took a while to catch up, but now they trumpet their ability to do the same, with terms like "digital phosphor," "persistence trace," and "digital persistence."

Single-shot capture: Digital scopes excel in single-shot capture. You can troll through the sampled data post-capture, either manually (by turning knobs for scrolling and magnifying) or with some helpful automated search tools (e.g., Tektronix's "Wave Inspector" – the name says it all).

Slow sweep: Analog scopes are hopeless when you want to view a waveform that takes many seconds; digital scopes couldn't care less. Use "rolling mode" at slow sweep rates.

Save/Recall: You can save one or more waveforms to memory, bring them back for comparison, etc. You can also save the scope's *state* (i.e., settings).

Measurements: The data's all there, so digital scopes have no problem measuring period, frequency, amplitude, time interval, duty cycle, etc. These measurements usually update continuously, and you can use settable horizontal and vertical cursors to define the measurement regions and intervals.

Math: Going further, digital scopes let you calculate products (e.g., to measure power from voltage and current), quotients (to normalize a waveform), jitter, histograms, frequency spectra, etc. Almost limitless possibilities,

[8] You can think of this as a "horizontal average" along one waveform capture, as compared with the "vertical averaging" of single sample points in successive stacked waveforms in *average* mode.

but you may prefer to extract the data and do the math offline.

I/O: You can send waveforms and data out (via connected Ethernet or whatever), and you can control the scope's operations remotely. A networked data-acquisition system!

Mixed-signal: Many digital scopes come with a bunch of 2-level channels (typically 16 or 32) along with the usual 2 or 4 full-resolution channels; so it works as a logic analyzer, but augmented by a few channels of clear waveform view. As with a traditional logic analyzer, you can trigger on a defined set of levels, and it can do bus decoding, bus triggering, and other fancy stuff for you.

Smart trigger: Good digital scopes let you trigger on just about any condition you can imagine: pulse width $<$, $>$, $=$, or $\neq$ to some value; runt pulses and glitches; setup or hold time violations; specified range of risetimes or falltimes; specified conditions or violations on serial buses; trigger after n events; and so on. (checkout the enjoyable reading in a datasheet from Tektronix, LeCroy, Keysight/Agilent, or Rohde & Schwarz)

Limit/Mask testing: You can set up a template and detect out-of-spec waveforms, for Go/No-Go testing on a production line; ditto for jitter and other measurable parameters.

Autoset: It's easy to get lost in this multi-dimensional wilderness; digital scopes provide a rescue button (autoset, autoscale, or some linguistic variant, depending on manufacturer), which will at least get something going on the screen (but see the Cautions, next).

Probe skew: When you're using several different probing systems (e.g., passive $10\times$ probe, active FET probe, current probe) the signal delays can vary by tens of nanoseconds or more, completely disrupting the fidelity of the multichannel display (hey, you may be tricked into thinking you've got a violation of causality – effect precedes cause![9]).

Probe readout, Probe power: Probes for use with contemporary digital scopes have extra connections by which they communicate their attenuation factor ($\times 1$, $\times 10$, $\times 100$) and other useful scale factor information (e.g., the amps/div of a current probe); they use such connections also to send power to the probe (needed, for example, with FET active voltage probes or Hall-effect current probes). This can be an annoyance, though, if your scope's input connectors are of the wrong format (which happens even within one manufacturer's scope offerings).

[9] As does the protagonist in Asimov's delightful short story from 1960: *Thiotimoline and the Space Age*.

O.2.2 Some cautions

There's not much not-to-like about digital oscilloscopes. But here are a few cautions, ways in which a digital scope can trick the unwary.

A. Aliasing

This one can fool even the experienced scope user: digital scopes are designed such that the *maximum* sample rate is always adequate for signals up to the scope's full bandwidth; but when you are running at a slower sweep rate in "sample" mode (i.e., one sample per displayed waveform point) the effective sample rate is much lower. So you may see some serious nonsense (jittery unstable signal, inability to trigger, weird change of waveform when sweep rate is changed, etc.) if there's a high frequency signal present.[10]

If you suspect aliasing, try speeding up the sweep, or changing to **PEAK DETECT** mode. Aliasing can be really annoying when you're dealing with signals that combine timescales (the classic one was analog television, with a 3.59 MHz color carrier on a $\sim$15 kHz horizontal line frequency).

B. Dead time

For human visual perception it's necessary to update the scope display at only $\sim$100 times per second or so. If the scope captures input waveform data only at that rate, the fraction of time that it's sensitive to important signal events (like a glitch or timing violation) may be exceedingly low. For example, at a middling sweep rate of 1 μs/div (thus 10 μs per sweep) a scope that's updating 100 times/second is active only 0.1% of the time.

When scope users became aware of this ugly secret, scope manufacturers went at it, and they now provide some measure of true update rate (usually in the form of "waveforms per second," typically in the range of 100,000–1,000,000). Be careful when evaluating such metrics, because there's more than a little "specsmanship" going on.[11]

[10] Dare we admit? One of the authors was testing a circuit that operates at frequencies in the kilohertz range, driving it with a digital function generator at 1.0 kHz. Go out to lunch, come back, look at the waveform – the scope is broken, won't trigger, jittery waveform sliding left and right. Weird. Tried everything. Then noticed that the generator had defaulted to its 1.0 *mega*hertz setting. Ha! Problem solved (and I won't ever tell anyone how dumb I was).

[11] Ask your scope salesperson, they love to flame the lying competition.

C. Lost in a multidimensional vector space

Analog scopes are simple, and you can see the full state of the instrument just by looking at the knobs and indicators. No such luck with the immense capabilities of digital scopes. Early digital scopes were particularly troublesome, lacking annunciators and (mostly) knobs. They've improved enormously,[12] but it's still awfully easy to be sitting in front of a scope that just isn't triggering, or showing significant vital signs. It takes some keen intuition to know which menu to pull down (horizontal? trigger? mode?) to get to the problem. It may be as simple as triggering on the wrong channel; or it might be that you've left the display in **AVERAGING** mode, and lack of a stable repetitive trigger produces a bunch of garbage. And you can even waste a minute not realizing that the thing is in **SINGLE-SWEEP** or **STOP** mode.

[12] We are particularly fond of the "QuickMenu" feature that was introduced by Tektronix in their original TDS3000-series "lunchbox" scopes. Inexplicably, this highly useful feature has been eliminated (despite our howls of protest) in Tek's successor scopes. We've been badgering them ever since.

D. The scope is lying to you

When observing signals with a digital scope, you may be victimized by a blessing (a vast array of measurement capabilities and settings) that becomes a curse (the scope's settings are not what you think). It's easy to forget some obscure but important settings that falsify the measurements you think you're making. For example, it's easy to forget (we've done it, often) that you've left an earlier **PROBE SKEW** compensation in effect, or that some channels still have **BANDWIDTH LIMIT** set. Such oversights corrupt your measurements in less-than-obvious ways that you may not notice for quite a while; and when you do, you're sentenced to serve time repeating the measurements properly.

ACRONYMS AND ABBREVIATIONS

APPENDIX **P**

Electrical engineers are fond (*too* fond, some would say) of acronyms and abbreviations, a familiarity that every educated circuit designer must necessarily acquire. To assist in that education, and for handy reference, we here provide a lightly annotated list of terms used in this book.

ac: literally "alternating current" (i.e., alternating *voltage*); more generally a varying signal
AC(T): advanced CMOS (logic family)
A/D: analog-to-digital
ADC: analog-to-digital converter
ADI: Analog Devices Inc.
AES: Audio Engineering Society
AFC: automatic frequency control
AGC: automatic gain control
AGF: all-glass fiber
AHC(T): advanced high-speed CMOS (logic family)
ALS: advanced low-power Schottky (logic family)
ALU: arithmetic logic unit (in a processor)
ALV: advanced low-voltage (logic family)
AM: amplitude modulation
ANSI: American National Standards Institute
APD: avalanche photodiode
APF: all-plastic fiber
ARM: a popular processor architecture from ARM Holdings
ASIC: application-specific full-custom integrated circuit
ASCII: American Standard Code for Information Interchange
ASF: all-silica fiber
ASSP: application-specific standard product
ATA: advanced technology attachment (a disk interface; see PATA, SATA)
ATAPI: ATA packet interface (a generalized ATA)
ATE: automated test equipment
ATM: asynchronous transfer mode
ATSC: Advanced Television Systems Committee (digital TV standards)
AUC: advanced ultra-low-voltage CMOS (logic family)
AVC: advanced (low)-voltage CMOS (logic family)
AVR: a microcontroller family from Atmel Corp.
AWG: American wire gauge
AZ: auto-zero

BBM: break-before-make (switch)
BCD: binary-coded decimal
BGA: ball-grid array (an IC package)
BJT: bipolar junction transistor
BNC: bayonet Neill–Concelman (connector)
BPSK: binary phase-shifting keying
BRT: bias resistor transistor
BSS: broadcast satellite services
BV: breakdown voltage
BW: bandwidth

C0G: low tempco (stable) ceramic dielectric
CA: conditional access
CAN: controller area network (bus)
CANH: controller area network high (a CAN signal)
CANL: controller area network low (a CAN signal)
CBR: constant bitrate (coding)
CCD: charge-coupled device
CCFL: cold-cathode fluorescent lamp
CCM: continuous-conduction mode (in a power converter)
CD: compact disc (optical storage)
CDMA: code-division multiple access
CDR: clock and data recovery
CES: Consumer Electronics Show
CF: compact flash (memory card)
CFB: current feedback (op-amp)
CHE: channel hot-electron
CLB: configurable logic block
CML: current-mode logic
CMOS: complementary metal-oxide semiconductor
CMRR: common-mode rejection ratio
codec: coder–decoder
cPLD: complex programmable-logic device
CPU: central processing unit
CR: carriage return
CRC: cyclic redundancy checksum
CRT: cathode-ray tube
CSMA: carrier-sense multiple-access
CSP: chip-scale package

CTR: current transfer ratio
CTS: clear to send (in a serial link)
CVBS: composite video, blanking, and sync
CVSD: continuously variable-slope delta-modulation

DA: dielectric absorption
D/A: digital-to-analog
DAC: digital-to-analog converter
DAQ: data-acquisition system
DBS: direct broadcast satellite
dc: direct current (i.e., a fixed voltage)
DCE: data communications equipment (in a serial link)
DCM: discontinuous-conduction mode (in a power converter)
DDR: double data rate (memory)
DDS: direct digital synthesis
DFC: digital frequency converter
DIN: Deutsches Institut für Normung (a German standards organization); a connector series
DIP: dual in-line package
DIR: direction (a control signal)
DMA: direct memory access
DMM: digital multimeter
DNL: differential nonlinearity
DPCP: display-port content protection
DPDT: double-pole double-throw (switch)
DRAM: dynamic random-access memory
DSBGA: die-size ball-grid array (an IC package)
DSL: digital subscriber line (for data over telephone line)
DSP: digital signal processing (or processor)
DSR: data set ready (in a serial link)
DTE: data terminal equipment (in a serial link)
DTH: direct-to-home (satellite TV)
DTL: diode–transistor logic
DTR: data terminal ready (in a serial link)
DUT: device under test
DVI: digital visual interface (for digital video)
DVM: digital voltmeter
DVR: digital video recorder

EAROM: electrically alterable read-only memory
EAS: avalanche energy specification
ECL: emitter-coupled logic
EEPROM: electrically erasable programmable read-only memory
EIA: Electronic Industries Alliance (standards and trade organization)
EMF: electromotive force (∼voltage)
EMI: electromagnetic interference
ENOB: effective number of bits
EPROM: erasable programmable read-only memory
eSATA: external SATA interface
ESD: electrostatic discharge
ESL: equivalent series inductance
ESR: equivalent series resistance
ETF: 8-to-14 (digital coding)

FCC: Federal Communications Commission
FDNR: frequency-dependent negative resistor
FED: field-emission display
FET: field-effect transistor
FFS: fringe-field switching (an LCD display technology)
FFT: fast Fourier transform
FG: floating gate
FGA: floating-gate array
FIFO: first-in-first-out (memory)
FIR: finite-impulse-response (filter)
FM: frequency modulation
F-N Fowler–Nordheim tunelling
FOT: fiber-optic transceiver
FPBW: full-power bandwidth
FPGA: field-programmable gate array
FR-4: "flame-retardant 4" (glass-epoxy PCB material)
FRAM (also FeRAM, F-RAM): ferroelectric random-access memory
FSB: front-side bus (of a computer processor)
FSE: full-sunlight equivalent (you saw it here first!)

GAL: generic array logic
GBP, GBW: gain-bandwidth product
GCC: GNU C-compiler
GCPW: grounded coplanar waveguide
GDT: gas-discharge tube
GIC: generalized impedance converter
GMR: giant magnetoresistance
GND: ground
GPIB: general-purpose interface bus
GPL: graphical programming language; general public license (in GNU)
GPS: global positioning system
GPU: graphics-processor unit
GUI: graphical user interface

HAPD: hybrid avalanche photodiode (detector)
HBM: human body model
HC(T): high-speed CMOS (logic family)
HDCP: high-bandwidth digital content protection
HDD: hard-disk drive
HDL: hardware description language
HDMI: high-definition multimedia interface (for digital display)
HDTV: high-definition television
HI: human interface

HP: Hewlett–Packard
HV: high voltage

IC: integrated circuit
ICSP: in-circuit serial programming
IDC: insulation displacement connector
IDE: integrated development environment (for coding)
IEC: International Electrotechnical Commission
IEEE: Institute of Electrical and Electronic Engineers
IF: intermediate frequency (in RF receiver)
IGBT: insulated-gate bipolar transistor
IGFET: insulated-gate field-effect transistor
IIC (I^2C): inter-integrated-circuit (a serial bus)
IIR: infinite-impulse-response (filter)
INA: instrumentation amplifier
INL: integral nonlinearity (in A/D conversion)
I/O: input-output
IP: Internet protocol; intellectual property
IPS: in-plane switching (an LCD display technology)
IR: infrared
ISA: International Society of Automation (a standard-setting organization)
ISI: intersymbol interference

JFET: junction field-effect transistor
JTAG: Joint Test Action Group (an IC interface)

KCL: Kirchhoff's current law
KVL: Kirchhoff's voltage law

LAB: logic array block (in programmable logic)
LAN: local area network
LCD: liquid- crystal display
LCX: low-voltage CMOS crossvolt (logic family)
LDO: low-dropout (linear voltage regulator)
LE: logic element (in programmable logic)
LED: light-emitting diode
LFSR: linear feedback shift register
LIFO: last-in first-out (memory)
Li-ion: lithium ion (battery)
LIN: local-interconnect network (bus)
LNA: low-noise amplifier
LNBF: low-noise block-downconverter plus feed (for satellite TV)
LO: local oscillator (in RF receiver)
LPF: lowpass filter
LPT: line printer (a parallel port)
LS: low-power Schottky (logic family)
LSB: least-significant bit
LSI: large-scale integrated circuit
LUT: lookup table
LV: low-voltage (logic family)
LVC: low-voltage CMOS (logic family)
LVDS: low-voltage differential signaling
LVPECL: low-voltage positive emitter-coupled logic
LVX: low-voltage crossvolt (logic family)
LXI: LAN eXtensions for Instrumentation

μC: microcontroller
MAC: multiplier-accumulator; media-access control
MBB: make-before-break (switch)
MCU: microcontroller unit
MDAC: multiplying digital-to-analog converter
MEMS: microelectromechanical system
MFB: multiple-feedback (active filter)
MIPS: mega-instructions per second
MLC: multilevel cell (in nonvolatile memory)
MMU: memory management unit
modem: modulator-demodulator
MOS: metal-oxide semiconductor
MOSFET: metal-oxide semiconductor field-effect transistor
MOV: metal-oxide varistor (surge protector)
MPTS: multiprogram transport stream (in digital TV)
MPU: microprocessor unit
MRAM: magnetoresistive random-access memory
MRI: magnetic resonance imaging
MSB: most-significant bit
MSI: medium-scale integration
MUX: multiplexer

NAN: not a number
NC: normally closed (switch)
NECL: negative emitter-coupled logic
NEMA: National Electrical Manufacturers Association (a standards-setting organization)
NIC: negative-impedance converter; network interface card
NiCd: nickel cadmium (battery)
NiMH: nickel metal-hydride (battery)
NIST: National Institute of Standards and Technology
NMI: nonmaskable interrupt
NMR: nuclear magnetic resonance
nMOS: n-type metal-oxide semiconductor
NO: normally open (switch)
NP0: low tempco (stable) ceramic dielectric
NRZ: nonreturn to zero (data code)
NRZI: nonreturn to zero inverted (data code)
NSC: National Semiconductor Corporation (now part of TI)
NTC: negative temperature coefficient
NTSC: National Television System Committee (analog TV standard)
NV: nonvolatile

NVM: nonvolatile memory

O/C: open-collector (logic output)
OCXO: oven-controlled crystal oscillator
O/D: open-drain (logic output)
OEM: original equipment manufacturer
OLED: organic light-emitting diode
op-amp: operational amplifier
OSI: open systems interconnection (network hierarchy)
OSR: oversampling ratio
OTA: over-the-air (broadcasting)
OTP: one-time-programmable NV memory

PAL: programmable array logic
PARC: Palo Alto Research Center
PATA: parallel ATA interface
PC: printed circuit; personal computer
PCB: printed circuit board
PCF: plastic-clad fiber
PCI: peripheral component interface (a computer bus)
PCIe (also PCI-E): extended peripheral component interface
PCM: pulse-code modulation
PCMCIA: Personal Computer Memory Card International Association (a card interface standard)[1]
PDA: personal digital assistant
PECL: positive emitter-coupled logic
PEN: polyethylene naphthalate (a capacitor dielectric)
PFC: power-factor correction (in ac-powered converters)
PFD: phase-frequency detector
PFM: pulse-frequency modulation
PGA: programmable gain amplifier
PID: proportional-integral-differential (in control systems); program identifier (in digital TV)
PIN: positive-intrinsic-negative (diode)
PI/PO: parallel-in-parallel-out
PIR: passive infrared (detector)
PIV: peak inverse voltage
PLA: programmable logic array
PLC: powerline cycles
PLD: programmable logic device
PLL: phase-locked loop
pMOS: p-type metal-oxide semiconductor
PMT: photomultiplier tube
POF: plastic optical fiber
POL: point-of-load
pp: peak-to-peak (voltage)
PPS: polyphenylene sulfide (a capacitor dielectric)
PRAM: phase-change random-access memory
PRBS: pseudorandom bit sequence
PROM: phase-change read-only memory
PSRAM: pseudostatic random-access memory
PSRR: power-supply rejection ratio
PTAT: proportional to absolute temperature
PUJT: programmable unijunction transistor
PV: photovoltaic (light detector)
PVC: polyvinyl chloride (insulator)
PVR: personal video recorder
PWM: pulse-width modulation

QAM: quadrature amplitude modulation
QPSK: quadrature phase-shift keying

RAM: random-access memory
RCO: ripple-clock output
RD: receive data (in a serial link)
RF: radiofrequency
RFI: radiofrequency interference
RG-nn: "Radio Guide" (coax cable designators)
RGB: red–green–blue (video signals)
RISC: reduced instruction set computing
RLL: run-length limited (digital codes)
rms: root-mean-square
ROM: read-only memory
RRI: rail-to-rail input
RRIO: rail-to-rail input and output
RRO: rail-to-rail output
RTC: real-time clock
RTD: resistance temperature detector (or resistive temperature device)
RTI: referred to the input
RTL: resistor–transistor logic; register-transfer level (in an HDL)
RTO: referred to the output
RTS: request to send (in a serial link)

SA: sense amplifier
SACD: Super Audio compact disc
SAD: silicon avalanche device (i.e., a zener TVS)
SAR: successive approximation register
SAS: serial attached SCSI (interface)
SATA: serial ATA (interface)
SAW: surface acoustic-wave
SBC: single-board computer
SC: subscriber connector (a fiber-optic connector)
SCPI: Standard Commands for Programmable Instruments
SCR: silicon-controlled rectifier
SCSI: small computer system interface
SD: secure digital (memory card)
SDI: serial data in

[1] Whose awkwardness spawned jokes like "Personal Computer Manufacturers Can't Invent Acronyms."

SDO: serial data out
SDR: single data rate (memory)
SDRAM: synchronous dynamic random-access memory
SDTV: standard-definition television
SE: single-ended
SED: surface-conduction electron-emitter display
SEPIC: single-ended primary-inductance converter
SERDES: serializer–deserializer
S/H: sample-and-hold
SHV: "safe high voltage" (connector)
SI: serial input
SIP: single in-line package
SMA, SMB, SMC: subminiature RF coax connector series
SMI: small media interface (a fiber-optic connector)
SMPS: switch-mode power supply
SMT: surface-mount technology
SMU: source-measure unit
SNR: signal-to-noise ratio
SO: serial output; small-outline (IC package)
SOA: safe-operating area
SODIMM: small-outline dual-in-line memory module
SOIC: small-outline integrated circuit
SOT: small-outline transistor
SPDIF: Sony–Philips Digital Interconnect Format (for digital audio)
SPICE: "simulation program with integrated circuit emphasis" (analog circuit simulator software)
SPDT: single-pole double-throw (switch)
SPI: serial peripheral interface (a simple IC bus)
SPL: sound pressure level
sPLD: simple programmable logic device
SPST: single-pole single-throw (switch)
SPTS: single-program transport stream (in digital TV)
SR: slew rate
SRAM: static random-access memory
SSD: solid-state drive (an NV memory)
SSH: secure shell (a network protocol)
SSP: synchronous serial port
SSR: solid-state relay
ST: straight tip (a fiber-optic connector)
STB: set-top box (for cable or satellite TV)
STP: shielded twisted pair (cable)
SWR: standing-wave ratio (on a transmission line)

T&M: test and measurement
TAC: time-to-amplitude conversion
TBH: take-back-half (a control algorithm)
TCP: transmission control protocol (an Internet protocol)
TCXO: temperature-compensated crystal oscillator
TD: transmit data (in a serial link)
tempco: temperature coefficient
THD: total harmonic distortion
TI: Texas Instruments
TNC: threaded Neill-Concelman (connector)
TO: transistor outline (e.g., TO-92, TO-220)
TSSOP: thin-shrink small-outline package
TTL: transistor–transistor logic
TVS: transient voltage suppressor
TWI: two-wire interface (a serial bus)

UART: universal asynchronous receiver-transmitter
UDP: user datagram protocol (an Internet protocol)
UHF: ultrahigh frequency; also a legacy coaxial connector
UL: Underwriters Laboratories (a safety certification company)
UPS: uninterruptible power supply
USB: universal serial bus (a data interface)
UTP: unshielded twisted pair (cable)
UV: ultraviolet

VBR: variable bitrate (coding)
VCO: voltage-controlled oscillator
VCVS: voltage-controlled voltage-source (active filter)
VCXO: voltage-controlled crystal oscillator
VDE: Verband der Elektrotechnik, Elektronik und Informationstechnik (a German organization whose activities include safety standards)
V/F: voltage-to-frequency (converter)
VFB: voltage feedback
VFD: vacuum fluorescent display
VGA: video graphics array (640×480 analog video)
VHS: video home system (video recording)
VLSI: very-large-scale integration
VME: VERSAmodule Eurocard Bus (a card interface)
VOD: video-on-demand
VOM: volt-ohm-millimeter
VSWR: voltage standing-wave ratio (on a transmission line)
VU: volume unit (an audio level)

WL: write latency (in computer memory)

X7R: a ceramic dielectric
XLR: a professional audio connector series
XO: crystal oscillator

Y5V: a ceramic dielectric
YIG: yttrium–iron garnet

Z5U: a ceramic dielectric
ZCS: zero-current switching
ZVS: zero-voltage switching